PREVENTIVE AND SOCIAL MEDICINE WORKBOOK

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MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

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LEVELS OF PREVENTION

LEVELS OF PREVENTION

Level	Definition	Phase

PATHOGENESIS OF DISEASE

METHODS TO ACHIEVE DIFFERENT LOPS

1. Primordial: .....

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2. Primary: A) Specific Protection (individual)

B) Health Promotion (community)

- ✦ Use of RO water in a family
- ✦ Safe water supply, sanitation
- ✦ Anaemia Mukt Bharat Program
- ✦ Useof IFA tab in pregnancy
- ✦ Use of Helmet while driving
- ✦ Law for use of helmet while driving

(Q): Inclusion of T.T. Vaccine in UIP

Question 1
Primordial
Primary
Secondary
Tertiary

Question 2
Primordial
Primary
Specific
Health promotion

3. Secondary: .....

4. Tertiary: .....

.....

ALGORITHM TO DIAGNOSE LEVELS OF PREVENTION

Features	Primordial	Primary	Secondary	Tertiary
Risk factor/ agent				
Disease				
Complications				
Disability				

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EXAMPLES OF DIFFERENT LEVELS OF PREVENTION

1. Insecticidal spray in malaria endemic area in Rainy reason
2. Use of Cervical Coller
3. Change of water in cooler every week - .....
4. Screening for colour blindness .....
5. Chemoprophylaxis - .....
6. Emergency Contraceptive (I-Pill) .....
7. Annual Health check up- .....
8. Encouraging kids for outdoor games. .....
9. Wound cleaning after dog bite .....
10. Wound cleaning after dog bite, Vaccine .....
11. Wound cleaning after dog bite, vaccine, IG .....
12. Disability Limitation / Prevention .....
13. Quit smoking- .....
14. Weight reduction- .....
15. Life style modification- .....

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VACCINES

SCIENTIST AND VACCINE

....

....

TYPES OF IMMUNITY

- Active
- Passive

Active and passive immunity can be given simultaneously in

Not given simultaneously .....

.....

.....

TYPES OF VACCINES

1. Live Attenuated Vaccine:
   (Antigenicity maintained, pathogenicity destroyed)
   Smaller doses- long lasting responses

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2. Killed vaccines:

Killed by chemicals and heat

Major antigens persist, smaller antigens destroyed.

Given as multiple doses, booster doses.

3. Both live and killed form:

Polio

Influenza

Typhoid

Cholera

Hepatitis A - .....

Japanese Encephalitis - .....

(Q): Killed vaccine is ?

(1)

- Hep B
- Measle
- BCG
- JE

(2)

- JE
- YF
- BCG
- Measle

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4. OTHERS:

a) Toxoids: .....

.....

b) Polysaccharide Vaccines:

- Pneumococcal vaccine: PCV23
- Meningococcal
- Hemophilus Influenza B (Hib)
- V i polysaccharide (killed typhoid)

These vaccines are C/I in children < 2 yrs

c) Conjugate:

Safely given in children <2 yrs

- PCV (PCV13, PCV14)
- Hib
- Killed Typhoid
- Meningococcal

d) Combined vaccines

- MMR
- DPT
- Pentavalent .....

e) Recombinant vaccines

- Hepatitis B- .....
- Rabies
- HPV

f) Proteinaceous

- Killed influenza
- Whole cell pertussis

g) Sub unit vaccine

- Hepatitis B
- Corona vaccine

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HPV Vaccine

Quadrivlent	Nanovalent
(6, 11, 16, 18)	(6, 11, 16, 18, 31, 33, 45, 52, 58)
Gardacil	Gardacil - 9
Cervavac	

0.5ml I,M

9-14 Yrs : 2 doses → 0, 6 month

15-26 Yrs : 3 doses → 0, 2, 6 month

STRAINS OF VACCINES

- BCG - .....
- Measles- .....
- Mumps- .....
- Rubella- .....
- Yellow fever - .....
- Chicken pox (varicella) - .....
- JE

- Polio
  - OPV - .....
  - IPV- .....

Switch day: .....

.....

.....

- National Validation Day
  - 9th May 2016
  - India certified tOPV free after confirmation of complete withdrawl of tOPV

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- Dengue: CYD-TDV
- Rota virus: RIX 4414
- Leprosy vaccine: MIP (Mycobacterium Indicus Pranii)
- HIV: mVA (modified vaccinia ANKARA)
- Malaria: RTS,S (ASO1), R-21/Matrix-M
- Influenza: A7/2009/California

UNIVERSAL IMMUNIZATION PROGRAM (UIP)

Started on Nov 19, 1985

✦ National Immunization Schedule (NIS)

At birth (hospital delivery) Home delivery	BCG Next visit	OPVo 0-15 days	Hep B b Not given				
6 weeks	Penta Valent 1	OPV1	RVV1	IPVf1	PCV1		
10 weeks	Penta Valent 2	OPV2	RVV2				
14 weeks	Penta Valent 3	OPV3	RVV3	IPVf2	PCV2		
9-12 months	Measles containing Vaccines MCV1 (MMR/ MR)	PCV B	JE1 (endemic area)	IPVf3	VIT A 1st dose (2nd -9th dose : .... Total dose : ..... <1 yr : ..... >1 yr : .... Total amount of Vit A :		

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1 ½ yr (16-24 months)	OPVB	DPTB1	MCV2	JE2				
5 yr	DPTB2							
10 yr	Td							
16 yr	Td							

Pregnancy vaccination:

- Primigravida: .....
- 1st dose - .....
  2nd dose -.....
- Minimum gap between 2 doses: .....
- Multigravida: Last delivery <3 yrs back- .....
  >3yrs back- .....

DELAYED IMMUNIZATION AS PER NIS

- Till 24hr : Hep B_B
- 0 - 15 days : OPV_O
- Till 1yr : Pentavatent, Hep B, IPV, PCV, BCG
- Till 5yr : OPV, MCV, Measle, Vit A
- Till 6yr : HIB
- Till 7yr : DPT
- Till 15yr : JE
- Till 16yr : Td
- Rotavirus: ..... 1st dose: .
  Last dose upto .....

(Q) : A mother brings child of 5 months age for first time to the pediatrician. What vaccine can be given?

Ans. .....
.....

☞ If there is a gap of >4 weeks in vaccine dose, ...

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DILUENTS OF VACCINE

1. NS (NaCl)
   - BCG Hib Dengue
   - YF JE .....

☞ Note: Dissolved vaccine - used within ....
.

2. Distilled Water
   - Measles
   - MMR

☞ Note: Vaccines available already in liquid form need no diluents.
Eg : DPT, Hep B, TT

DOSES OF VACCINES

- BCG - .....
- IPV- .....
- OPV: .....
- RV: .....
- Vit A .....
- (DPT, Hep B, Hib, Measles, MMR, PCV, JE) .....

VARIOUS ROUTES OF ADMINISTRATION

.....

.....

.....

.....

.....

.....

- I/M: <2 yrs, I/M vaccine given in the anterolateral aspect of middle one third of thigh, not in shoulder. DPT, HepB, Hib, Pentavalent JE (Killed) IPV, Rabies, Tetanus, Td

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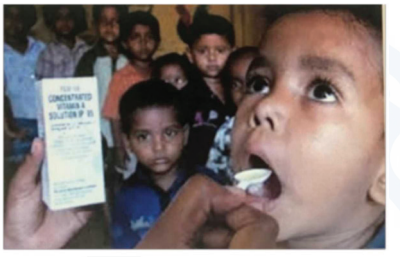
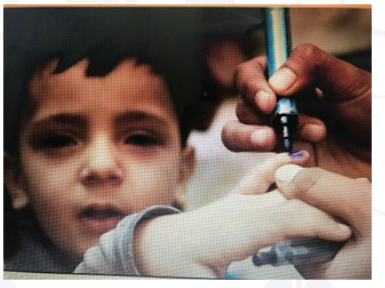
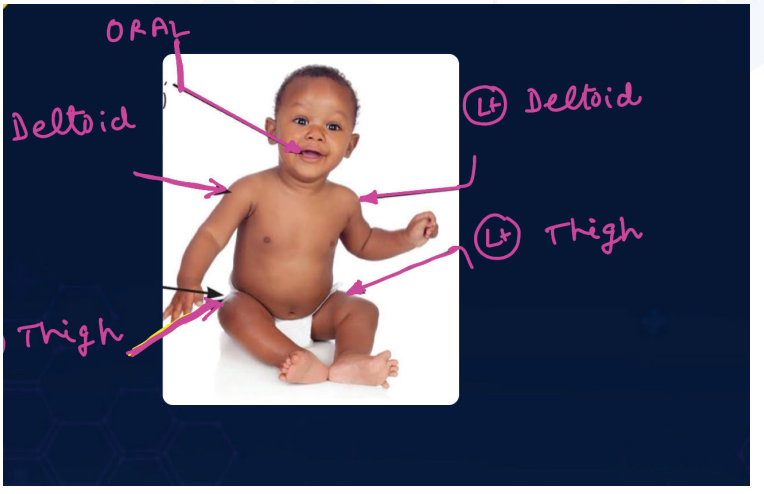
❑ IMAGE BASED QUESTIONS:







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CONTRAINIDICATIONS OF VACCINE

1. Pregnancy

(Q): VACCINES C/I IN PREGNANCY

Question 1	Question 2	Question 3
Hepatitis B	Tetanus	Tetanus
Rabies	Yellow fever	Yellow Fever
Yellow Fever	Measles	Measles
Tetanus	Rabies	Rubella

2. Patient is on corticosteroids

3. Patient is HIV + ve

4. H/O anaphylaxis

5. H/O fever: .....

6. H/O egg allergy:

7. presence of active neurological diseases- .....

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ADVERSE EVENTS FOLLOWING IMMUNIZATION

1. OPV- .....
2. BCG- .....
3. Hep B- .....
4. DPT- .....
   -
   -
   -
   -
   -
5. TT- .....
   .....
6. RV- .....
7. Measles-.....
8. MMR- Thrombocytopenia
9. Influenza (killed)-.....
10. Varicella- Febrile seizures.
11. YF- Vaccine Ass. Viscerotropic Ds.

☞ NOTE: Keep the baby for 30 min after vaccine administration to observe for any side effects.

STORAGE AND COLD CHAIN

Cold Chain: Transfer of vaccine from manufacturer to recipient

Temp: .....

(Q): Reverse Cold Chain: .....
.....

Temp: + 2 to + 8 degrees Celsius.

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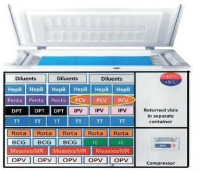
Instruments required for storage of vaccines



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- ILR (Ice Line Refrigerator): .....

  .....
- Temperature is monitored by : .....
- Long Term Storage : .....
  - Room Temp: .....
  - <0 degrees Celsius (-15'C to -25'C): .....
  - + 2 to + 8 degrees Celsius: DPT, TT, Hep B, JE, BCG, PCV, DILUENT.
- Lowest storage center : .....

  AT PHC All vaccines including OPV, Measles, Rotavirus, are stored in ILR. Stored upto 1 month
- Deep Freezers: Used in Ice pack preparation in PHC
- Maximum chances of cold chain failure : .....
- Transportation Boxes :

....

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Type of Box	Vaccine carrier	Day carrier
Number of Ice Packs		
No of vials		

(Q) 1: most heat sensitive: .....

2: most freeze sensitive/least heat sensitive: .....

Vaccine Vial Monitor

✦ GRADES OF VVM:

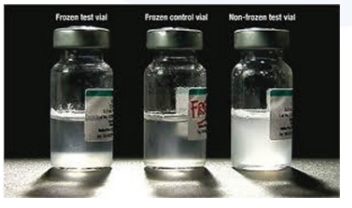
I: Inner square white

II: Inner square colour lighter than outer colour

III: same colour

IV: darker than outer one

Shake Test



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OPEN VIAL POLICY

- Criteria for open vial policy
  - Used .....
  - Cold chain maintained
  - Vaccine withdrawal aseptic
  - Vial should not merged in water
  - Written date of opening.
- Applicable on Multidose vial – .....

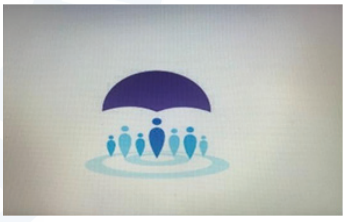
  .....

  Not applicable on .....

  .....

RING IMMUNIZATION

Any suspected case in center,
vaccinate the susceptible children
within ..... of infected case.



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HERD IMMUNITY

- Break in chain of disease transmission

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- No role of herd immunity in: .....

  .....

- Role of Herd Immunity

DS	Threshold
• Measles, pertussis, chicken pox -	.....
• Polio, mumps-	.....

INTENSIFIED MISSION INDRADHANUSH

- ✦ Started in 2017
- ✦ Goal: To vaccinate >90% (upto 100%) for 12 Vaccine preventable diseases by Dec 2018

1. TB	5. PERTUSSIS	9. Hib
2. POLIO (OPV, IPN)	6. TETANUS	10. PCV
3. HepB	7. MEASLES	11. RUBELLA
4. DIPHTHERIA	8. ROTAVIRUS	12. JE

- ✦ Survey : .....
- ✦ Technique : .....

SEROCONVERSION TIME

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EXTRA EDGE

- Vaccination Status
  1. Full immunization : Child who had received BCG, 3 doses of OPV, 3 doses of pentavalent and one dose of MCV (MR / MMR) by 1yre of Age
  2. FIC plus : Child who had received all vaccine given under UIP by 1 yr of Age
  3. Complete Immunization : child who had received all vaccines given under UIP by 2 yr ofAge
  4. Drop out : Child who had received few vaccines but did not complete schedule as per age under UIP
  5. Left out : Child who had not received any vaccine as per UIP till that Age
  6. Zero dose child : Child who had not received first doseof pentavalent vaccine by 1 yrofAge

- SAFE-VAC : Software for reporting AEFI

SAFE-VAC
Surveillance and Action For Events following vaccination

✦ **U win** : digital portal for tracking vaccination record of pregnant women children 0 5Yr

U-WIN

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- **e VIN** : electronic vaccine intelligence network
  vaccine stocks cold chain monitoring logistic

- **COVINET** : global network of corona virus.
  Aim facilitate and coordinate global expertise and capacities for early and accurate detection monitoring and assessment of SARS-COV2, MERS COV and NOVEL CORONA virus of public Health importance

- 5 saal 7 Bar initiative

- CONDITIONING OF ICE PACK

  Before keeping ice packs in cold boxes after removal from deepfreezer (as temp <0°c) keep for some times outside to bring temp to 0°c to avoid freeze damageof vaccine

  - Check point ->
    1. Water starts to form and ice starts melting inside
    2. On shaking ice pack ice core should move inside

- HOLD OVER TIME

  Time taken by equipment to increase inside cabinet temperature fromtemperature at time of power cut to maximum temperature limit of equipment (+8°C)

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- Types ofVVM
  - ✦ VVM 2, VVM7, VVM 14, VVM30
  - ✦ Number denotes time in days taken by inner square to reach discard point if vial exposed to 37°C constantly
  - ✦ Chemical : Paratoluene sulfonate

- Storage of diluent
  - ✦ at +2-+8°C in ILR
  - ✦ In case of Non availability ofspace in ILR, can be kept outside ILR but always remember to cool for at least 24hr in ILR before use

- Adult BCGvaccination
  - ✦ Prevent TB in adult as a part of NTEP
  - ✦ High Risk adult
    - a) Smokers
    - b) Diabetic
    - c) Exposed to people with TB
    - d) Who are unsure if they Have received BCG before
    - e) >18yr
    - f) Prior TB ds
    - g) under wt

- BRIDGE (Boosting Routine immunization demand generation Expansion)
  Trainning for frontline workers to enhance inter personal communication skill

- NCCMIS (National cold chain Mx information system)
  Monitoring functionality of cold chain equipments

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CONCEPT OF HEALTH AND DISEASE

❑ WHO DEFINITION OF HEALTH

INDICATORS OF QUALITY OF LIFE

PQLI (PHYSICAL QUALITY OF LIFE INDEX)		HDI (HUMAN DEVELOPMENTAL INDEX)	
1. Components		1. Components	
•		•	
•		•	
•		•	
		(a) ..... (b)	
2.	Range :	2.	Range : .....
3.	Value :	3.	Value : .....
4.	Use for Internation Comparison	4.	
5.	Calculation		

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HDI Calculation

- = $\sqrt[3]{\text{Life I.} \times \text{Edu I} \times \text{Income I.}}$
- Index = $\frac{\text{Actual Value - Min. Value}}{\text{Max Value - Min Value}}$

Dimention	Max	Min.	India
1. Life Expectancy			
2. Mean years of schooling			
3. Expected yr of schooling			
4. Per capita Income (PPP &)			

- HDI < 0.55 = Low
  0.55 - 0.699 = Medium
  0.70 - 0.799 = High
  ≥ 0.8 = Very High

Other Indicators

1. Global Hunger Index

Indicator 1) Undernourished population
2) Underwt children (0-5yr)
3) Child mortality (0-5yr)

Score : ≤ 9.9 = Low
10.0 - 19.9 = Moderate
20.0 - 34.9 = Serious
35.0 - 49.9 = Alarming
≥ 50.0 = Extremly alarming

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2. Multi Dimentional Poverty Index

Dimention	Indicator
1. Education -	A. School attendance
	B. Years of schooling
2. Health -	A. Child mortality
	B. Nutrition
3. Living standard -	A. Water
	B. Electricity
	C. Fuel
	D. Sanitation
	E. Housing
	F. Assets

Interpretation (Deprivation%)

- 20 - 33.3% - Vulnerable to poverty
- > 33.3% - Poverty
- > 50% - Severe poverty

Range -> 0 to 1

INDICATORS OF HEALTH

- Mortality indicators
- Morbidity indicators
- Disability indicators

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MORTALITY INDICATORS

❑ 1. Crude death rate (CDR):

- CDR = total no of death due to all causes / 1000 Mid year Population

- For International comparison

  1.) Calculate age specific death rate (standardization)

  2.) Difference in age group population.

❑ Standardization

1. Direct :

- Need Age specific Death Rate
- Standard population
- Calculate expected Death Rate
- S. Death Rate $= \frac{\text{Total expected Death}}{\text{Standard pop}^n} \times 100$

2. Indirect

- Used when Age specific Death Rate NA
- Standardized Mortality ratio $= \frac{\text{Observed Death}}{\text{Expected Death}} \times 100$
- SMR > 100 = Unfavourable
- < 100 = Favourable

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In a village expected death rate is 3/1000, but observed death rate rate is 6/1000, Calculate Standerdized motality ratio

a. 1 b. 2

c. 100 d. 200

❑ **2. Infant mortality rate (IMR):**

- IMR = No of infant death (0-1 yr.) / 1000 Live Birth
- (Q) Very good indicator of health status of a country

❑ **3. Maternal mortality ratio (MMR):**

- MMR = No of maternal deaths / 100,000 Live Birth
- (Q) Very good indicator of availability and utilization of maternal component of MCH services

❑ **4. Life expectancy:**

- Only positive mortality indicator

MORBIDITY INDICATORS

- ✦ Incidence of disease
- ✦ Prevalence of disease
- ✦ Hospital admission rate
- ✦ Bed occupancy rate
- ✦ Bed turnover ratio

DISABILITY INDICATORS

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❑ **1. Disability adjusted life years (DALY)**

- Years lost due to premature death
- Years lost in life due to disability
- 1DALY = 1 year of loss
- Best indicator of burden of DS

❑ **2. Quality adjusted life years (QALY)**

- Life lived in healthy state with or with out adjustment
- Effectivity of intervention
- 1 Qaly = 1yr of life x Utility Value

❑ **3. Health adjusted life expectancy(HALE) :**

- Life expected to live in healthy state with or with out adjustment depending on current mortality and morbidity parameters.
- (Q) Best indicator of Disability

Indicators of burden of ds:

1. DALY
2. Disease prevalence
3. Proportional mortality rate

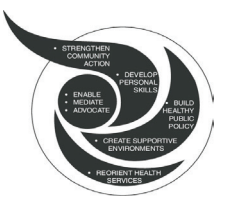
$$PMR = \frac{\text{Total no of death due to specific cause}}{\text{Total no of death due to all causes}} \times 100$$

(Q) Total death reported were 100 in population of 10,000 ; JE deaths were 10 ; calculate PMR of JE

HEALTH PROMOTION

1. ALMA-ATA Declaration 1978
2. OTTAWA Charter 1986
3. JAKARTA Declaration 1997
4. MDG 2000
5. SDG 2016

OTTAWA CHARTER



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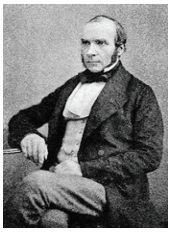
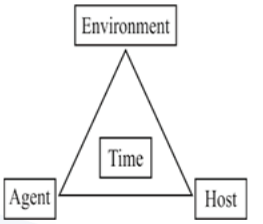
THEORIES OF DISEASE CAUSATION

Epidemiological theory – .....





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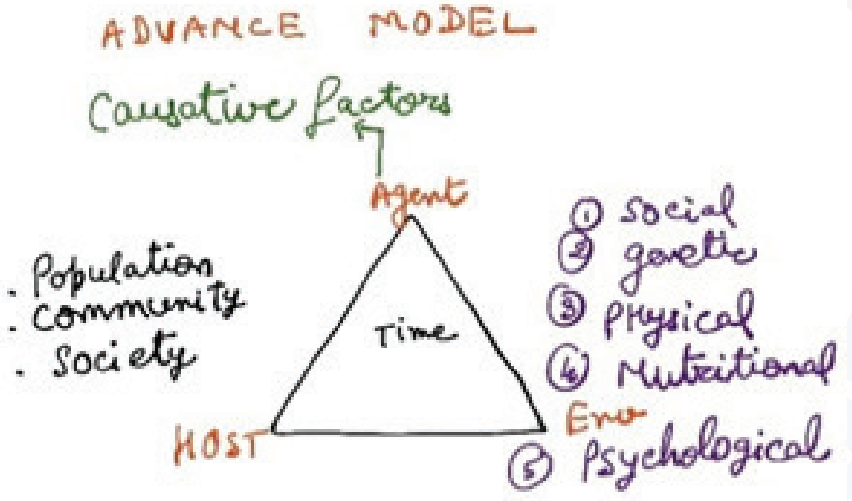
✦ ADVANCED MODEL OF EPIDEMIOLOGICAL THEORY



MANAGEMENT OF DISEASE

1. Control of disease

- Reduction in incidence
- Reduction in prevalence
- Reducing the duration of disease
- Reduction of the patient suffering
- Reducing the financial burden

Eg: AIDS, DM, HTN, CANCERS control

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2. Elimination

- Break in chain of disease transmission
- No new case appear
- Disease agent persist
- Continue surveillance activity

Egs: Diseases eliminated from India

- Dracunculosis (2000)
- Leprosy (2005)
- Yaws (2006)
- Polio (2014)
- Neonatal tetanus (2015)
- Trachoma (2024)

❑ Diseases to be eliminated from India

1) TB .....
2) Measles, Rubella
3) Filariasis .....
4) Leishmaniasis
5) Malaria .....

Eradication

- Complete removal of causative agent from environment
- Stop surveillance activity
- No agent persist

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❑ **Diseases eradicated till now**

- Small pox .....

Virus:

- Polio virus .....
  .....

❑ **Diseases to be eradicated:**

- Polio
- Measles

COMPLICATIONS OF DISEASE

..... (disturbance at cellular level)

.....

..... (anatomical, physiological, psychological loss of any body part or organ)

.....

..... (loss of function i.e. normal for age)

.....

..... (loss of social function)

.....

DEATH CERTIFICATE

It is having 4 lines

- Ia -immediate cause of death .....
- Ib -underlying cause of death .....
- Ic -Main underlying cause of death .....
- II -Associated disease .....

MOST IMPORTANT LINE IS .....

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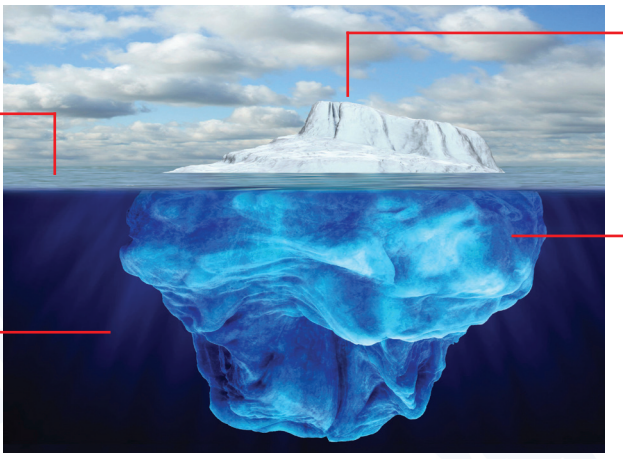
ICE BURG PHENOMENON



❑ **Disease showing this phenomenon (Subclinical cases, carrier present)**

- HIV, Malaria, typhoid, cholera, diphtheria, polio

❑ **Disease not showing phenomenon (Subclinical cases, carrier absent)**

- Rabies, Tetanus, Measles, pertussis

SURVEILLANCE AND MONITORING

❑ **Monitoring is** .....

❑ **Surveillance -** .....

✦ TYPES

1. Active surveillance-Health worker is visiting every home and find out disease and provide treatment to them

   Ex.- Malaria by .....,

   Filariasis, leishmaniasis, leprosy, Tuberculosis by .....

2. Passive surveillance-diseased personal going to hospital to get treatment

   Ex.-noncommunicable diseases, cancers

Active space

3. Sentinel surveillance-mainly for .....

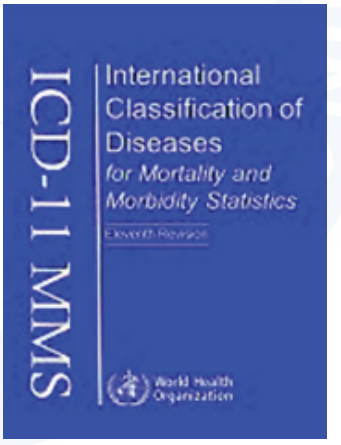
   ✦ Advantages

      - Identifying the missing cases
      - Supplement notified cases

ICD 11

- International classification of diseases
- Currently running 11th revision
- Volume -
- Chapter -
- Alpha numerical classification

Important disease	Chapter
1) Mental, behavioural, psych. ds	: 6
2) Sexual Health	: 17
3) Unclassified Unknown etiology DS	: 21
4) Traditional Medicine	: 26



Active space

EXTRA EDGE

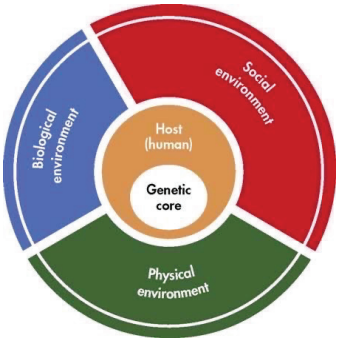
- Theories of Ds causation

1. Germ theory
2. Multifactorial theory
3. Web of causation
4. Epidemiological
5. 'BEINGS' model

This concept postulates that human disease and its consequences are caused by a complex interplay of nine different factors.

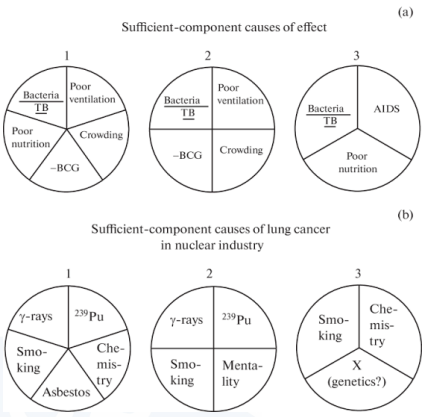
- Biological factors innate in a human being
- Behavioural factors concerned with individual lifestyles
- Environmental factors as physical, chemical and biological aspects of environment.
- Immunological factors
- Nutritional factors
- Genetic factors
- Social factors
- Spiritual factors and
- Services factors, related to the various aspects of health care services.

6. 'WHEEL' of DS Causation



Active space

7. Sufficient component cause model
   k/as ROTHMAN PIE



1. Necessary Cause

   A component cause that is a member of every sufficient cause.

2. Component cause

   Any one of the set of conditions which are necessary for the completion of a sufficient cause. (This is a piece of the pie.)

3. Sufficient cause

   A set of conditions without any one of which the disease would not have occurred. (This is one whole pie.)

Health Index of India (NITI AAYOG)

- Indicators (Total 24)

1. Health outcome 70% (10)
2. Governance and information 12% (3)
3. Key input and processes 18% (11)

- AIM : A tool to track health performance ofstates and union Territories

Fiscal Health Index (NITI AAYOG)

- AIM : Assess financial health of states which help policymakers understand state fiscal health and inform policy interventions

Active space

- ✦ Currently for 18 states which hasmajor contribution in Indian economy concept of well being

Concept of well being

A) Objective :

- Standard of living
- Level of living (United Nation)

(Health, food consumption, education, occupation, working condition, Housing, social security, clothing, leisure and recreation

B) Subjective : Quality of life

Disability Rates

A) Event type indicator

- Number ofdaysof restricted activity
- Bed disability days
- Work loss days (school loss days) within a specified period

B) Person type indicator

- Limitation ofmobility
- Limitation of activity

Disease : Physiologic psychological dysfunction

Illness : Subjective state of person who feels aware of not being well

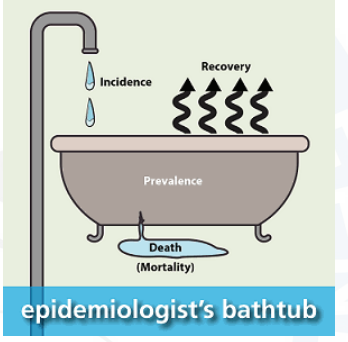
Sickness : State of social dysfunction

Active space

EPIDEMIOLOGY

INTRODUCTION OF EPIDEMIOLOGY

- Father of modern epidemiology – .....
- Father of Public Health - .....



epidemiologist's bathtub

COMPONENTS OF EPIDEMIOLOGY

1. Frequency (Incidence, prevalence)
2. Distribution (time, place, person)
3. Determinant (epidemiological studies)

FREQUENCY: INCIDENCE, PREVALENCE

FEATURES	INCIDENCE	PREVALENCE
Definition		
Expressed as		
Calculated by		
Effect of duration		
Special feature		

Active space

(Q) : Effect of Incidence, prevalence of disease

1. New vaccine in market: .....
2. New antiTB drug for MDR cases which shortens duration of therapy: .....
3. High number of new cases of short duration with 100% mortality: .....

(Q) : Calculate Incidence and prevalence if HIV old cases are 10 and new cases are 2 for year 2024 for Population of 10,000

DISTRIBUTION OF THE DISEASE

❑ **PLACE :**

1. Urban – Non communicable diseases (high)
   Rural – Infectious disease, skin infections, nutritional disease
2. State:
   HIV – southern state
   Filariasis – Bihar, Kerala, UP, AP, TN, WB, Maharashtra, Odisha, Gujrat.
   JE - Assam, Karnataka, Western UP, Bihar, WB, TN, AP, Manipur, Goa.
   Leishmaniasis : WB UP, Bihar, Jharkhand.
   Leprosy : UP, MP, Chattigarh, Bihar, Jharkhand, WB, Odisha, Maharashtra.
3. Country:
   Developing – Infectious disease, nutritional disease
   Developed – Non communicable disease

Active space

PERSON :

1. Age- Measles <5 yrs
   HIV- 30-35 yrs
2. Sex – Lung ca: M>F
   Osteoarthritis: F > M

❑ **TIME :**

- Endemic – ....
- Epidemic – .....
  .....
  .....
  .....
- Pandemic – .....
- Sporadic .....

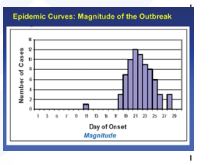
✦ Fluctuation/epidemic –

1. Short epidemic (hour to week)
2. Periodic (recurrence at frequent interval)
3. Long term – decades

➢ 1: Short term -

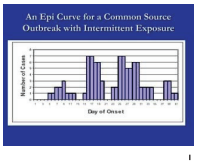
(a) Common source single exposure (point source epidemic)

Eg :
- Food Poisoning
- Bhopal gas tragedy (Gas - methyl isocyanate)
- Vizag gas leak (Gas - Styrene)



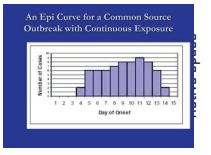
(b) Common source multiple exposure

Eg: Sex worker



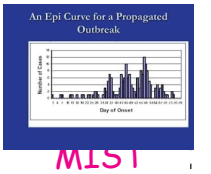
(c) Common source continues exposure

Eg: Water intake from infected well
Milkman



(d) Propagative type

Eg: All water borne infections
Sexually transmitted diseases

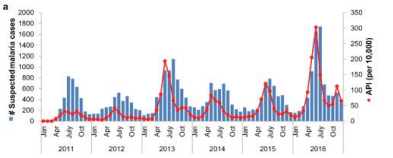


MIST

➢ 2. Periodic

(a) Seasonal variation - change in epidemiological tetrad characteristic

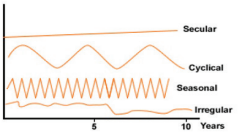
Eg: Dengue, malaria: rainy season
Measles: winter
Dehydration – summer



(b) Cyclical

i. Antigenic variation
Eg : influenza(yearly)

ii. Human behavior
Eg : Road traffic accidents(RTA)- weekends



➢ 3. (Q) Long term (secular trend) - changes seen in decades in one direction only

DETERMINANT

Epidemiological study

Observational		Experimental / Interventional
DESCRIPTIVE Hypothesis formation (time, place, person)	ANALYTICAL Hypothesis testing i) Cohort ii) case control iii) cross sectional iv) ecological	Hypothesis confirmation Randomized study • Clinical trial • Vaccine trial • Preventive trial • Risk factor trial • Cessation trial • Evaluation of health services Non random trial • Uncontrolled trial • Natural experiments • Before /after comparison trial

Active space

STUDY DESIGN OF EPIDEMIOLOGICAL STUDIES

CASE CONTROL STUDY (CASE REFERENCE)

COHORT STUDY (PROSPECTIVE STUDY) (FOLLOW UP STUDY)

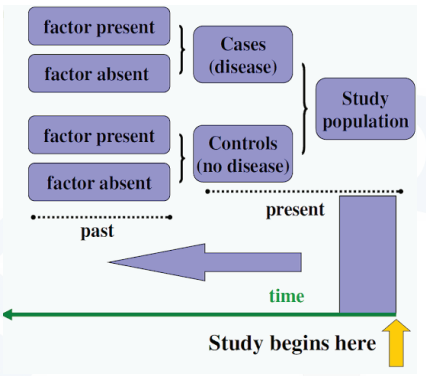
RETROSPECTIVE COHORT STUDY: (HISTORICAL COHORT / NON CONCURRENT COHORT STUDY

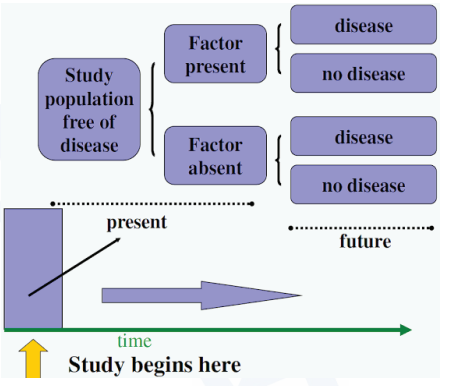
CROSS SECTIONAL STUDY (PREVALENCE STUDY)

Active space

ECOLOGICAL STUDY (CORRELATION STUDY)

VARIOUS TYPES OF EPIDEMIOLOGICAL STUDIES





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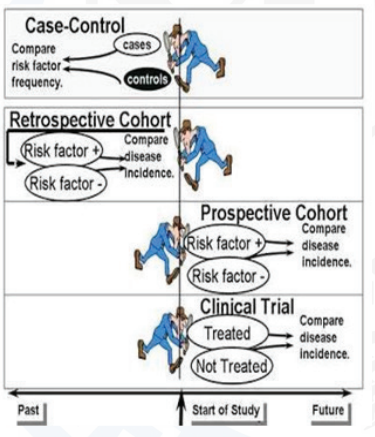

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ANALYTICAL STUDY COMPARISION

FEATURES	COHORT	CASE CONTROL	CROSS SECTIONAL	ECOLOGICAL
Direction	.....	.....	.....	.....
Importance	.....	.....	.....	.....
Unit of study	.....	.....	.....	.....
Presence of disease at start	.....	.....	.....	.....
Parameters				

Active space

IDENTIFICATION OF STUDY

Type of study	Direction	Presence of outcome at start of study
Case control		
Prospective cohort study		
Retrospective cohort study		
Cross sectional study		

❑ EXAMPLES :

1. Study started in 2012 in Delhi university students in a group of 5000-6000 about role of physical activity and results calculated in 2024 about its impact on cardiovascular diseases. Identify type of study?

2. H/O vaccination was studied in 1300 kids with autism and 4400 kids without autism and it was found that there was no association between autism and vaccination, type of study:

3. Study started on post natal mother and their antenatal wt goin correlated with new born birth wt

4. A study done in 3 states to see mean blood pressure in each community. Health workers assigned and visited each houses of these communities and mean blood pressure was compared. what is study design

5. To know average alcohol consumption in a city, a study conducted where total alcohol sale per day was calculated from data recorded from various sources and total population of that city information was taken from SRS data, identify type of study:
   .....

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CASE CONTROL AND COHORT STUDY

CASE CONTROL STUDY :

- Backward study
- Retrospective study
- Outcome to exposure study
- Disease to risk factor study
- Effect to cause

COHORT STUDY :

- Incidence study
- Forward study
- Prospective study
- Exposure to outcome study
- Risk factor to disease study
- Cause to effect

FEATURES	ADVANTAGES OF CASE CONTROL	DISADVANTAGES OF COHORT
Duration		
Cost of the study		
Ethical Risk		
H/o Multiple RF		
Rare disease		
Attrition		
	DISADVANTAGES OF CASE CONTROL	ADVANTAGES OF COHORT
Parameter		
Recall bias		
Multiple Outcome		

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PARAMETERS

Case Control Study

	Ds + nt	Ds -nt
R/F + nt	.....	.....
R/F -nt	.....	.....
TOTAL	.....	.....
TOTAL	.....	.....

In a study it was reported that Hypertension was present in 500 out of 800 pt of myocardial infarction while it was seen in only 100 out of 500 Non MI individual, calculate ODD Ratio

	Ds + nt	Ds -nt
R/F + nt	.....	.....
R/F -nt	.....	.....
TOTAL	.....	.....
TOTAL	.....	.....

COHORT STUDY

1. Incidence
2. Relative risk (risk ratio) = ...
3. Attributable risk (risk difference) = ...
4. Population attributable risk = ...

Ex:

	MI	NON MI
Obese (1000)	120	880
Nonobese (1000)	60	940

Calculate:

1. Incidence

Active space

2. Relative risk

3. Attributable risk

4. Population attributable risk

Strength of association between disease and risk factor:

- RR
- ODDS

  1 :

  >1 :

  <1 :

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CROSS SECTIONAL STUDY

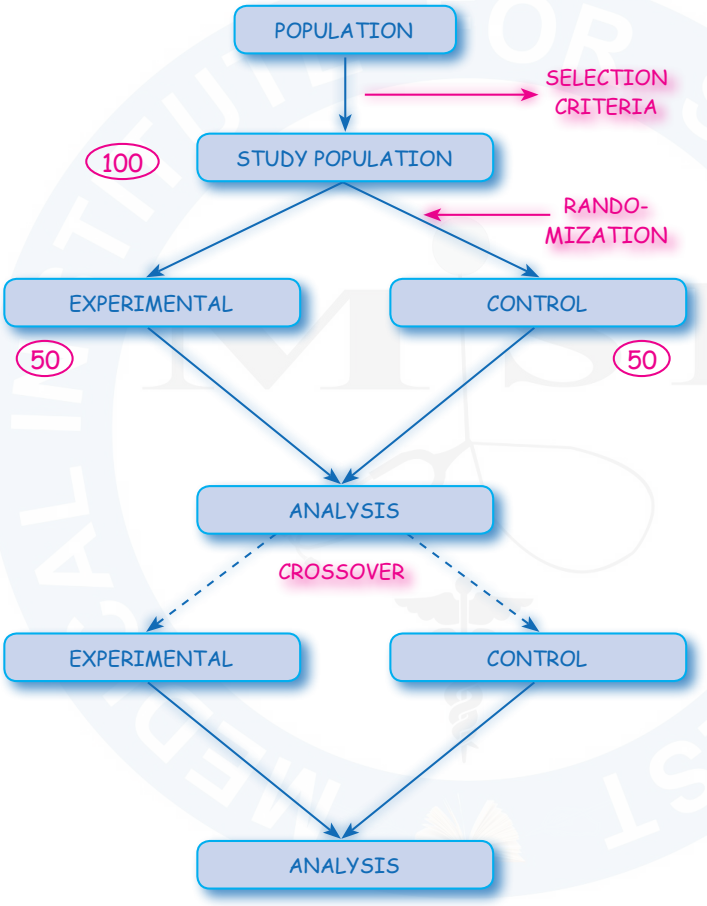
- Aka snapshot study / prevalence study
- PARAMETER : .....
- Based on Primary data
- Study done at point

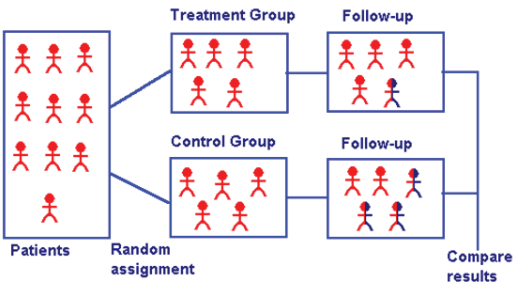
ECOLOGICAL STUDY

- Based on secondary data
- Used for nutritional survey
- Unit of study- population

RANDOMIZED CONTROL TRIAL

Study design:





RCT

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1. Most important step of RCT: .....
2. Randomization done after selecting patients from population before making groups.
3. Advantage of randomization:
   (i) Removing selection error
   (ii) Removing confounding error (both known and unknown)

EPIDEMIOLOGICAL ERRORS

SYSTEMIC	SAMPLE
Individual	Study done on sample population
Types : i. Subjective ii. investigator iii. Analyzer	.....

I. Subjective Bias

1. Recall error : .....
2. Attention .....
   - Change in behavior without information
   - Cohort study
3. Apprehension bias :
   - Anxiety related
   - Cohort study

II. Investigator Bias

Investigator Error :

1. Interviewer error :
   difference in time given for interaction
2. Selection error :
   Errors in selection

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III. Analyser Bias:

Errors during analysis

Other Types of Error

❑ CONFOUNDING ERROR:

Confounders are factors

1. Unequally distributed in study and control group.
2. Independent effect on outcome

❑ BERKESONIAN ERROR :

- Type of selection error
- Different rate of hospital admission

❑ ERROR CORRECTION:

1. Blinding
   a) Single : .....
   b) Double : .....
   a) Triple : .....

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2. Matching : .....

3. Randomization : .....

.....

PHASES OF CLINICAL TRIAL

PHASE		CONDUCTED ON	PURPOSE
1.	Preclinical	Animals	Safety profile in animals
2.	Phase 0 (micro dose trial)	Healthy person	Safety profile of drug in small dose
3.	Phase I	• Healthy person (mainly) • Terminally ill patients	Safety profile in full therapeutic doses
4.	Phase II	Patient	Effect of drug (max failure)
5.	Phase III	Patients	Comparison of new drug with existing
6.	Phase IV	Patients	Long term side effects of drug

- Clinical phase

  i. On healthy person –

  ii. On patients –

  iii. Longest phase –

  iv. Post marketing phase-

  v. Drug launched in market:

  vi. maximum drugs failure

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NESTED CASE CONTROL STUDY

- Indicated for
  - Disease is rare
  - Investigation is expensive
- Type of cohort study
- In a group we do case control

MISCELLANEOUS POINTS

- Best Study for rare diseases :
- Best study for rare DS with expensive investigations : .....
- Study with attrition :
- Best study for natural disease course:
- Famous cohort study :
- Nested case control study :
- First step of epidemiological investigations :
- How long to continue study :

.....

.....

.....

.....

- Best epidemiological study :
- Best analytical study :

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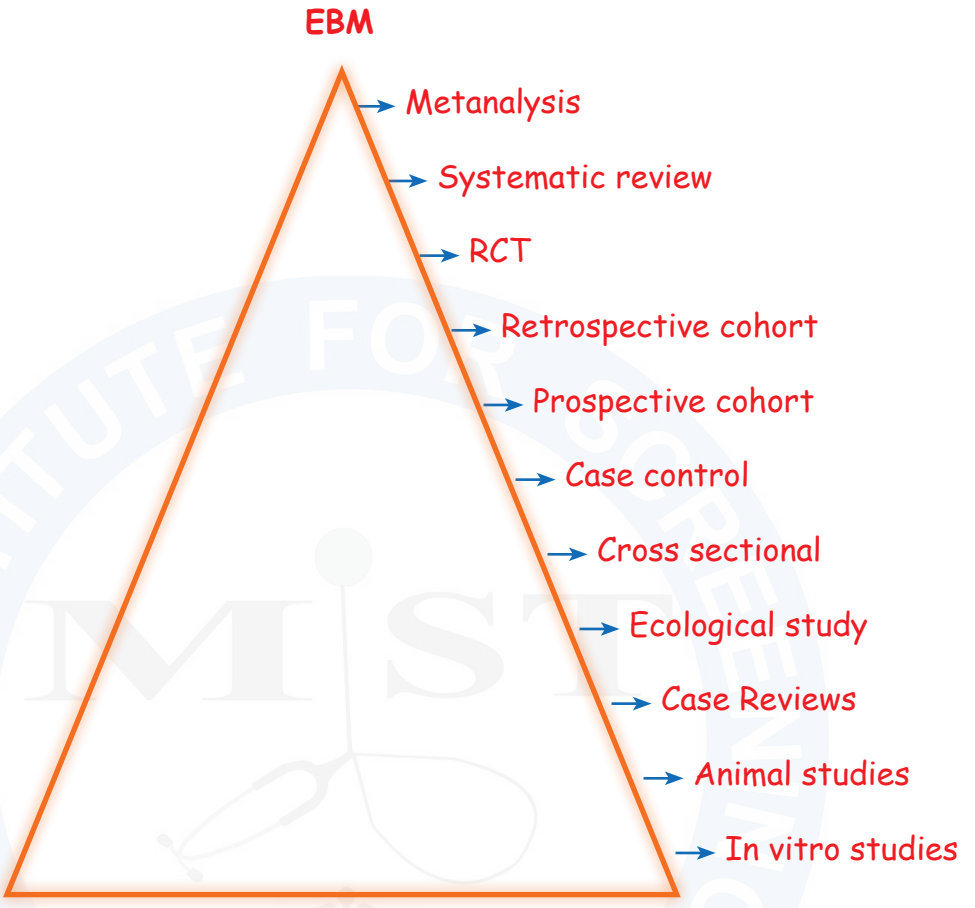
IMPORTANCE OF VARIOUS STUDIES



HILL CRITERIA

Also known as surgeon general criteria for causality

Best Criteria: .....

(Risk factor should always present before disease)

Best study: .....

ASSOCIATION & CAUSATION

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EXTRA EDGE

1. Removal of confounding By
   1. Matching : (MC / simplest)
   2. Stratified randomization (Best)
   3. Randomization (2nd Best)
   4. Stratification
   5. Restriction
   6. Statistical Modelling / multivariate Analysis

2. Bias
   1. Lead time Bias
      Apparently looking increased survival Rate due to early Detection ofcases by screening
   2. Neyman Bias (Incidence - Prevalence Bias)
      Error in Detection of Incidence prevalence due to high case fatality, early recovery Asymptomatic cases
   3. Pygmelian Bias
      High Motivation
   4. GOLEM Bian
      Demotivation

3. Attrition Bias : < 5%
   Retention Rate : > 95

4 Open Study - Absence of blinding

5 Intention to treat TRIAL
Result of RCT not affected by attrition

6 Ratio of case & control in case control study

	Cases	Control
i) Ideal	1	4
ii) Minimum	1	1

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7) ARR (Absolute Risk Reduction)

ARR = CER - EER

CER = Control event rate

EER = Experimental event rate

7 NNT = 1/ARR

NNT (Number needed to treat)

Lower NNT indicate more beneficial treatment

Ideal NNT = 1

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SCREENING

TYPES OF SCREENING

1. Prescriptive	2. Prospective
benefit	benefit .....
level of prevention	level of prevention .....
Eg : Cancer Cervix, breast cancer, diabetes mellitus, hypertension	Eg : sex workers, immigrants, corona screening at airport
3. Mass screening	4. High risk screening
Large group of individuals	E.g.: Diabetes Mellitus
Eg: eye checkup in school	
5. Multiphasic screening	6. Multipurpose screening
>1 test for a single disease	Screening for >1 disease at a time
Eg: Diabetes Mellitus – blood and urine tests	E.g.: antenatal screening (HIV, Hep B, Hep C, Thyroid, Syphillis)

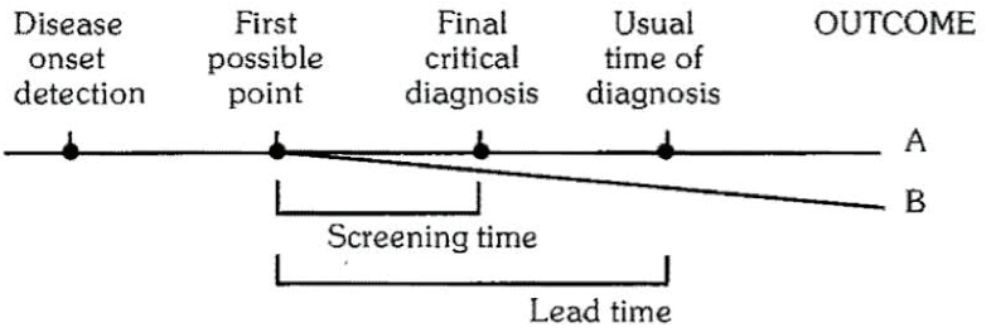

FIG.1
Model for early detection programmes

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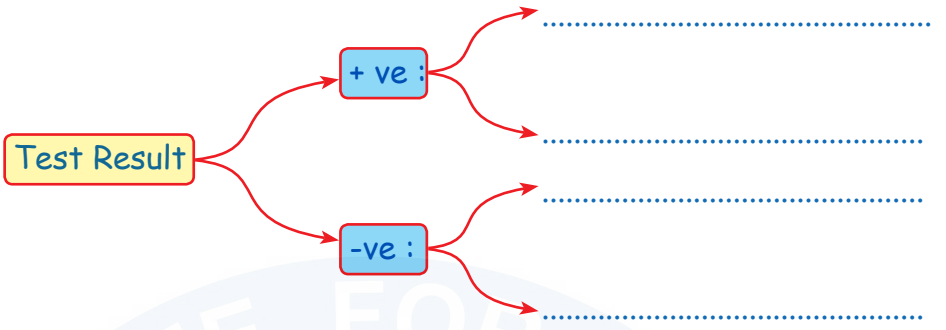
RESULT OF SCREENING TEST



(Q) : True positivity - .....

True negativity – .....

Test results	Disease + nt	Disease –nt
+ ve	.....	.....
-ve	.....	.....
Total	.....	.....

Sensitivity : .....

Specificity : .....

PPV : .....

NPV : .....

Active space

Identify the formula for Sensitivity

(i) FP/TP + FP (ii) TP/TP + FP

(iii) FP/TN + FP (IV) TP/TP + FN

Total population of 2000 was assessed for HIV infection. out of these 200 were diagnosed as HIV positive. A new ELISA screening test was applied on same group and result says 260 positive, out of which only 130 had the disease calculate

☞ Calculate:

1. Sensitivity of test, Specificity of test, PPV, NPV

Results	Disease + nt	Disease -nt
Test + ve	.....	.....
Test -ve	.....	.....
Total	.....	.....

Sensitivity- a/a + c = ...

Specificity- d/b + d = ...

PPV- a/a + b = ...

NPV- d/c + d = ...

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RELATIONSHIP BETWEEN VARIOUS PARAMETERS

1. Disease prevalence to predictive value	
2. Sensitivity and specificity of test	
3. FP and FN	

- Most important quality of a screening test is = ...
- Usefulness of screening test depends on: .....
- High FP cases in a study signify
  - .....
  - .....

- Same test applied in different community, more FP cases:
  - .....
  - .....

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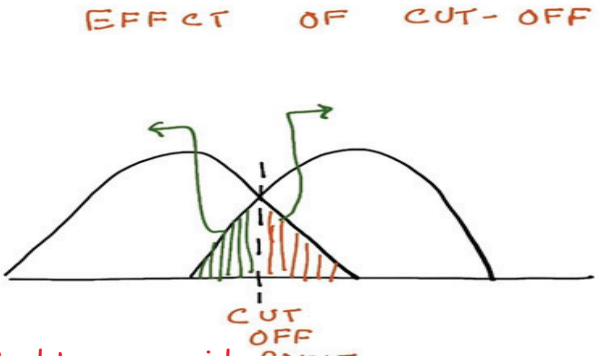
✦ Predictive value of test (diagnostic power of a test) depends on:

- .....
- .....
- .....

TEST IN SERIES	TEST IN PARALLEL
Specificity –	Specificity -
Sensitivity –	Sensitivity -
PPV-	PPV –
NPV-	NPV -

EFFECT OF CUTOFF LEVEL



If cutoff shifted to upper side:

If cutoff shifted to lower side:

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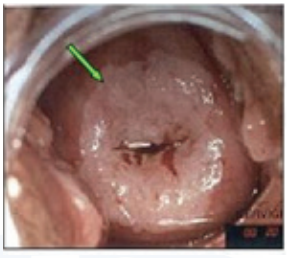
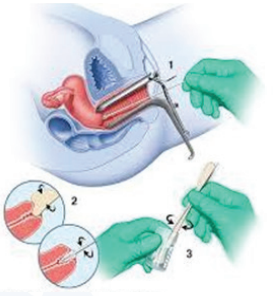
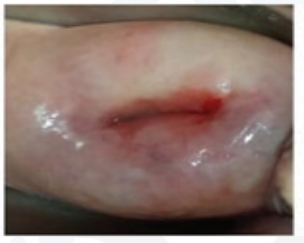
DIFFERENT SCREENING TESTS

1. Cancer Cervix:







I. VIA and VILI test:

Preferred in periphery at the level of PHC/CHC

cost effective.

VIA: visual inspection of cervix after acetic acid

aceto white patch is suspecious

VILI: Visual Inspection of cervix after lugol's iodine

colourless patch is suspecious

II. PAP smear

District hospital onwards

- confirmatory test for Cancer Cervix : colposcopy guided biopsy

Active space

2. Breast Cancer:

   i. SELF BREAST EXAMINATION

   Best – mammography

   MRI > USG

   Confirmation: Biopsy

Self Breast Examination

3. Cancer prostate:

   PSA (prostate specific antigen)

4. Oral Cancer-

   Bimanual oral examination

5. Colorectal Cancers:

   Occult blood in stool

   Early detection –very good prognosis

6. Neonatal Hypothyroidism:

   Neonatal blood day 3-5 for TSH

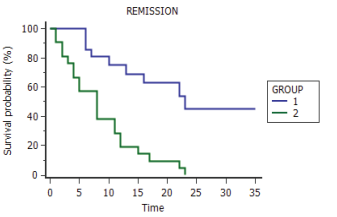
7. DM: RBS

Active space

EXTRA EDGE

1. Kaplan Meier Curve

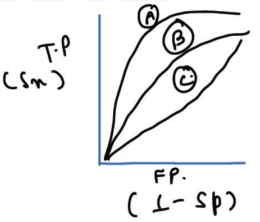
   For survival Analysis



2. Pre test Probability of DS same as Ds Prevalenc

   Post test Probability of DS same as PPV.

3. ROC (Receiver operator curve)



- ✦ Relationship b/n
  - TP and FP
  - Sn and (1-SP)
- ✦ Best curve : (A)

4. LR (Likelyhood Ratio)

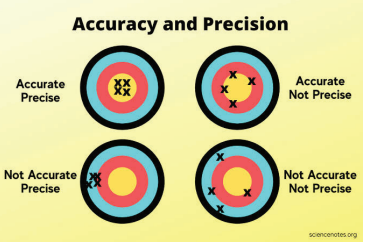
$$LR^{+} = \frac{Sn}{1-Sp} = \frac{TP}{FP.}$$

$$LR^{-} = \frac{1-Sn}{Sp} = \frac{FN}{TN}$$

5. Post Test ODD = Pre test ODD x LR

6. Precision / Repetability : Same result repetedly

   Accuracy / Validity : Result near to actual value



Active space

7. Kappa Score (Cohen's Kappa)

A measure of inter-rater reliability, specifically designed to assess the agreement between two or more raters on categorical data.

$$= \frac{\text{Agreement observed - Agreement expected by chance}}{\text{1- Agreement expected by chance}}$$

Cohen's Kappa	Interpretation
0	No agreement
0.10 - 0.20	Slight agreement
0.21 - 0.40	Fair agreement
0.41 - 0.60	Moderate agreement
0.61 - 0.80	Substantial agreement
0.81 - 0.99	Near perfect agreement
1	Perfect agreement

Active space

DEMOGRAPHY

INTRODUCTION

Parameters affecting: demographic process.

1. Mortality
2. Morbidity
3. Marriage
4. Migration
5. Fertility
6. Social Mobility

INDICATORS OF DEMOGRAPHY

VITALS Indicators

✦ 1. CBR (Crude Birth Rate) :

$$CBR = \frac{\text{No of live births}}{\text{.....}} \times 1000$$

✦ In a population of 30,000 live births are 500. Calculate crude birth rate per 1000 population:

✦ 2. CDR (Crude Death Rate):

$$CDR = \frac{\text{No of death}}{\text{.....}} \times 1000$$

✦ 3. GFR (General Fertility Rate):

$$GFR = \frac{\text{Total no of live births}}{\text{.....}} \times 1000$$

Active space

✦ 4. GMFR (General marital fertility rate):

$$GMFR = \frac{\text{Total no of live births}}{\text{.....}} \times 1000$$

✦ 5. TFR (Total Fertility Rate): Total no of live children born to a woman in her entire reproductive period known as completed family size

✦ 6. GRR (Gross Reproduction Rate): Total no of girl children born alive to a lady in her entire reproductive period.

✦ 7. NRR (Net Reproductive Rate): Total no of girl children remaining alive till their own reproduction.

Most important demographic vital indicator

Only demographic fertility indicator which includes mortality

☞ Current Values:

Eg : Obstetric history

1. FTVD – Male baby of 20 yr old (Alive + healthy)
2. FTCD – Female died after 2yr

Calculate TFR, GRR, NRR

✦ 8. GROWTH RATE:

In area with CBR 18.7, CDR 7.2 calculate growth rate

Active space

☞ DOUBLING TIME:

✓ = $\frac{70}{GR}$ (YR)

Growth Rate	Doubling Time
Slow: (0-0.5) %	>140 yrs
Moderate: (0.5-1.0) %	70-140 yrs
Severe: (1.0-1.5) %	47-70 yrs
Very Rapid: (1.5-2.0)%	35- 47 yrs
Explosive: >2.0%	<35 yrs

☞ Doubling Time of India:

☞ Other Points:

- Crude Birth Rate = (TFR x 8) + 1
- NRR/GRR = TFR/2

☞ Goal for:

- NRR = ...
  ↑
- TFR = ...
  ↑
- (Couple protection rate) = ...

  CPR = ...

✦ Eligible couple (EC)

✦ Eligible couple register (EGR) :

- maintained by .....
- promotion of contraceptive use

Active space

- 150-180 EC / 1000 population
- Current Couple Protection Rate = ...

Population Indicators

1. Sex ratio
   - Sex ratio (Birth)

     .....
   - Child sex ratio (0-6 yrs)

     .....

     .....

     .....

   Eg: Total population = 10,000

   Female birth are 40% of Total live birth

   Sex ratio = ? > 1000

   < 1000

   1000

   None of above

2. Literacy Rate :

   = total no of literate × 100

   .....

   In 2011, 74% : Male – 82%

   Female -65%

   Eg : Total population = 20,000

   <7 yrs = 5000

   Literate people : 50% of Total population

   LR = 25%, 50%, 67%, 75%,

Active space

3. Dependency Ratio (DR)

$$= \frac{\text{non-working population}}{\text{working population}}$$

As per 2011, = 54.4

Eg : In a family there is Mr X 69 years with wife Mrs X 58 years. Their son 30 year with his wife 28 years and 5-year grand child. Calculated dependency ratio for this family:

33%

50%

66%

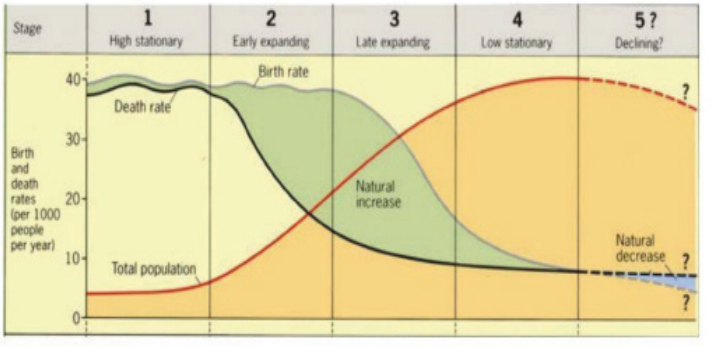
75%

4. Population density :

No of person per sq km of land area

In 2011 = 382

PHASES OF DEMOGRAPHIC CYCLE / DEMOGRAPHIC TRANSITION



Phases Of Demographic Cycle

PHASE	BIRTH RATE	DEATH RATE	POPULATION SIZE	D. GAP.
I : High Stationary	Very high	Very high	Stationary	Constant

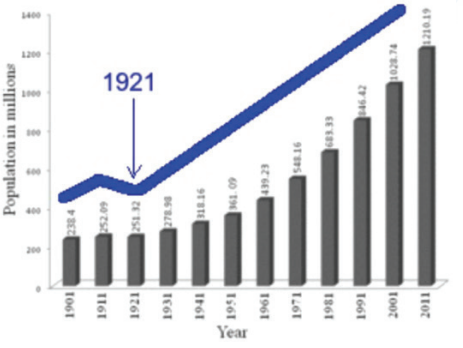
Active space

II : Early Expansion	Very high	Starts declining	Starts increasing at fast rate	1. Starts ↑ sign in beginning 2. Max at end of phase II
III : Late Expansion	Starts declining	Continue to decline	Increasing population Slow rate	Starts declining
IV : Low Stationary	Very low	Very low	Stationary	Constant
V : Reversal	Almost Zero	Natural death continue	Decreasing	Reversal

DATA COLLECTION METHODS

CENSUS :

- Conducted every .....
- Ministry .....
- 1st -.....
- 1st disability census -.....
- Last census (15th) – .....
- Next due -.....
- Population – 00.00 hr of 1st March
- Year of big divide -.....
- Control by Census Commissioner



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NATIONAL FAMILY HEALTH SURVEY (NFHS) :

- Every 5 years
- 6 rounds of NFHS

  (i) 1992-93, (ii) 1998-99, (iii) 2004-2005,

  (iv) 2015-2016 (v) 2019-2020 (vi) 2023-24
- Nodal agency → Int institute of population science, Mumbai
- Ministry → Health and family welfare

CIVIL REGISTRATION SYSTEM (CRS) :

aka -VITAL REGISTRATION SYSTEM (VRS)

1. *Birth
2. *Death
3. Marriage
4. Hospital responsible for registration
5. Home delivery

6. Delivery outside India

SAMPLE REGISTRATION SYSTEM (SRS) :

- DUAL survey

  (i) Continuous reporting of birth, death, marriage (CRS)

  (ii) Survey by an independent officer
- Every

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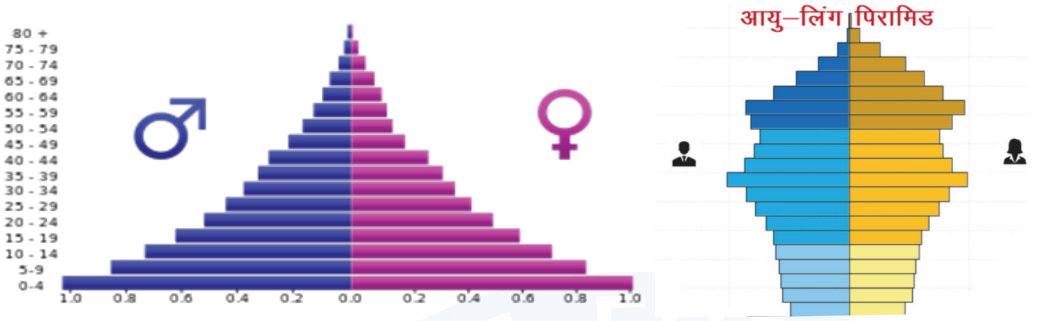
POPULATION PYRAMID



Population Pyramid

- Double histogram
- Features :

A: Ht

B: Shape

India- broad base, narrow tip

Developed country –spindle shape

C : Symmetry

Active space

EXTRA EDGE

1. Total world population (2024) : 8195 million
2. Annual growthRate world (2024) : 0.87%
3. Most populated country in world (2024) : INDIA
4. Total INDIA population (2024) : 1451 million (145.1 Crore)
5. CBR (INDIA) (2024) : 16
6. CDR (INDIA) (2024) : 7
7. Annual growth rate (INDIA) (2024) : 0.9%
8. Demographic divident : Reduction in fertility Rate leading to increase in working population providing an opportunity for accelerated economic growth.
9. 17.5% of world population living in INDIA on 2.4% of world land area
10. Most populous state of INDIA : UP (16.7% India population)
11. Highest TFR state : Bihar (3.3)
12. Dependency Ratio : 48.17%
    Young dependency Ratio : 38.0%
    old age dependencyRatio 10.1%
13. Demographic bonus : decline in dependency ratio (mainly young dependency Ratio) due to decline in fertility
14. Demographic burden : increase in dependency Ratio due to increase in old age dependency Ratio
15. Population Density : 464 /sq km.
16. TFR of INDIA (2024) : 2.0

Active space

CONTRACEPTION

TYPES OF CONTRACEPTION

TYPES OF COUPLE

Eligible couple :

Target couple :

NATIONAL FAMILY WELFARE PROGRAMME (NFWP)

- 1977(Rename)
- Cafeteria approach
- Extensive education approach
- Home delivery of contraception

जोड़ी ज़िम्मेदार
जो प्लान करे परिवार

Active space

- GATHER approach of counseling
- NISHCHAY kit

.....

.....

- Fixed day surgical approach

- PP IUCD (Post-partum IUCD insertion)

- CHAYA / CENT CHROMAN /SAHELI
  - Non hormonal oral contraceptive tab
  - Adv : 1. Can be given just after delivery (lactation)
    2. Quick return of fertility
  - Ormeloxifene 30 mg tab
  - SERM (Selective estrogen receptor modulator)
  - Doses:
    (i) Twice/wk -3months
    (ii) Once/wk-as long as required
  - Disadvantage:
    Spotting → amenorrhea
- ANTARA :

  Injectable hormonal preparation
- KAP gap - .....

  K-knowledge

  A-attitude

  P-practice

Active space

- UNMET NEED OF CONTRACEPTION 9.4%
- MISSION PARIVAR VIKAS

  Accelerate use, awareness of family planning services in the High TFR district

- CRITERIA FOR FP SURGERY
  - Min 1 child ; >1 yr
  - Female : 15-49 yr
  - Male <60 yr
  - Verbal consent of female
- INCENTIVE (To The Patient):
  - VT-1100 Rs
  - Lap TT-145 Rs
  - Mini laparotomy-600 Rs
  - CuT-75 Rs
- INCENTIVE (To The Motivator):

  VT-200 Rs

  TT-150 Rs

❑ **METHODS of Contraception**

TEMPORARY	PERMANENT
Natural	Tubectomy (female)
Barrier	Vasectomy (male)
Hormonal	
IUCD	

Active space

NATURAL METHODS

1. Calendar(rhythm) -

   .....

   .....

   .....

2. Basal body temp –

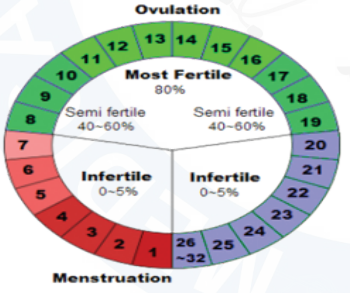
   Increase in body temp ≥ 0.5 degrees Celsius: impending ovulation (due to release of Progesterone)

3. Cervix mucus method (Billing method)

   Estrogen- thin, copious, ferning pattern + nt

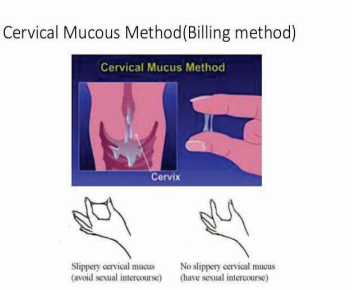
   Progesterone –thick, scanty, loss of ferning pattern

4. Lactational Amenorrhoea



Calendar Methods

Basal Body Temperature



Billing Methods

BARRIER METHODS

Goal – prevent fertilization

Advantages – .....

Efficacy increases by – spermicidal gel

spermicidal agent - .....

Best Methods

Active space

❑ Methods :

- Condoms :

	Male	Female
1. Ring		
2. Shape		
3. Material		

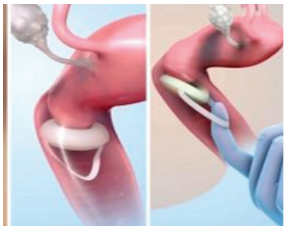
- Cervical cap(diaphragm)
- Vaginal sponges (foam tablet)

  (Available in market as Today tablet)
- Vaginal ph regulatory gel : Phexxi gel

Condoms

Female Condoms

Spermicidal Gel



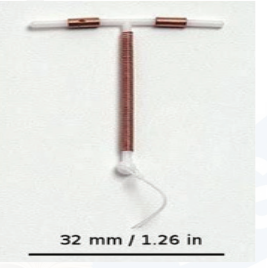
Vaginal Foam Tablet

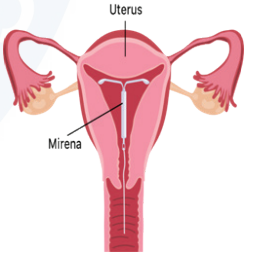
Active space

IUCD (INTRA UTERINE CONTRACEPTIVE DEVICE)

Types of IUCD

Generation	Device	MOA
1. 1st generation (Inert)	Lippi's Loop Grafenberg Ring	aseptic inflammation
2. 2nd generation (Cu)	CuT 380A CuT 220B ML 375 ML 250	.....
3. 3rd generation (Hormonal)	MIRENA (LNG-20)	• inhibit ovulation • cervical mucus thick • Endometrium out of phase
4. 4th generation (frameless)	Flexi Guard FIBRO PLANT	prevent fertilization same as 3rd gen.





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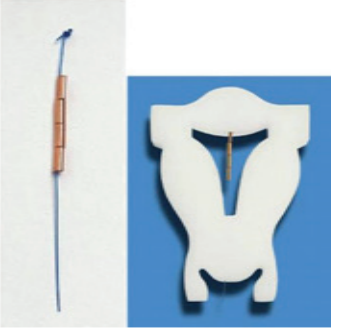
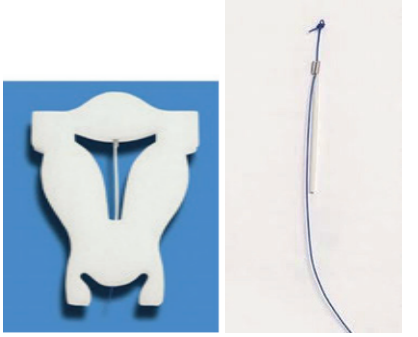


Flexi Guard



Fibro Plant

TIME OF INSERTION:

- Best:
- Preferred:

SHELF LIFE:

- CuT 380A –
- MIRENA (LNG-20) –
- ML 375 -

SIGNIFICANCE OF NUMBER: Surface area of copper in sq mm:

☞ Best Method of Contraception for .....

✦ Non contraceptive benefit

CONTRA INDICATION:

Absolute	Relative
1. PID	1. H/O PID
2. Undiagnosed bleeding P/V	2. Anaemia
3. Suspected pregnancy	3. H/O multiple partners
4. Distorted Cavity; fibroid; anomali; End Ca	4. Not able to come for follow-up
5. Wilson DS : Cu T	5. Nulligravida (for Cu T)
6. H/O Breast Ca : LNG	6. H/O ectopic pregnancy

Active space

MIST

SIDE EFFECTS:

1. Menstrual irregularity (MC)

   .....

2. Pain – infection

   .....

3. Failure :

   .....

   .....

4. Ectopic pregnancy

   .....

5. Perforation / Displacement

   .....

   .....

6. Anemia

   .....

HORMONAL METHOD

Different Methods:

- Oral
- IUCD
- Injection
- Implants
- Vaginal ring
- Skin patch

Active space

ORAL Hormonal Methods:

Types:

COP (E + P)	POP (P)
C/I in lactation	in lactation
Cyclical manner 3 wks on 1 wk off	Continuously
MALA – N (free)	CERRAZET tab
MALA – D (purchase)	75 mcg of Desogestrel/Tablet

Composition of MALA – N:

(21) Active - Ethinyl Estradiol .....
Levonogestrel .....
(7) Iron Tablet - ..... ferrous fumarate (19.5 mg elemental iron)

When to start:

Day 3-5th of cycle

MISSED TABLET :

✓ One Tablte

- Take missed tablet as soon as possible
- continue Rest of Tablet as per schedule

✓ ≥ 2 Tablet

1. In 1st / 2nd wk of cycle
   - Take 1 tab as soon as possible
   - continue Rest of Tablets as per schedule
   - Back up contraceptive for 7days
   - If sexual relationship in last 72hr. take emergency contraceptive pills

Active space

2. In 3rd wk of cycle
   - Take 1 pill as soon as possible
   - Take rest of hormonal pills as per scheule
   - Discard Non Hormonal pills
   - Start newpack immediately after Hormonal pills

3. In 4th wk (Non Hormonal pill)
   - Discard missed Non Hormonal pills
   - Continue taking rest of Non Hormonal pills
   - Start new pack as per schedule

MOA

.....

OCP with Cancers:

Increased	Decreased

NON CONTRACEPTIVE BENEFIT:

- Treatment : acne, hirsutism
- Treatment : functional ovarian cyst
- Treatment : fibrocystic breast disease
- Treatment : fibroadenosis
- Treatment : PCOD
- Treatment : AUB

Active space

CONTRA INDICATION:

Absolute	Relative
1. DVT	1. female >40 yrs (non - smoker)
2. Cardiovascular disease	2. female >35 yrs (smoker)
3. Liver disease	3. migraine
4. Family H/O breast Ca	4. Lactation
5. Congenital hyperlipidemia	5 DM and HTN

SIDE EFFECTS:

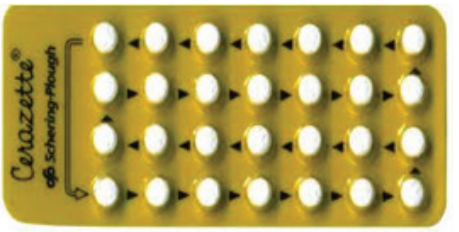
- DVT
- Cancer – Breast, cervical, hepatocellular
- CVD
- Migraine
- Wt gain
- HTN
- DM
- Vaginal candidiasis
- PID : chlamydia

BEST METHODS:

Mala-N

Mala-N

Active space



Progesteron Only Pills

HORMONAL IUCD:

MIRENA (LNG-20)

- MOA : ovulation inhibition, prevent implantation
- Total Hormon : Levonogestrel 52 mg
- Daily release : .....
- Duration: .....
- S/E: Spotting – amenorrhea

HORMONAL INJECTIONS:

MOA : Ovulation Inhibition

I. DMPA

Antara (DMPA)

Depot Medroxy Progesterone Acetate

.....

.....

II. NET-EN

Nor Ethisteron Enanthate

.....

.....

III. DMPA S/C 104

.....

.....

Active space

Disadvantage:

(i) Menstrual irregularity : spotting- amenorrhea

(ii) Delay in return of fertility

IMPLANTS:

- Norplant 6 (6 Rods): 5 yrs
- Norplant 2 (2 Rods): 3 yrs
- Implanon / Nexplanon (single, BaSO4): 3 yrs

Implants

SKIN PATCH:

- O + P combination
- 1 patch – 7 days
- 3 patch – pack
- MOA : ovulation inhibition

Skin Patch

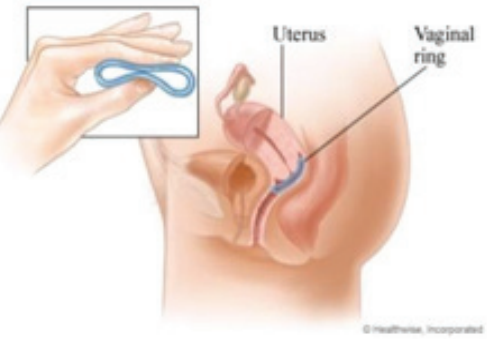
Different Sites for Skin Patch

Active space

VAGINAL RING:

- O + P combination
- 21 days (3 wk on)
- Remove for 7 days (1 wk off)

Vaginal Ring



Insertion of Vaginal Ring

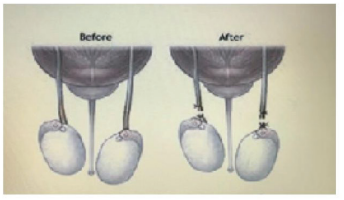
PERMANENT METHODS

TUBECTOMY (female)	VASECTOMY (male)
Less cost effective	More cost effective
Not required	Additional 3 months contraception (Barrier)
MCC of failure : wrong identification of structure	MCC of failure : wrong identification of structure

Active space

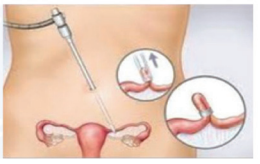
TUBAL LIGATION:

SURGICAL			NON SURGICAL (Essure)
Methods	Mini laparotomy	Laparoscopic	Essure implant • Hysteroscopically inserted • B/L tubal cornual ends • MOA : inflammation fibrosis • Additional barrier method for 3 months
method of ligation	.....	.....	
indications • Patient with cardiac and respiratory disease • Post-partum period • Interval period	....	....	

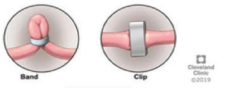


VASECTOMY

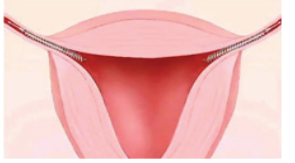
Mini laparotomy



Laparoscopic Tubectomy



Laparoscopic Tubectomy



Essure Implant in Place

Essure Implant

MIST

Active space

EMERGENCY CONTRACEPTIVE

- IUCD – .....
- LNG tablet(1.5mg) – .....
- ULLIPRISTAL(30 mg) – .....
- RU-486 (MIFEPRISTONE, 10 mg) – .....

- YUZUPI regime
  1. OCP with < 50 mcg EE - 4 tabs stat → Repeat 4 Tab after 12 hr. [within 72 hr.]
  2. OCP with ≥ 50 mcg EE - 2 Tablet stat → Repeat 2 Tablet after 12 hr. [within 72 hrs.]

EFFFICACY OF CONTRACEPTIVE METHODS

1. Pearl index – .....
2. Life table analysis .....

PEARL INDEX:

- No of accidental pregnancy .....
- 1200 month exposure
- PI: $\frac{\text{no of accidental pregnancies}}{\text{total months of exposure}} \times 1200$

(Q): 500 women used OCP for 5 years and accidental pregnancy reported in this duration – 5; calculate PI

Active space

Pearl Index Value of Different Methods

- VT/TT : 0.1 - 0.15
- MIRENA : 0.2
- DMPA : 0.3
- OCP : 0.1-0.5

- IUCD : 0.5-2.0
- SAHELI : 1.8-2.8
- CONDOM : 2-14
- DIAPHRAGM : 12
- SPONGES : nulliparous – 10-20
  Multiparous – 20 -40

LARC (LONG ACTING REVERSIBLE CONTRACEPTIVE)

1. NON HORMONAL
2. HORMONAL

CONTRACEPTION OF CHOICE

1. Young adolescent: Barrier
2. Newly married couple
   A) living together : OCP
   B) living a part : Barrier
3. Birth spacing : IUCD
4. Lactating women : POP (Best)
   : PPIUCD (Preferred)
5. AUB : Mirena
6. Pt on Anticoagulant : Progesterone only method
7. Epilepsy, Sickle Cell : DMPA
8. Endometrios,
   Obesity, > 40yr : Progesterone only method
9. Pt with SLE : Mirena > Implant
10. CHD, pulmonary
    Vascular ds : Implanon > Mirena
11. HIV patient : IUCD with Barrier
12. PCOD with Hirsutism : 3rd /4th generation progesterone OCP
13. Unmarried
    sexually active : Barrier (preferred)
    Best : OCP with Barrier

Active space

EXTRA EDGE

1. National Family Planning Indemnity scheme (NFPIS)

F P operation claim	Amount
A. Death in Hospital / with in 7days of discharge	2,00,000/-
B. Death 8-30days of discharge	50,000/-
C. Expense for Rx of medical complication	25,000/-
D. Failure of F.P. operation	30,000/-
E. Doctor / facility covered for Litigation upto 4 cases/ including defence cost	2,00,000/- (Per case)

2. Contraceptive after delivery

	BF mother	Non BF Mother
1. OCP	avoid as long as Lactating (min 6 month)	after 3 wk of delivery
2. POP	within 3wk of delivery	within 3 wk of delivery
3. CHAYA	within 4wk of delivery	within 4 wk of delivery
4. ANTARA	after 6wk	within 3 wk
5. IUD	within 48hr or after 6wk	same as BF
6. Mini Laparotomy	within 1 wk or after 6wk	same as BF

Active space

HEALTH CARE SYSTEM OF INDIA

ELEMENTS OF PRIMARY HEALTHY CARE

E : Health Education

L : Locally endemic disease prevention and control

E : Essential drugs

M : MCH care

E ~ I : Immunization

N : Nutrition

T : Treatment of common disease

S : Safe water supply and sanitation

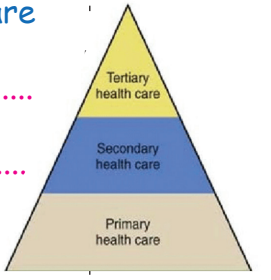
PRINCIPLES OF PRIMARY HEALTH CARE

1. Equitable distribution
2. Community participation
3. Intersectional coordination
4. Appropriate technology

LEVELS OF HEALTH CARE

1. Primary level care 2. Secondary level care 3. Tertiary level care

- ....
- ....
- ....



SERVICES PROVIDED AT FRU :

1. 24 hr delivery [Normal Assisted]
2. Emergency obs care [Include Surgical intervention]
3. New born care
4. Emergency care of sick children
5. Full range of FP services [Including Lap surgery]
6. Safe abortion services
7. Rx of RTI/STI
8. Blood storage facility [24 x 7]
9. Essential Lab services
10. Referral [Transport services]

FEATURES OF HEALTH CENTERS

POPULATION	SUBCENTER		PHC		CHC
Plain	.....		.....		.....
Hilly	.....		.....		.....
BED	TYPE A	TYPE B	TYPE A	TYPE B	: .....
	Non MCH	MCH	Delivery / Month		
NUMBER AND TYPE OF STAFF	TYPE A (3)	TYPE B (4)	TYPE A (13)	TYPE B (14)	46-54
	CMO - 1 MPWF - 1 MPWM - 1	CMO - 1 MPWF - 2 MPWM - 1	MO - 1 HAF - 1 HAM - 1	MO - 2 HAF - 1 HAM - 1	Specialist

Active space

GRASS ROOT LEVEL WORKERS (VILLAGE WORKER)

1. ASHA (Accredited social health activist)
2. Anganwadi worker (AWW) at ICDS (Integrated child development Services)
3. TBA (Traditional birth attendant)
4. VHG (Village Health Guide)

	TBA	VHG	ASHA	AWW
Location	Village	Village	Village	*ICDS center (anganwadi)
Population				• AWC • Mini AWC Anganwadi on demand • > 40 Children • < 6yr.
Education				
Training duration	30 days (3 months)	200 hr (3 months)	.....	4 month
Training by	• MPW – F • HA – F • Medical officer	• Medical officer	• AWW • MPW – F • HA – F • Medical officer	• Mukhya sevika • Child development project officer

Active space

ASHA

- ASHA part of
- Ap 2005(2005-2012) – NRHM (ASHA)
- 2013 – NUHM (*USHA) (1000-2500)
- 2013 (NHM) (combination of NRHM and NUHM) (PMSSM)

☞ *USHA – Urban social health activist

Selection criteria

Function of ASHA Workers

1. A part of JSY .....
   .....
2. Home delivery of contraception
   .....
3. Active disease surveillance
   - TB
   - Leprosy
   - Filariasis
   - Leishmaniasis
   - .....
4. Home based new born care
   - Home delivery .....
   - Hospital vaginal .....
   - Hospital operative .....
5. Home Based care of young child
   .....
6. Treatment of minor problems

Functions of ASHA Workers

Active space

7. IFA tablet
8. DOTS agent
9. Health Education
10. Member of village health sanitation and nutrition committee (VHSNC)

    ASHA: (i) gram panchayat

    (ii) superwise toilet construction

☞ NOTE: ASHA never conducts delivery

10. Mobilization of community to ICDS center for nutrition day (once/ month)
11. Impact Indicator :
12. Birth Wt Record
13. Birth/death Registration
14. Reporting of DS out break

ANGANWADI WORKER (Heart of ICDS center)

❑ **Beneficiary:**

(i) .....

(ii) .....

(iii) .....

(iv) .....

❑ **Functions of AWW:**

1. Growth monitoring
2. Immunization
3. Mid-day meal
4. Referral services
5. Non formal school education
6. Health education

Active space

- Ministry of ICDS .....
- **MID DAY MEAL PROGRAMME / PM POSHAN YOJNA**

  - Composition of food:

AGE	PROTEIN	K CALORIE
0-6 yrs	12 – 15 gm	500
Pregnant / lactating	18 – 20 gm	600
Malnourished child	20 – 25 gm	800

SCHEMES AT ICDS CENTER

1. KISHORI SHAKTI YOJNA
   - Beneficiary : 11 - 18 Yr
   - Focus on self development, Nutrition, Health status, literacy, vocation skill
2. Nutrition Program for Adolescent Girls
   - 11 - 19 yr (wt <30kg : 11-15 yr)
     (<35kg : 15-19 yr)
   - 6 kg food grain / month
3. Rajiv Gandhi scheme for empowerment of Adolescent girls (SABLA)
   - 11 - 18 Yr
   - Improve nutrition, health status
   - Self development & empowerment
   - Awareness about health, Hygiene, nutrition, adolescent reproductive & sexual health (ARSH), family & child care

Active space

TBA:

Conduction of home delivery under supervision of MPW – F

VHG:

- Functions:
  - Health education
  - DOTS agent
  - Treatment of common disease
  - Contraceptive distribution

MULTIPURPOSE WORKERS

MPW – F:

- Functions:
  - MCH care
  - Conduction of delivery : Sub center
    Home
  - Td vaccination
  - IFA tablet
  - IUCD insertion
  - Ante natal vist
  - Post natal visit
  - Contraception advice
  - Identification of cases for MTP/ tubal ligation and referral
  - Identification of high risk pregnancy cases and referral ; high risk cases at time of delivery and referral
  - Treatment of common disease
  - Test : Hb%, Blood sugar, Urine : Albumin, sugar
  - Record maintenance

Functions of MPW-F

Active space

MPW – M:

✦ Functions:

1. Treatment of common disease
2. Immunization
3. Chlorination of well water
4. Active malaria surveillance
5. Part of *IMNCI programme

Functions of MPW-M

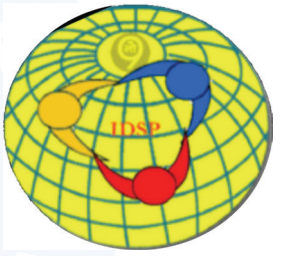
*IMNCI: Integrated management of neonatal and childhood illness

Early detection and timely referral to avoid preventable mortality

6. Part of IDSP project :

Integrated disease surveillance project

Early detection, timely intervention to avoid upcoming epidemic



The core conditions under surveillance in IDSP are as follows :

(i) Regular surveillance :

- Vector borne disease : Malaria
- Water borne disease : Acute diarrhoeal disease (cholera)
  : Typhoid
- Respiratory diseases : Tuberculosis
- Vaccine preventable diseases : Measles
- Diseases under eradication : Polio

  Other conditions : Road traffic accidents (Link-up with police computers)
- Other international commitments : Plague
- nusual clinical syndromes (causing death/ hospitalization) : Meningoencephalitis/Respiratory distress, Haemorrhagic fevers. other undiagnosed conditions

Active space

(ii) Sentinel Surveillance :

- Sexually transmitted diseases /blood borne : HIV/HBV, HCV
- Other conditions : Water quality monitoring
  : Outdoor air quality (Large urban centres)

(iii) Regular periodic surveys :

NCD risk factors : Anthropometry, physical activity, blood pressure, Tobacco, Nutrition etc.

(iv) Additional state priorities :

Each state may identify upto five additional conditions for surveillance.

7. Contraceptive advice
8. Record maintenance
9. Sputum collection of suspicious TB cases and transfer to DMC (designated microscopy center)

MEDICAL OFFICER

1. In charge of PHC
2. Treatment of common disease
3. School health check
4. Minor surgery under LA
   - .....
   - .....
   - .....
5. Training of staff
6. Conduction of meeting of staff members once a month
7. Supervision of all national program activity at PHC level

HEALTH ASSISTENT – M / F

1. Supervising function of MPW
2. Visit once/wk – respective sub center
3. Randomly visit .....
4. Record maintenance
5. HA – F : conducting meeting with MPW – F, NGO in that area

Active space

MIST

COMMUNITY HEALTH CENTER

- Specialist:
  - Physician
  - Paediatrician
  - Surgeon
  - Obstetrician and Gynaecologist
  - Ophthalmologist
  - Anaesthetist
  - Dental surgeon
  - AYUSH doctor
  - Public health programme manager
- Major surgery : .....
- Ophthalmologist – .....
- All other specialist - .....
- Blood transfusion
- AYUSH : Ayurveda, Yoga, Unani, Siddha, Homeopathy

URBAN HEALTH CARE SYSTEM

- **Primary level care:**
  - U PHC
  - Health and wellness centers
  - Poly Clinic
- **Secondary level care:**
  - U CHC : Metro –

    Non metro –
- **Tertiary level care:** medical college super specialty hospital

Active space

INDIAN PUBLIC HEALTH STANDARDS (IPHS)

	RECOMMENDED	CURRENT
Doctor	1/1000	.....
Nurse	3/1000 3/1 doctor	.....
Bed	1/1000	.....
ASHA	2/1000	.....
MPW - F	1/5000	.....
MPW - M	1/5000	.....
HA - M	1/30,000	.....
HA - F	1/30,000	.....
TBA	1/1000	.....
VHG	1/1000	.....
Lab technician	1/10,000	.....
Pharmacist	1/10,000	.....

☞ ASHA, AWW - Never conduct delivery

☞ ANM are replaced by MPW (F)

ANM (Auxillary nurse midwife)

Active space

EXTRA EDGE

ICDS

1. Fund : Central state -> 60:40 (state/UT)
   90:10 (NE state/Himalaya state)
   100:0 (UT with out Legislation)
2. >1 meal in ICDS : Morning snacks, Hot meal (cooked)
3. Children <3yrs, pregnant, lactating mother : take home ration
4. Scheme universal (No BPL/SC/ST Criteria)
5. Administrative unit of ICDS
   - Rural area → Community development block (1,00,000)
   - Tribal area → Tribal development block (35,000)
6. Trainning : 1 CDPO → 4 Mukhya Sevika → 100 AWW
7. Ministry : Women and child development
8. Poshan Tracker
   1. Anganwadi workers
   2. Beneficiaries
      Pregnant, Lactating women
      0-6yr children, Adolescent girls

9. Poshan Vatika
   1. Vegetable Garden
10. Saksham Anganwadi
    Improve infra structure of ICDS Center

Active space

11. Fully operational FRU

	Plain	Hilly /NE except Assam
	(Minimum C-section / yr)	
1) SDH/CHC	60	36
2) DH	120	72

Functional 24x7 PHC

- 10 Delivery / month
- 5 Delivery / month (Hilly, NE except assam)

12. IPHS Guidelines (2022)

IPHS guidelines (2022)

- Health and wellness centers [HWC-SHC, UHWC, HWC-PHC, UPHC] in both rural and urban area are k/as AYUSHMAN AROGYA MANDIR and will provide Primary health care services.
- multispeciality polyclinic under UPHC nearer to community will provide ambulatory specialist services particularly in urban area
- Community health center in rural area can be either FRU / NON FRU depending on range of services available. In urban area all CHC will be providing services at par with FRU
- CHC, District hospital, sub district hospitals will provide secondary care services.
- all Rural and urban HWC should have National Identification number [NIN-Id] and register on AYUSHMAN BHARAT portal

Active space

13. Form under IDSP

Form -'S'

State_____ District_____ Block_____ year_____

Name of the Health Worker/Nolunteer/ Practitioner	Name of the Supervisor	Name of the Reporting Unit

ID No./Unique Identifier (To be filled by DSU) Reporting week From dd mm yy To

	a	b	c	d	e	f	g	h	i	j	k	l	m
	Cases							Deaths					
	Male			Female			Total	Male			Female		
	<5yr	5yr	Total	<5yr	5yr	Total		<5yr	5yr	Total	<5yr	5yr	Total
1. Fever													
Fever < 7 days													
1 Only Fever													
2 With Rash													
3 With Bleeding													
4 With Daze/Semiconsci-ousness/Unconsciousness													
Fever> 7 days													
2. **Cough with or without fever**													
< 2 weeks													
> 2 weeks													
3. Loose Watery Stools of Less Than 2 Weeks Duration													
With Some/Much Dehydration													
With no Dehydration													
With Blood in Stool													
4. Jaundice Cases of Less Than 4 Weeks Duration													
Cases of acute Jaundice													
5. Acute Flacid Paralysis Cases in Less Than 15 Years of Age													
Cases of Acute Flacid Paralysis													
6. Unusual Symptoms Leading to Death or Hospitalization that do not fit into the above.													

Date: ***Signature***

FORM P
(Weekly Reporting Format –IDSP)

Name of Reporting Institution:		I.D. No.:	
State:	District:	Block/Town/City:	
Officer-in-Charge	Name:	Signature:	
IDSP Reporting Week:-	Start Date:-	End Date:-	Date of Reporting:-
	__/__/_____	__/__/_____	__/__/_____

S.no	Diseases/Syndromes	No. of cases
1	Acute Diarrhoeal Disease (including acute gastroenteritis)	
2	Bacillary Dysentery	
3	Viral Hepatitis	
4	Enteric Fever	
5	Malaria	
6	Dengue / DHF / DSS	
7	Chikungunya	
8	Acute Encephalitis Syndrome	
9	Meningitis	
10	Measles	
11	Diphtheria	
12	Pertussis	
13	Chicken Pox	
14	Fever of Unknown Origin (PUO)	
15	Acute Respiratory Infection (ARI) / Influenza Like Illness (ILI)	
16	Pneumonia	
17	Leptospirosis	
18	Acute Flaccid Paralysis < 15 Years of Age	
19	Dog bite	
20	Snake bite	
21	Any other State Specific Disease (Specify)	
22	Unusual Syndromes NOT Captured Above (Specify clinical diagnosis)	
	Total New OPD attendance (Not to be filled up when data collected for indoor cases)	
	Action taken in brief if unusual increase noticed in cases/deaths for any of the above diseases	

FORM L
(Weekly Reporting Format – IDSP)

Name of the Laboratory:		Institution:	
State:	District:	Block/Town/City:	
Officer-in-Charge:	Name:	Signature:	
IDSP Reporting Week:-	Start Date:-	End Date:-	Date of Reporting:-
	__/__/_____	__/__/_____	__/__/_____

Diseases	No. Samples Tested	No. found Positive	
Dengue / DHF / DSS			
Chikungunya			
JE			
Meningococcal Meningitis			
Typhoid Fever			
Diphtheria			
Cholera			
Shigella Dysentery			
Viral Hepatitis A			
Viral Hepatitis E			
Leptospirosis			
Malaria		PV:	PF:
Other (Specify)			
Other (Specify)			

Line List of Positive Cases (Except Malaria cases):

Name	Age (Yrs)	Sex (M/F)	Address: Village/Town	Name of Test Done	Diagnosis (Lab confirmed)

Active space

PREVENTION IN OBSTRETRIC AND PEDIATRICS

PREVENTION IN OBSTETRICS

- **Definition of MMR:**
  - No of maternal death / .....
  - Any death from point of conception till 42 days after delivery due to any cause directly related to pregnancy; aggravated by pregnancy but not due to accident / Incidental cause.
- **MMR Rate/Ratio**
- **Period included:**
  - .....
  - .....
  - .....
- **CAUSES OF MMR:**

  Direct: .....

  Indirect: .....
- **RHIME METHOD:** (verbal autopsy)
  - Representative, resampled, routine household interview and medical evaluation
- **Values of MMR:**
- **Goal:** .....

  .....

  .....

Active space

In population of 1,00,000 reported live births are 4000. Causes of maternal deaths are as follow:

Ectopic-2, PET-2, abortion-2, APH-2, PPH-2, electrocution -2
Calculate MMR

a. 250/1 Lac

b. 300/1 Lac

c. 10/1 Lac

d. 12/1 Lac

- Total Numbers of pregnancy

  .....

  .....

- Numbers of pregnancy registered at a time

  .....

  .....

METHODS FOR REDUCTION IN MMR

A. ANTENATAL PERIOD :

- Registration
- Regular antenatal checkup :

  (a) Average : .....

  Every month – till 7 months

  Every 15 days – 8th

  Every wk – till delivery

  (b) Minimum : .....

  1st – In first trimester

  2nd – 14 -26wks

  3rd – 28 -34wks

  4th – 36 wks till delivery

Active space

(c) Td vaccine :

(d) IFA tablets

Composition (adult) :

- elemental iron .....
- folic acid .....

Minimum: 100 tabs in pregnancy

Pregnancy – 4th month onwards till delivery (180 days)

(e) Calcium tab :

4th month onward till delivery

(g) Investigation

- Hb%
- Urine : A & S
- Blood sugar
- Blood group
- VDRL
- Hep B
- HIV

(h) Detection of high risk cases and referral

(f) Single tab 400mg Albendazole in 2nd trimester

B. NATAL CARE :

- Conduct institutional delivery
- Home delivery by trained person TBA in presence of MPW F with use of disposable delivery kit
- By 5 clean methods : .....

  .....

  .....

  .....
- Early detection, timely referral of high risk cases to avoid complication

Active space

C : POSTNATAL CARE :

- Number of Visits
  - Average – 5-6
  - Min : Hospital – .....
    Home – .....
- IFA Tab (as long as lactating) minimum 180days)
- Calcium (as long as lactating) minimum 180days)
- Contraception
- Breast feeding
- Vit A – .....

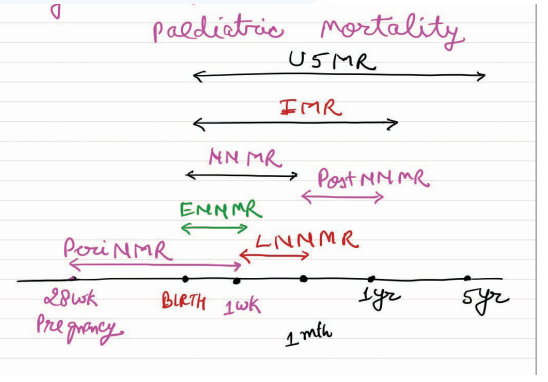
PREVENTION IN PEDIATRICS

⇨ U5MR =
⇨ IMR =
⇨ NNMR =
⇨ ENNMR =
⇨ LNNMR =
⇨ Post NNMR =
⇨ Peri natal MR =

.....

.....

.....



Pediatrics Mortality

Active space

Eg : In a given population, total birth-4050, still birth-50, death 0-7 days-150, death 7-28 days - 50. calculate neo natal mortality rate?

a. 62.5

b. 50

c. 49.4

d. 12.5

METHODS OF PEDIATRIC MORTALITY REDUCTION

1. Breast feeding :

❑ **When to start:**

- .....
- Max within (vaginal delivery) .....
- Max within (operative) .....

❑ **How long to continue:**

- Exclusive : .....
- Supplementary : .....

COMPARISON OF BREAST MILK AND COW MILK

	Breast milk	Cow milk
Protein	1.1 gm %	3.3 gm % (3x)
Cysteine Taurine		
Fat	(3.4 gm %)	(4.1 gm %)
DHA		
Lactose	(7.4 gm %)	(4.4 gm %)
Ig		
Na +		

Active space

Calcium		
Phosphorus		
Ca/P ratio :		
Iron		
Vitamin deficient		
Energy	65-70 kcal/100 ml	67 kcal/100 ml
Infection	Very low	Very high
Whey: Casein ratio	60 : 40	20 : 80

- **MAA** : .....
- **Baby friendly hospital concept (1991)** : keep baby by side of mother
- **Breast feed awareness wk** : .....

2. Immunization

Already discussed in the chapter of vaccination

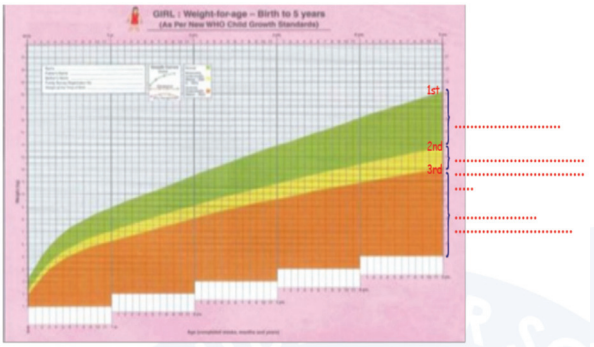
3. Growth monitoring

- Anganwadi center (ICDS)
- Anganwadi workers
- Parameter : Wt for age .....
  Wt for ht .....
  Ht for age .....

Active space

- ICDS GROWTH CHARTS

  MGRS : Multicentric growth reference study Charts



ICDS Growth Charts

- Birth wt :
  - Indicator of chances of survival, development of newborn
  - Average birth wt (INDIA) – .....
  - LBW - .....
  - Birth Weight is measure in Rural areas .....
  - Within 1 hr of delivery
  - Best Indicator of maternal nutrition

Salter Scale

Weighing Scale

4. School health :

Health checkup Medical officer - dental disease

Eye checkup : teacher (1/150)

Vision eye center

Referral if .....

Deworming : Albendazole 400 mg

Active space

IFA tab ⟶ .....

mid day meal (1/2 protein ; 1/3 calorie) .....

Maximum Numebrs of the students – .....

Space – .....

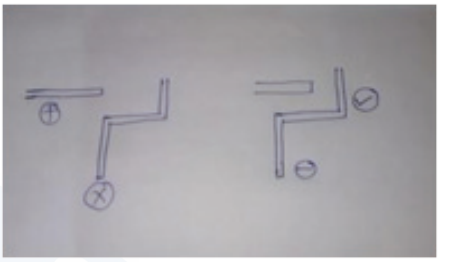
type of desk .....

Urinal – .....

Toilet – .....

Light should be – Left side

Ventilation – .....



Minus Type of Desk

✦ Applied nutrition programme

where land available in school UNICEF is funding for seed, manure, water supply equipment for developing school gardens which can be utilized in school feeding programme, and for nutritional education

✦ School Health and wellness programme under AYUSHMAN BHARAT

Health & wellness Ambassadors .....

Health & wellness Messenger .....

School activity

a. Wkly : classroom interaction by Health Ambassador

: weekly IFA Tab Tuesday

b. Monthly / Bimonthly

: Thematic school assembly

: Question box response

c. Quarterly

: Thematic Activity

: parent teacher meeting

d. Bi annual

: Albendazole Tablet

Active space

NATIONAL HEALTH PROGRAMS

NATIONAL HEALTH MISSION (PMSSM)

✦ NHM includes

- Reproductive and child Health
- National TB elimination Program
- National Vector born disease control Program
- National Program for control of blindness and visual impairment
- Mental Health
- Non communicable disease screening DM, HTN

☞ NOT INCLUDED:

- National AIDS control Program
- Cancer Program

1. RMNCAH + N

Reproductive, maternal, newborn, childhood & adolescent health plus nutrition strategy

❑ **ACTIVITY**

1. Registration of every pregnant woman
2. Mother and child tracking system (MCTS)
3. National mobile medical units (NMMU)
4. MCH wing
5. PM Surakshit matratva abhiyan (Vande Mataram Scheme) :.....

.....:

NMMU

PM Surakshit Matratva Abhiyan Logo

Active space

✦ Types of Stickers

Green: .....

Red: .....

Blue: .....

Yellow: .....

.....

PM Matra Vandana Yojna Logo

6. Matra Vandana Yojna

A. 1st – Rs 3000 on registration and atleast one Ante notal visit in 6 month from LMP Date

2nd – Rs 2000 [6 weeks after delivery]

- Hospital delivery
- 1st dose of baby vaccination
- Contraception

B. Single installment of 6000/- 6wk after dilivery if 2nd baby is also female

7. SUMAN :

.....

.....

Surakshit Matratva Ashwasan

8. Kilkari :

.....

.....

.....

LaQshya Logo

9. LaQshya : .....

.....

10. Dakshta : .....

.....

Active space

11. ARMMAN : .....

.....

.....

Armman Logo

3. JSY - Janani Suraksha Yojna

- Purpose .....

Cash incentive in Rs

	RURAL			URBAN		
	Patient	ASHA	Total	Patient	ASHA	Total
LPS	1400	600	2000	1000	400	1400
HPS	700	600	1300	600	400	1000

- LPS – low performance states; all delivery, all women

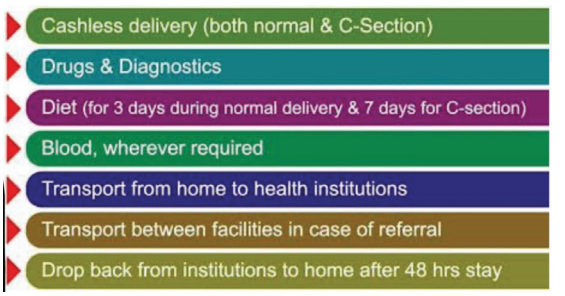
  MP, Chattisgarh, UP, Uttarakhand, Odisha, Rajasthan, Assam, J & K

- HPS – high performance states; BPL women; SC / ST, upto 2 delivery

4. JSSK: Janani Shishu Suraksha Karyakram

- Started in 2011
- Maternal as JSY Scheme
- Neonate (infant upto 1 yr)

Free Entitlements for Pregnant Women



Active space

- **1. HBNBC**
- **2. HBYC**
- **3. IMNCI (Integrated management of Neonatal and Childhood Illness)**
  - Worker .....
  - Beneficiaries :
  - Aim: early detection and timely referral to avoid preventable mortality
  - Disease: diarrhea ARI, Measles, malnutrition, malaria
  - Categories:
    - Green: .....
    - Yellow: treatment and follow up

      Improvement → .....

      No improvement → .....
    - Pink: .....
      .....
- **4. F-B NBC (facility based newborn care)**
  - (i) NBCC (newborn care corners)
    - At each delivery point
    - Newborn resuscitation
  - (ii) NBSU (newborn stabilizing unit)
    - From CHC onwards
    - 6 beds : 4 baby, 2 stepdown beds

Janani Shishu Suraksha Karyakram Logo

Active space

MEDICAL INSTITUTE OF SCREENING TECHNOLOGY MIST

(iii) Special newborn care unit (SNCU)

- From district hospital onwards
- 16 beds : 12 ICU, 4 stepdown beds

❑ 5. Nutritional Rehabilitation Centers (NRC)

- 0-5 yrs
- Severe malnutrition
- Medical and nutritional care

5. Rashtriya Bal Swasthya karyakram (RBSK) (2013)

Beneficiaries- .....

0-6 yrs = ...

6- 18 yrs = ...

Detection of

- Birth defect
- Diseases
- Developmental delay
- Deficiency

RBSK Logo

6. Rashtriya kishore swasthya karyakram 2014 (RKSK)

✦ Beneficiary:- .....

✦ Functions

- Sexual health(ARSH CLINIC)
- Menstrual hygenie
- Substance abuse
- Non communicable disease

RKSK Logo

Active space

- Mental health
- Violence, injury
- Nutritional health

7. Navjat Shishu Suraksha Karyakram (2009) (NSSK)

- Training of health workers for basic newborn resuscitation
- 2 days duration
- Aim - Decreased IMR

Active space

EXTRA EDGE

- Reproductive female & children <15yr population is 57.5% of Total population
- Reproductive female & children <5yr population is 32.4% of Total Population
- 3rd ANC visit by medical officer at PHC (28-34wk)
- Average wt gain in pregnancy : 9-11kg
- Wt gain in undernourished women : 5-6 kg
- Extra calorie required in pregnancy : 60,000 k cal
- Kangaroo Mother Care : for LBW Baby

  Component :

  1. Skin to skin contactof baby to motherchest
  2. Adequate nutrition through breast feed
  3. Ambulatory care (early Hospital discharge)
  4. Support of mother by her family for baby care

- Wt recording of baby in ICDs center
  1. Monthly from birth upto 1 Yr
  2. Monthly in second year of life
  3. Every 3 monthly b/n 2-5yr

- Inspire (Seven strategies for ending violence against children)
  1. Implementation and enforcent of law
  2. Norms and value
  3. Safe environment
  4. Parent and care giver support
  5. Income and economic strengthening
  6. Response and support services
  7. Education and life skill

Active space

✦ NHM (PM Samagra Swasthya Mission)

Population Norm

- N RHM → < 50 000
- NU HM → ≥ 50,000
- Mahila Arogya Samiti → 250-500 (50-100 House Hold)
- ASHA → 1000
- USHA / Link worker / community Health worker
  → 1000-2500 (200-500 House hold)
- ANM → 10,000
- MPWF → 5,000

✦ Rogi Kalyan Samiti (Patient Welfare Committee)

(Hospital Management Committee)

- Registered society act as groupof trustee for Hospital to manage hospital affairs
- Members → Local elected member, NGO, Government official, Panchayat Raj Institution members.
- Core unit of planning, budgeting, implementation of NRHM → DISTRICT
- Emergency Response service vehicle
  - 102 → Transportof pregnant women, children
  - 108 → Emergency patient transport
- Ayushman Bhav campaign

  Ensure reach of health care services to last mile and enable access of everyone in society to these services.

  1. Ayushman Apke Dwar
  2. Health mela at Ayushman Arogya mandic

Active space

✦ Delivery Points

L1 : Minimum 3 delivery / month

L2 : Minimum 10 delivery / month with Mx of complication

L3 : Minimum 20-50 delivery / month with C - section

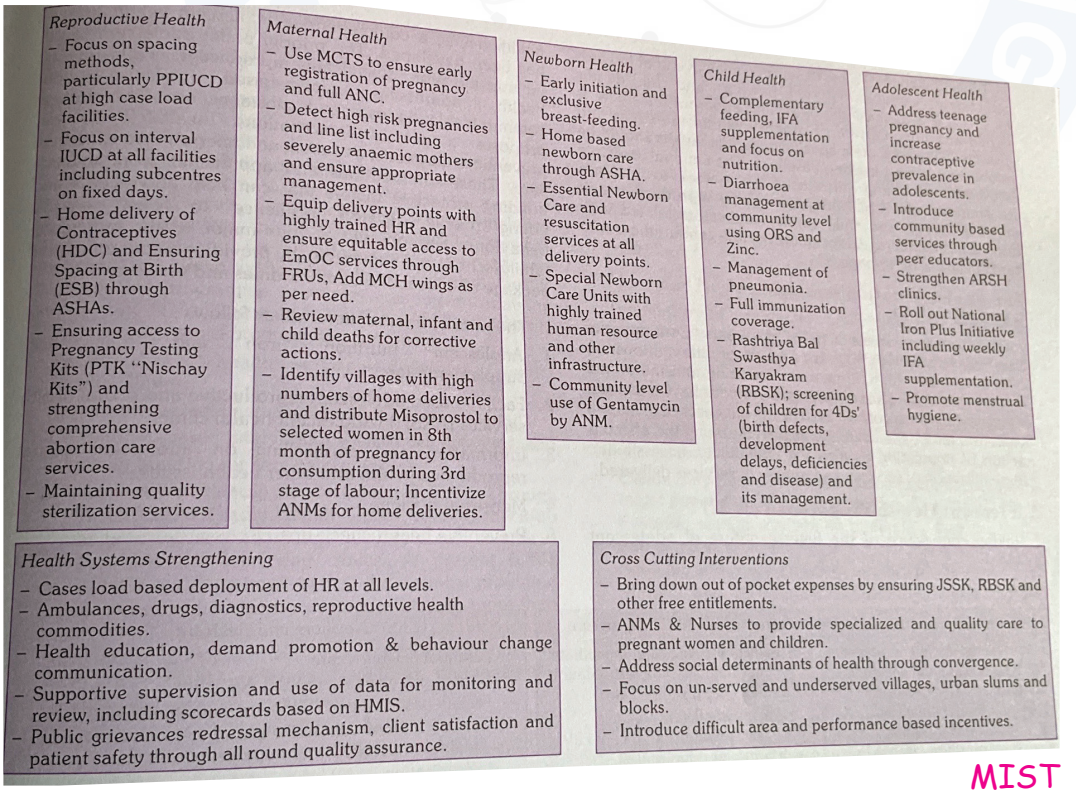
✦ RMNCHA+N 5x5x5 matrix

- 5 High impact intervention in

A) 5 thematic area (Reproductive, Maternal, newborn, child, adolescent Health)

B) 5 cross cutting intervention

C) 5 Health system strengthening



Active space

NUTRITION

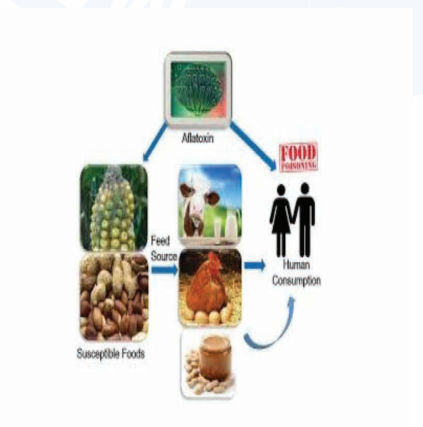
FOOD ADULTERATION

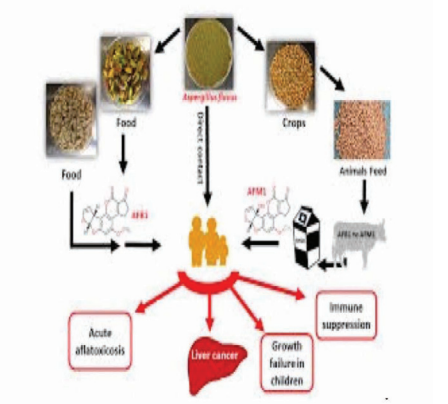
Lathyrism

Epidemic dropsy

Ergotism

Endemic Ascites





Aflotoxicosis

Active space

DISEASE	TOXIN	AGENT	FOOD STUFF
Lathyrism	.....	Lathyrus sativus	Yellow pulse
Epidemic dropsy	.....	.....	Mustard seed
		Sensitive Test to detect contamination : paper chromatography (0.0001% contamination)	
Endemic ascites	.....	Crotorella seed (jhunjhuniya seeds)	Millet
Aflotoxicosis	Aflatoxin (hepatotoxic)- hepatic cancer	Aspergillus flavus	Grains (high, moisture during storage)
Ergotism	Ergot alkaloid (clavine)- vasospasm- gangrene	Claviceps fusiformis	Flowering plants

MILK

- **Deficiency:** .....
- **Pasteurization:**
  - Partial boiling- sudden cooling
  - Kill pathogenic organism (not spores, ova, cyst)

MIST

Active space

- Organism killed:
  - Coliform group, Typhoid bacilli
  - Brucellosis, Q fever
- Methods of Pasteurization:

METHODS	TEMPERATURE	DURATION
Holder(VAT)	.....	.....
High temp short time (flash)	.....	.....
Ultra high temp method	.....	.....

- Test applied before pasteurization: .....
  Purpose: to know of degree of contamination
- Test applied after pasteurization:
  To check efficacy of procedure
  (1) Phosphatase test .....
  (2) Coliform count = ...
  (3) Standard plate count .....
  .....

Methylene Blue Test

BALANCE DIET

Energy from
Protein: .....
Fat: .....
Carbohydrates: .....

CALORIC REQUIREMENT (KCAL PER DAY)

Work	MALE	FEMALE
SEDENTARY	.....	.....
MODERATE	.....	.....
HEAVY	.....	.....

Active space

✦ EXTRA CALORIC REQUIREMENTS

1. Preg : .....

   .....

2. Lactation

   0-6 month: .....

   >6 month: .....

   Eg.: Calculate the Caloric requirement of Female Moderate Worker H/O delivery 5 month back

❑ DAILY REQUIREMENT DOSE

Age Group	Category ofwork	Body wt. (kg)	Protein (g/d)	Calcium (mg/d)	Iron (mg/d)	Iodine (mcg/d)	Folate (mcg/d)	Vit. B12 (mcg/d)	Vit. A (mcg/d)	Vit. D (IU/d)
Men	Sedentary									
	Moderate	65	54.0	1000	19	140	300	2.2	1000	600
	Heavy									
Women	Sedentary									
	Moderate	55	46.0	1000	29	140	220	2.2	840	600
	Heavy									
	Pregnant Women	55 + 10	+ 9.5 (2nd trimester) + 22.0 (3rd trimester	1000	27	220	570	2.45	900	600
	Lactation 0-6 m 7-12 m		17.0 13.0	1200	23	280	330 330	3.2	950	600

1. EAR : estimated average requirement = ...
2. RDA : Recommended Dietary Allowance (Mean + 2SD) = ...

MID-DAY MEAL PROGRAM / PM POSHAN YOJNA

Poshan Shakti Nirman

- Ministry of .....
- Composition .....
- Composition of food

Primary (1-5): 450 kcal 12g of protein

Middle (6-8): 700 kcal 20g of protein

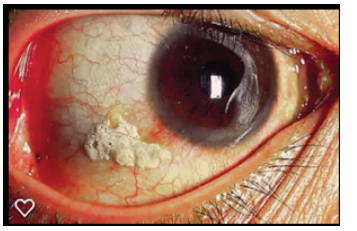
MIST

Mid-day Meal Program

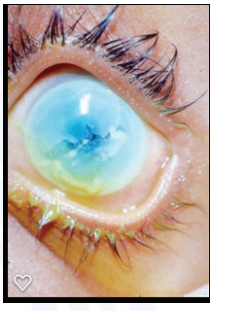
Active space

VITAMIN-A PROPHYLAXIS PROGRAM

- **Deficiency:** .....
  - WHO: PRIMARY SIGN
    - X1A: .....
    - X1B: .....
    - X2: .....
    - X3A: .....
    - X3B: .....
  - Secondary features:
    - N: .....
    - XF: .....
    - XS: .....



BITOT spot



Keratomalacia

- **1st symptom:** .....
- **1st sign:** .....
- **Most common clinical sign for screening:** .....
- **Most sensitive screening test:** .....
- **MAJOR PUBLIC HEALTH PROBLEM:**
  - Prevalence of:
    - NB - .....
    - Bitot Spot - .....
    - Corneal Scar - >0.05%
    - Corneal xerosis - >0.01%
    - Serum retinol (<10 mcg/ml) - .....
- **SOURCES:**

Animal: (richest) .....

(2nd richest) .....

Vegetable: carrot, mango

Papaya, colourful fruit, veg.

Active space

❑ **Treatment of Vit A deficiency:**

3 doses of Vit A Amount

- ✦ All children with measle : give one dose of Vitamin A if not received with in 1 month
- ✦ Severe malnurished children : one additional dose of Vitamin A

❑ **PROPHYLAXIS** .....

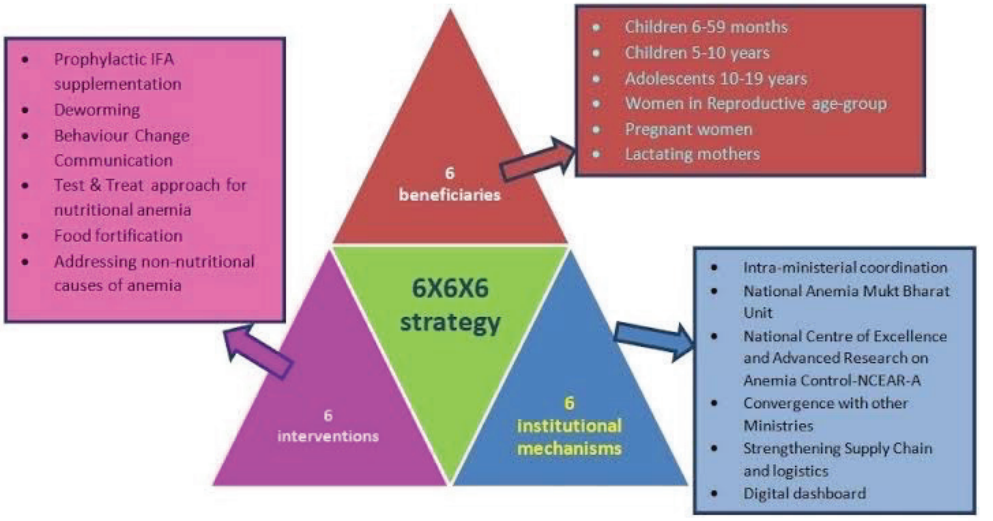
INTENSIFIED IRON PLUS INITIATIVE (ANEMIA MUKT BHARAT)

GOAL

↓ Prev of anaemia by 3%/yr in children, adolescent, Rep ♀ (2018-22)

:

Anemia Mukt Bharat



❑ **STRATEGY:**

Test, Treat, Talk

Active space

❑ PROPHYLAXIS

BENEFICIARIES	IRON	FOLIC ACID	FREQUENCY
6 months - 59mth	20 mg	100 mcg	
5 – 9 yrs	45 mg	400 mcg	
10 – 19 yrs	60 mg	500 mcg	
20 – 49 yrs	60 mg	500 mcg	
Pregnancy	60 mg	500 mcg	
Lactation	60 mg	500 mcg	

❑ Cut off level of anemia

1. In male – :
2. In non-pregnant female : .....
3. Female (pregnant) : .....
4. Children (6 months- 6 years) : .....
5. Children (6 yrs – 14 yrs) : .....

❑ Most sensitive indicator of iron deficiency

- .....

  N – 20-160 mcg/L

  <10 mcg/L – complete absence of iron store

❑ Continue iron preparation After Normal Hb %

❑ Good Veg sources

- Pumpkin seeds
- Date
- Jaggery
- Pista

❑ Animal Sources

Red Meat

Active space

NATIONAL IODINE DEFICIENCY DISEASE CONTROL PROGRAM (NIDDCP)

❑ PREVENTIVE MEASURES

1. Iodised salt

   Production level .....

   Consumption level .....

2. Life saving DOTS

   (Jeevan bindi) – Iodine skin patch

Jeevan Bindi

❑ Major public health problem:

Prevalence of goitre .....

❑ INDICATORS

1. Prevalence of neonatal hypothyroidism
2. Urinary iodine level excretion

NATIONAL PROGRAM FOR PREVENTION AND CONTROL OF FLUOROSIS

- N level: ..... (mg/L)
- Max upper tolerable limit: .....
- Flourosis .....
  - (Dental fluorosis) .....
  - Skeletal: .....
  - Crippling .....
- DENTAL CARIES .....

  Treatment: fluoride containing tooth paste
- Deflouridation of water:

Active space

NATIONAL NUTRITION MISSION (POSHAN ABHIYAN)

- 1. Goal
- 2. Poshan Maah

MACRONUTRIENTS

A. Protein :

- Quality

  DIAAS (Digestible Indispensible AA score)

  PDCAAS – protein Digestibility corrected AA score)

  AAS – Amino acid score

  NPU – net protein utilization

  NPU – amount of protein retained in body after digestion and absorption

  Best quality indicator for dietary protein supplementation

	EGG	SOYA
NPU	.....	.....
WT/WT	.....	.....

- Daily req :

  0.83 gm/day/kg

- Limiting AA

Active space

Marasmus	Kwashiorkor

- DEFICIENCY:
  1. Marasmus
  2. Kwashiorkor

B. FAT

- Essential fatty acid:
  1. Linoleic acid (most important) :

     Richest source: .....
  2. Linolenic acid

     Richest source: .....
  3. DHA (decosahexanoic acid)

     Brain development

Phrenoderma

- Deficiency of EFA: .....
- Types of Fatty Acid (based on carbon bond)

	SATURATED FATTY ACID	UNSATURATED FATTY ACID	
		MUFA	PUFA
Source	.....	.....	....

Active space

MIST

C. CARBOHYDRATES:

- Major source of energy
- Glycemic index : ability of food to increase blood sugar level

  ≤ 55: .....

  56-69: .....

  ≥ 70: .....

D. FIBER:

- Intestinal function
- 30 gm/day
- Soluble : pectin
- Insoluble : cellulose

MICRONUTRIENTS

MINERALS

1. Calcium :

- Vit required for Ca metabolism: .....
- Grain which is richest in Ca concentration: .....

2. Zinc :

- Requirement : 5-10 mg/day
- Deficiency : hair loss
  Infertility
  Delayed wound healing
  Diarrhoea

Figure 1: Acrodermatitis Enteropathica on 15 month old female characterised by periorificial and acral dermatitis and alopecia. There are annular, well demarcated, erythematous, crusted plaques and vesicles distributed symmetrically around orifices and hands.

- ACRODERMTITIS ENTEROPATHICA: .....
  .....

Active space

3. Selenium :

- Antioxidant
- Other dietary antioxidant:

- Requirement : 50-60 mcg/day
- Deficiency : .....

❑ **4. Iron**
❑ **5. Iodine** } Already discussed above
❑ **6. Fluorine**

VITAMINS

1. VIT A: Already discussed above

2. VIT B COMPLEX

❑ **B1 (Thiamine)**

Deficiency: .....

.....

.....

Prevention: .....

.....

.....

❑ **B2 (Riboflavin)**

Deficiency: .....

.....

Active space

- **B3 (Niacin)**

Cassle Neck

- **B5 (Pantothenic acid)**

  Deficiency: .....

- **B6 (Pyridoxine)**

  Deficiency: .....

  Supplement B6 with INH

  B6 – Treatment of emesis gravidarum

- **B7 (Biotin)**

  Deficiency: .....

- **B9 (Folic Acid)**

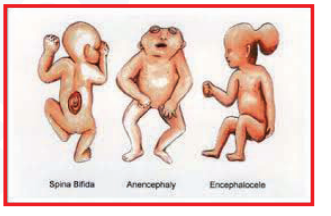
  Deficiency: .....

  .....

  .....

  .....

  .....



NTD

- **B12 (Cyanocobalamine)**

  Deficiency: .....

  .....

  .....

Active space

3. VIT C (ASCORBIC ACID)

Deficiency: .....

Requirement: 40 mg/day

Richest source: .....

.....

4. VIT D

Most active form: .....

(1, 25 dihydroxy cholecalciferol)

Deficiency: .....

.....

Richest source: Halibut oil

Cod liver oil (2nd richest)

Rickets

5. VIT E (Tocoferol)

Antioxidant (Strongest)

Requirement : 8-10 mg/day

Richest source : .....

Deficiency : haemolytic anemia of newborn

6. VIT K

Deficiency: .....

Administration of vit K to each and every newborn after delivery

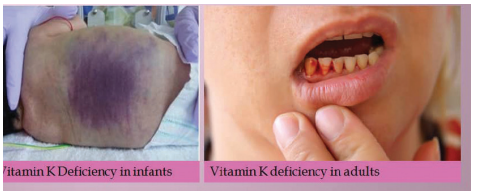
Natural form (fat soluble):

K1: phyloquinone (diet)

K2: menaquinone (intestine)

Synthetic Form (Water soluble)

K3: menandione



Vitamin K Deficiency in infants | Vitamin K deficiency in adults

Active space

FOOD STANDARDS

- **PFA STANDARDS :**
  - Prevention of food adulteration act (1954)
  - Minimum quality requirement under Indian conditions
- **BIS (BUREAU OF INDIAN STANDARDS) :**
  Degree of excellence over PFA standard
- **FSSAI (FOOD SAFETY STANDARDS AUTHORITY OF INDIA)**
  Licensing to businessmen
- **CODEX ALIMENTARIOUS :**
  Standards for international market
- **AGMARK**
  Certification of agricultural products

BIS Logo

Agmark Logo

FSSAI Logo

Active space

EXTRA EDGE

1. MY Plate for day 2000 kcal

MY Plate for day 2000 kcal

Food Pyramid for balanced Diet 2000 kcal

- Nutritional Assessment
  1) Clinical examination
  2) Anthropometry
  3) Lab investigation Biochemical
  4) Functional assessment
  5) Dietary intake assessment
     a) Raw Food wt (one dietary cycle -> 7days)
     b) Cooked food wt
     c) Oral questionnair
  6) Vital statistic
  7) Ecological study

Active space

BIOSTATISTICS

TYPES OF DATA

❑ 1. Based on Measurements

QUANTITATIVE	QUALITATIVE
.....	.....
Hb %	Rh
Blood sugar	Blood Group
BP	Presence of disease
Cholesterol	

❑ 2. Based on Possible numbers

CONTINUOUS	DISCRETE
....	....
Hb%	Rh
Length	Blood group
Weight	Presence of disease
Blood sugar	
Serum cholesterol	

Eg: Cholesterol is what kind of data? : .....

DATA PRESENTATION METHOD

A: Quantitative Data

1. Histogram – .....
2. Frequency polygon
3. Frequency curve

Active space

4. Line chart
   - Trends of events over time
   - Relationship between two parameters
5. Scattered diagram
   - .....

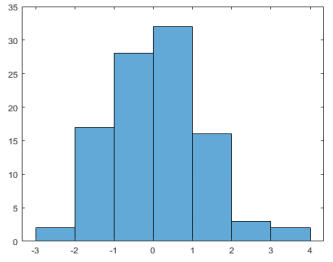
     Degree of association between two parameters

     0 = ...

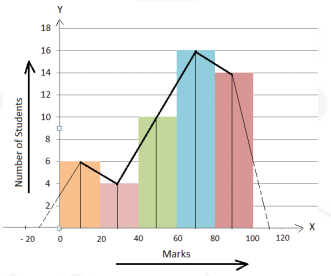
     0 to + 1 = ...

     -1 to 0 = ...

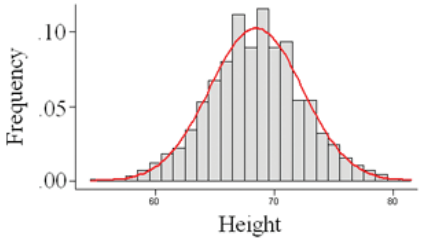
✦ Histogram



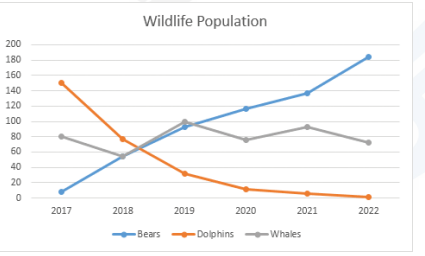
✦ Frequency polygon



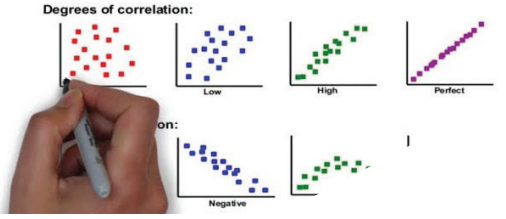
✦ Frequency curve



✦ Line chart



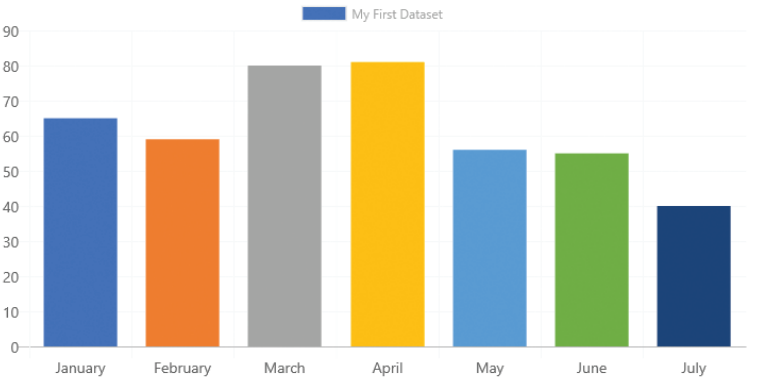
✦ Scattered diagram

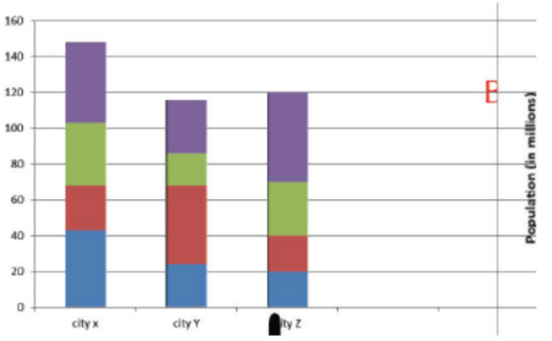


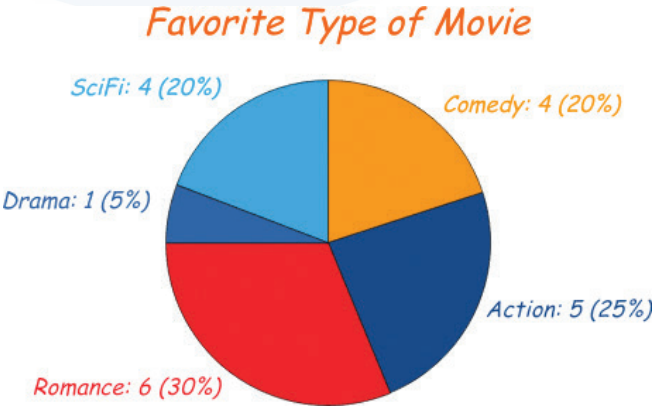
Active space

B: Qualitative Data:

1. Bar chart
2. Pie chart (Total value 100%)
3. Pictogram : (with picture)
4. Spot map analysis :







Pie Chart

Active space

Varities of Apples in a food store	
Red Delicious	
Golden Delicious	
Red Rome	
McIntosh	
Jonathan	

= 10 apples = 5 apples

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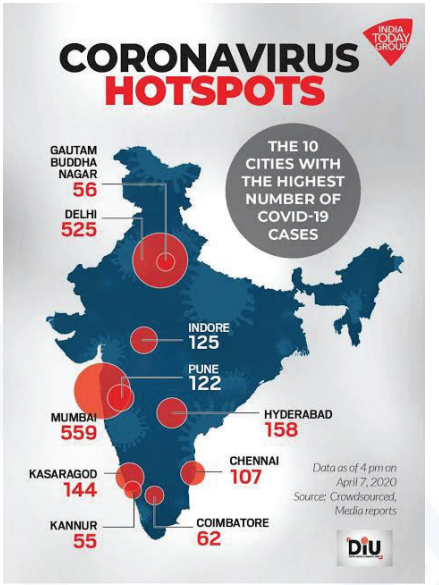
Spot Map Analysis



Pictogram

DATA CALCULATION

CENTRAL TENDENCY	DISPERSION
Mean	Range
Median	Standard deviation
Mode	Relative deviation
	Standard error
	Variance
	Variability
	Coefficient of variation
	Precision

CENTRAL TENDENCY

MEAN

Calculate mean for : 3 4 5 6 7,8 9

Active space

MEDIAN

Middle value (Odd No)/Average of middle two values (even no) arranging in ascending/descending order

Eg: (a) 3, 4, 9, 6, 8, 7, 5 calculate the median

(b) 3, 4, 9, 6, 7, 5 calculate the median

MODE

Most frequently occurring value

Eg: (i) 3, 3, 4, 4, 3, 5, 8 = ...

(ii) 3, 4, 3, 4, 5, 8 :

(a) Uni model : .....

(b) Multi model : .....

Calculate Mode

1) 3	1) 3.5	1) 3, 4
2) 3.5	2) 3, 4	2) 3. 5
3) 4	3) Any of above	3) 3
4) 5	4) None of above	4) 4

✦ Central tendency parameter value more than one

☞ OUTLIERS:

Extreme value for given data

Eg : 40, 45, 48, 50, 52, 100 kg

Most affected – .....

Least affected – .....

Preffered - .....

Active space

☞ Preference:

Mean: .....

Median: .....

Mode: .....

DISPERSION

- **Range :** value between two extreme ends
- **NORMAL DISTRIBUTION**

Features:

1. Bell shaped (B/L symmetrical)
2. Mean = median = mode
3. 50% - above mean
   50% - below mean
4. Mean ± 1 SD
5. Mean ± 2 SD
6. Mean ± 3 SD
7. SD = 1

Active space

✦ SKEWED DEVIATION

Eg: Mean Cholesterol level of 100 person is 140 mg% with SD of 5mg% Calculate

(a) Cholesterol range (b) % of population covered

✦ STANDARD DEVIATION

Calculation

1 Mean

2 Deviation of each value from mean

3 Square Deviation

4 Add square Deviation

5 Divide by Total number value

6 Square Root

Ex.: On 15th August 2017, 10 babies delivered at LHMC, Delhi. Each weighs 2.8 kg. Calculate standard deviation

Active space

✦ RELATIVE DEVIATE

Z score: deviation of a value from mean in terms of SD

RD = Z score

Upto ± 2 = ...

Beyond ± 2 = ...

Eg: Mean wt of 100 students in a class is 56kg with SD of 2.5kg. Calculate relative deviate of 'X' with wt of 60kg and comment if normal / abnormal

✦ STANDARD ERROR OF MEAN

- Measure of difference b/w sample and population value
- SE of mean = ...
- Depends on:

  .....

  .....
- Measure of chance variation

  Ex. Hb survey done on 10,000 students. Mean Hb was 10.5 gm% With of SD of 0.5 gm %.Calculate standard error of mean

STATISTICAL ERRORS

❑ NULL HYPOTHESIS

TRUE	FALSE
Type 1 error (most dangerous)	Type 2 error

- p = ...
- β = ...
- α = ...

Active space

MIST

✦ Significance of p value

- P = < 0.05 - .....
- P = < 0.01 - .....

TEST OF SIGNIFICANCE

PARAMETRIC	NON PARAMETRIC .....
1. Student t test :	1. Sign test .....
Compare same group before & after intervention / Compare 2 different groups	
2. ANOVA test	2. Chi Square test

✦ Examples

- study done on doctors, jr, nurses of delhi and mumbai about Prevalence of anoemia. Best test for analysis is
  a. Chi square test
  b. Student T test
  c. Unpaired student t test
  d. Anova test

- BP of 100 Hypertensive patients attending medicine opd in safdarjang hospital was compared before and after giving a new anti hypertensive drug. Test used for comparison is
  a. Student t test
  d. Student paired t test
  c. Student unpaired t test
  d. Chi square test

Active space

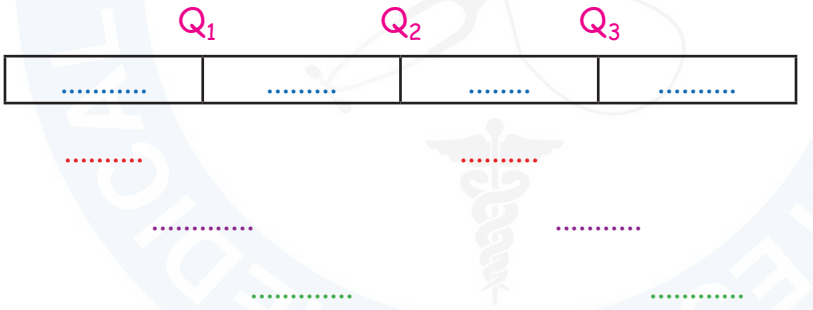
- Base line BP of Delhi people compared ẽ Mumbai people test used for comparison
  a. Paired T test
  b. Unpaired T test
  c. ANOVAS test
  d. Chi square test

DISTRIBUTION OF DATA

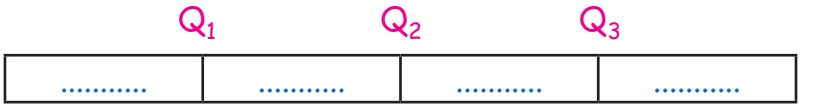
1. Centile distribution

   Eg : Height of 200 individuals was recorded in a camp. Mean Ht was 160cm Ht 'X' comes at 95th percentile. Calculate % of individuals having height above 'X'

2. Quartile distribution :



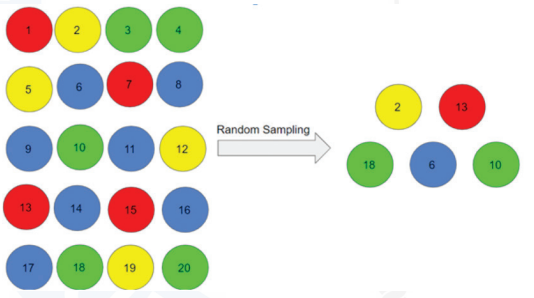
Eg: mean height of 200 students in a school was 145 cm. height of a student 'X' is at level of 2nd quartile. Calculate % of students having heights less than 'X'



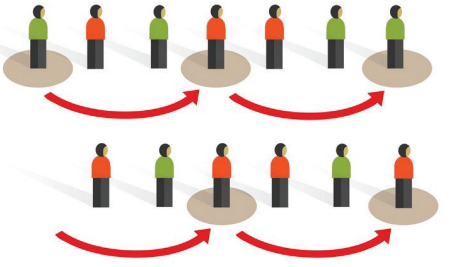
Active space

SAMPLING METHODS

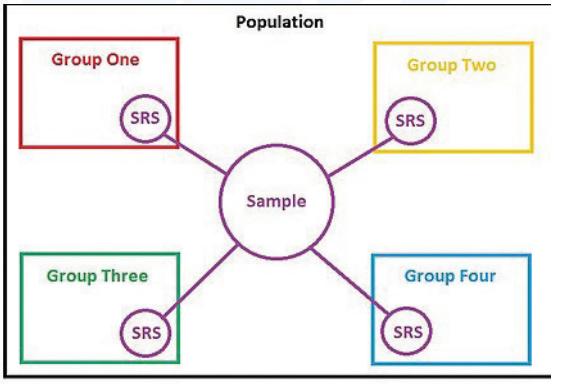
1. Random sampling methods –
   1) Stratified random : divide the values – groups – randomization
   2) Systematic random : sample interval
   3) Simple random: lottery system
   4) Cluster random: 30x7
2. Non random sampling methods :
   i) Voluntary
   ii) Quota
   iii) Snowball
   iv) Convenience



Simple Random

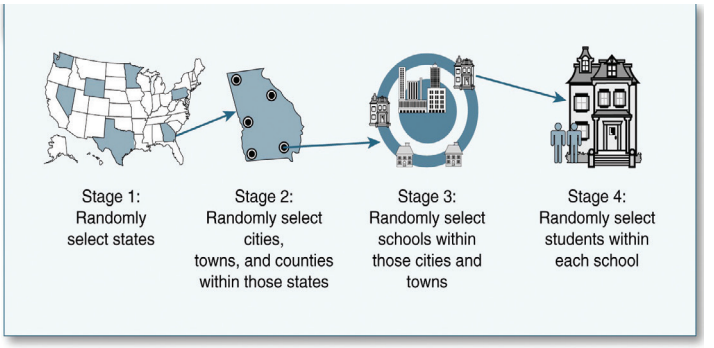


Systematic Random

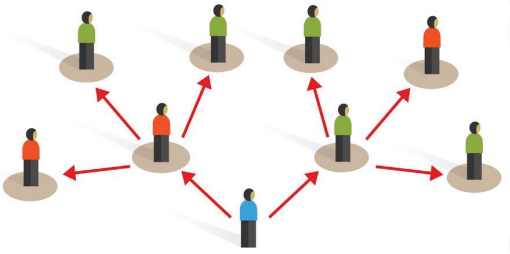


Stratified Random

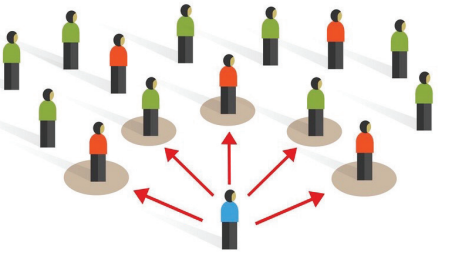
Active space



Cluster Random

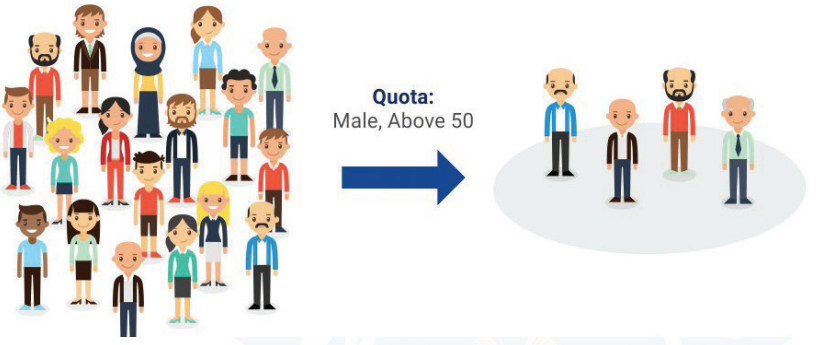


Snowball Sampling



Convenience Sampling

Quota Sampling



Quota Sampling

Active space

ENVIRONMENT & HEALTH

WATER

- Most purest form of water .....
- Most important source of water supply in rural area .....
- minimum distance between sanitary well to source of contamination .....

Sanitary Well

WHO CRITERIA FOR DRINKING WATER:

1. Free from pathogenic organism (E coli)
2. Chloride content : <200 mg/L
   Max upper limit – upto 600 mg/L
3. TDS (Total Dissolve Solid)
   <500 mg/L
   Max upper limit – 1000 mg/L
4. Nitrate - <45 mg/L
5. Nitrite - <3 mg/L
6. pH : 6.5 – 8.5
7. gross radiological activity
   α = <0.5 βq/L
   β = <1.0 βq/L
8. colour - <15 TCU (true color unit)
9. turbidity - <1 unit
   NTU (nephlometric turbidity unit)

Active space

HARDNESS OF WATER

1. TEMPORARY (Carbonate)
   : Ca/Mg Bicarbonate
2. PERMANENT (Non Carbonate)
   : Ca/Mg Chloride, Nitrate, Sulfate

LEVELS OF HARDNESS:

Soft - .....

Moderately hard – .....

Hard – .....

Very hard - .....

☞ Softening of water recommended:

.....

Purification:

1. Storage :

   5-7 days: 90% of suspended impurities are removed

2. Filteration

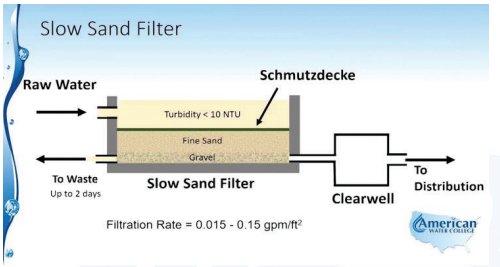
	Slow sand filter	Rapid sand filter
Rate	Slow	Rapid
Size (sand particle)	.....	.....
Type of filtration	Mechanical + biological	Mechanical
Shumutz Deck Layer	Present	Absent
Quality	Better	Low
Venturiometer	.....	.....
Washing	Scraping	Back Washing.
Operation	Less Skilled	Highly Skilled

Active space

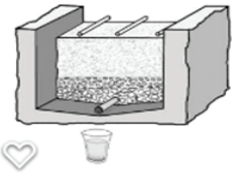
Shumutzdeck Layer :

- Algae and Planktons

❑ **Venturiometer:** measures resistance of head of filter



Slow Sand Filter



Rapid Sand Filter

Venturiometer

DISINFECTION:

- Chlorination :
- Ozone
- UV rays

⇨ DISINFECTION AGENT

☞ Kill only pathogenic organism

Not ova, cyst, spores

⇨ CHLORINE DEMAND:

- Amount of chlorine required to kill pathogenic organism
- By : .....
- Indicator : .....

HORROCK'S apparatus

Active space

Amount of bleaching powder (g) = ...

n = no of 1st cup showing colour change

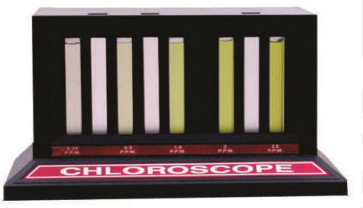
Ex.: In HORROCK'S apparatus, 4th cup onwards colour change is noted. Calculate amount of bleaching powder required to disinfect 455 L of water.

⇨ BREAK POINT:

.....

⇨ FRC:

- N drinking water : .....
- Disaster : .....
- Swimming pool : .....
- Instrument : .....



Chloroscope

☞ ORTHOTOLUIDINE TEST (OT): .....

.....

.....

ORTHOTOLUIDINE ARSENITE TEST (OAT) : .....

- No effect of Iron, Magnese, Nitrite

.....

.....

BACTERIOLOGICAL INDICATOR OF WATER CONTAMINATION

FECAL	NON FECAL
E coli	Klebsiella .....
Fecal streptococci	
Cl perfringens	

Active space

INSECTS

Disease Caused by Various Insects

❑ **1. House fly :**

.....

- Trachoma
- Conjunctivitis
- Typhoid
- Cholera

❑ **2. Mosquito :**

(i) Anopheles : .....

(ii) Culex :

(iii) Aedes :

(iv) Mansoni :

.....

.....

.....

.....

.....

❑ **3. Sand fly :**

.....

- K – .....
- O – .....
- S – .....
- O – .....
- Chandipura Virus encephalitis

Active space

❑ **4. Tse-Tse fly :**

Sleeping sickness (African Trypanosomiasis)

❑ **5. Black fly :**

Onchocerciasis (river blindness)

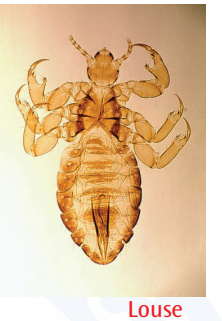
❑ **6. Louse :**

E - .....

T - .....

R - .....

Pediculosis



Louse

❑ **7. Flea :**

- Endemic typhus
- Plague .....

.....

❑ **8. Mite :**

.....

❑ **9. Soft tick :**

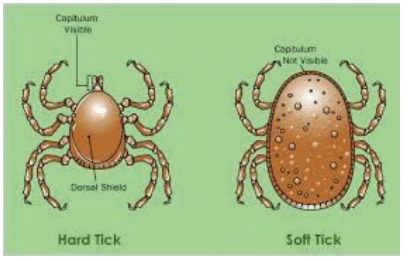
- R - .....
- Q - .....

.....

.....

- KFD (Kyasanur forest disease)

Outside India



Tick

Active space

❑ **10. Hard tick :**

- KFD (India)
- RMSF (Rocky mountain spotted fever)
- Indian tick typhus
- Tularemia
- Babesiosis

❑ **11. Reduwid bug :**

Chaga's disease

.....

TYPES OF MOSQUITO:

FEATURES		ANOPHELES	CULEX	AEDES	MANSONI
A : Adult					
1.	Sitting position	Acute angle with surface	Parallel with surface	Parallel with surface	Parallel with surface
2.	Wing	Spotted	—	—	—
3.	Body	—	—	White spots	—
4.	Fly range	3-5 km	9-11 km	100 mt (100-200 mt)	3 Km.
5.	Breeding site	Fresh water	Dirty water	Artificial water collection	Water with aquatic plants
6.	Eggs	Single, boat shaped	Clusters (irregular)	Single (cigar)	Clusters(star shaped)
7.	Larva	Parallel to surface	Suspended from surface	Suspended from surface	Attach to Roots of Aquatic plants

Types of Larva

Active space

CONTROL OF MOSQUITO

❑ **1. Larva control:** .....

i) Environmental: avoid breeding .....

ii) Chemicals:

- Paris green dust (stomach poison) : .....
- Temephos : .....
- Insect growth regulator

iii) Biological:

- Gambusia fish
- Anti larva bacteria - B Thuringiensis Subunit Israelensis [Bti], B Sphaericus

iv) Mechanical: removal of water plant (mansonian);

❑ **2. Control of adult:** .....

A. Insecticidal agent:

(i) DDT

- Synthetic
- Residual effect (6-12 months)
- Every 6 months spray
- Organochlorine (least toxic)
- 70-80% para-para isomer
- DDT + pyrethrum – Synergistic action

(ii) Melathion :

- Synthetic
- Residual (3-6 months)
- Every 3 months spray
- Organophosphate group (Alkyl) : least toxic

  Aryl: parathion
  Chorthion
  Diazenon
  Paraoxol

- Ultralow volume spray

Active space

(iii) Pyrethrum :

- Natural – flowers of chrysanthemum
- Non residual
- Space spray
- With DDT as synergistic action

(iv) Pyrethroid (synthetic)

- Deltamethrin
- Permethrin
- Allethrin
- Cyclethrin

❑ **3. Use of mosquito net:**

- ITBN (Insecticide treated bed net) : 6 month pyrethroid
- LLIBN (long lasting insecticide bed net) : 3 yrs

❑ **4. Mosquito Repellants:**

- Cream, Lotion, Liquid, Skin patch
- DEET (diethyl toluidine) : synthetic pyrethroid

❑ **5. Attractive Toxic sugar bait [ATSB]**

AIR

1. Air pollution controlled by .....

   (Ministry of environment and forest and climate change)
2. Most important air pollutant – .....
3. Best Indicator of air pollution – .....
4. Biological indicator of air pollution – .....
5. Corrected effective temperature of comfort zone .....
6. Instrument :

   i) Kata thermometer: .....

   .....

   .....

   .....

Kata thermometer

Active space

ii) Globe thermometer: ..... iii) Sling psychrometer: .....

....

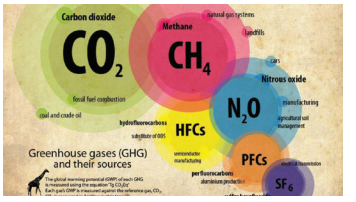
iv) Anemometer : ..... v) Wind wane :

....

7. P4SR (predicted 4 hr. sweat rate)
   - Max upper limit at temperature of comfort zone : 3 L
   - Mc Ardle Index : Max upper limit of P4SR: 4.5 L

8. • PM2.5 – particulate matter 2.5 micron
   Upper limit - mcg/dl^3
   • PM10 (10 micron)
   Upper limit - mcg/dl^3

9. GREEN HOUSE GASES:
   - Global warming
   - Content of Green House Gases
     - Water vapours
     - CO_2



Active space

- Methane
- N_2O
- CFC (chlorofluorocarbon)
  - Hydrofluorocarbon
  - Perfluorocarbon
  - Hexafluoride

- Agreements to control effect of greenhouse gases:

1. Kyoto protocol (2005)
2. Paris agreement (2015) : CoP21

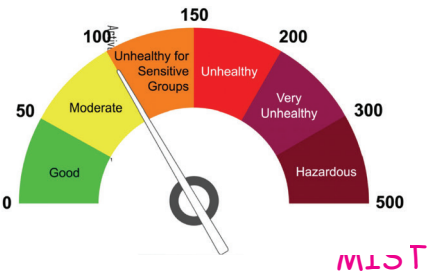
....

Keep rise in global temp <2°C (upto 1.5°C)

3. Dubai (Cop28) 2023
4. BAKU [AZERBAIJAN] (Cop29) 2024
5. BELEM [BRAZIL] (Cop30) 2025

10. SOILING INDEX:
    - Indicator of air pollution
    - Measurement of air pollutant

11. AQI: (AIR QUALITY INDEX) Parameter PM2.5, PM10, CO, SO_2, NO_2, Methane O_z, Lead



Active space

NOISE

- Max upper limit without any problem – .....
- rupture of TM with sudden exposure of sound - .....
- Permanent hearing loss with continuous exposure - .....
  .....
- Sound permitted in ICU/Library – .....

HOUSING/VENTILATION

- Space ..... sq feet/person (.....Cu feet/person)
- Ventilation (door + window) - .....
  .....

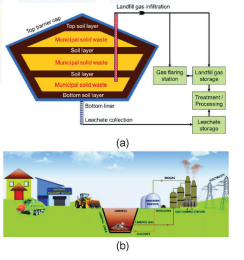
RADIATION

- Max permissible upper limit : ..... rad/person/year
- Use of lead apron (prevention)
- .....

WASTE DISPOSAL

- Refuse – solid domestic waste
- Sullage – liquid waste (without fecal matter)
- Sewage – liquid waste with fecal matter

Refuse disposal:



1. Sanitary land filling (controlled tipping) –
2. Composting
3. Incineration

- Land area required for disposal of refuse generated by 10,000/Yr populotion by sanitary land filling.....
- Method of refuge disposal in small camp/tracker .....
  .....

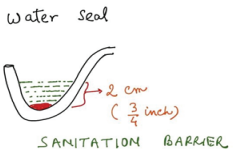
Active space

Sewage:

Composition:

Strength:

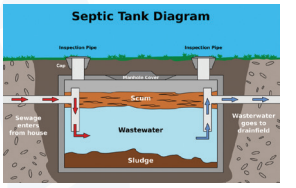
1) BOD (biochemical oxygen demand) : >300 mg/L – strong)
2) Chemical oxygen demand
3) Suspended solid



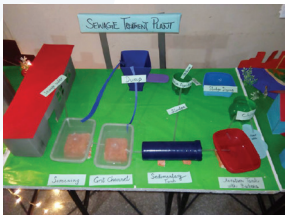
Water seal : The minimum hight should be

Sewage disposal:

1. Septic tank (unsewered area)
   - 24 hr. stay
   - Septic tank – .....
   - Subsoil (outside) – .....



2. Sewage treatment plants (sewage system)
   A. primary treatment:
      (i) Screening – large floating object
      (ii) GRIT chamber : heavy solid –sand
      (iii) Primary sedimentation chamber :
      (5-7days) – light suspended particles



   B. secondary treatment :
      a. Oxidation
         Aeration tank (Heart ofsewage treatment plant)

      By (i) Trickling filter method
      (ii) Activated sludge process

Active space

b. Sec Sedimentation

c. Sludge Digestion

Oxidation Ponds :

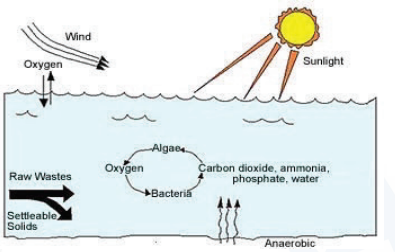
(i) Sewage Lagoons/Stabilization Ponds.

(ii) Treatment of sewage, industrial waste

(iii) Open shallow pool of 3-5 feet deep.

(iv) Day

(v) Night



Light :

Measurement unit

Discription	Quantity Measured	Unit
1. Poit source brightness	Luminous intensity	Candela
2. Flow of light	Luminous flux	Lumen
3. Light amount reaching surface	Illumination	Lux
4. Light re-emitted by surface	Luminance	Lambert

$$\text{Day light factor} = \frac{\text{Illumination Indoor}}{\text{Illumination Outdoor}} \times 100$$

- Living Room DF = 8
- Kitchen DF = 10

Active space

INTERNATIONAL HEALTH

QUARANTINE

Quarantine: Separation of .....

Quarantine period:

YF: ..... days (IP: ..... days)

Corona disease: ..... days (IP: ..... days)

plague 14th century

Validity of YF certificate:

Start: .....

Last: .....

✦ Control of mosquito for Prevention of Yellow Fever in India:

1. Regular insecticidal spray
2. Area free from breeding sites .....
3. Breteau index : <5%
4. Aedes Aegypti index : <1%

☞ Quarantine replaced by: .....

BIOTERRORISM

Category A agents:

- Anthrax: .....
- Plague: .....
- Botulism: .....

Active space

- Small pox: .....
- Ebola
- Tularemia
- Viral hemorrhagic fever

Category B agents:

- Brucellosis
- Q fever
- Cl perfringens
- Epidemic typhus
- Viral meningitis

Category C agents:

- Nipah
- Hunta
- Corona

INTERNATIONAL HEALTH REGULATION

✦ Inform WHO within 24 hr of occurrence

INTERNATIONAL AGENCY

1. WHO

   Head Office : .....

   Started : .....

   WHO day : .....

   WHO Day Theme for 2025 :

   Healthy beginnings, Hopeful futures

WHO Logo

Active space

MIST

2. UNICEF

UNICEF Logo

Head Office : .....

Beneficiaries : .....

Functions :

G : growth monitoring

O : ORS

B : breast feeding

I : immunization

F : female food

F : female education

F : fertility control

3. FAO (food and agriculture organism)

Head Office : .....

Increase production of food to keep pace with ever growing world population to avoid malnutrition

4. ILO (international labour organization) :

Head Office : .....

Improve working & living condition of working population all over world

5. UNDP : United Nation Developmental Program

It is the main source of funds for technical assistance.

UNDP projects cover virtually every economic and social sector - agriculture, industry, education and science, health, social welfare, etc.

6. USAID (united states agency for international development) : malaria

The US has been assisting in a number of projects designed to improve the health of India's people.

Active space

These are : (1) malaria eradication; (2) medical education; (3) nursing education; (4) health education; (5) water supply and sanitation; (6) control of communicable diseases; (7) nutrition; and (8) family planning. The recent trend in assistance from the USA is increasingly in the support of agricultural and family planning programmes, with some reduction in aid in the general public health field

7. DANIDA (Danish International Developmental Agency) : Blindness
8. SIDA : Swedish International Developmental Agency : TB
9. World Bank : Economic Growth
10. UNFPA : United Nation Fund for Population

    Activity : Reproductive health
11. Colombo Plan : Industrial, economic development

    The bulk of Colombo assistance goes into industrial and agricultural development, but some support has also been given to health promotion, mostly through fellowships.

    The contribution of Canada in supplying Cobalt Therapy Units to medical institutions in India was an important item of aid under the Colombo plan
12. Rockefeller Foundation

    Its purpose is to promote the well-being of mankind throughout the world. In its early years, the Foundation was active chiefly in public health and medical education. The advancement of life sciences, the social sciences, the humanities and the agricultural sciences.

    The establishment of the 1. All India Institute of Hygiene and Public Health at Kolkata. 2. Development of medical college libraries;

    3. Assistance to research projects and institutions (e.g., National Institute of Virology at Pune and more recently the setting up of a field demonstration area (Ballabhgarh) in connection with a department of preventive and social medicine, as well as to the All India Institute of Medical Science. 4 Support to the improvement of agriculture, family planning and rural training centres as well as to medical education.

Active space

13. Ford Foundation

Ford Foundation has been active in the development of rural health services and family planning.

The Ford Foundation has helped India in the following projects :

1. Orientation training centres: Training courses in public health for medical and paramedical personnel from all over India.

2. Research cum - action projects: Aimed at solving some of the basic problems in environmental sanitation, e.g., designing and construction of hand-flushed acceptable sanitary latrines in rural areas

3. Pilot project in rural health services : Develop and operate a coordinated type of health service which will provide a useful model for health administrators in the country.

4. Establishment of NIHAE: Establishment of the National Institute of Health Administration and Education at Delhi.

5. Calcutta water supply and drainage scheme :

6. Family planning programme :

14. CARE

Care (Cooperative for assistance and Relief Everywhere) Care provides emergency aid and long. term development assistance. for war victims.

Care-India focused its food support in the ICDS programme and in development of programmes in the areas of health and income supplementation

15. International Red cross.

1. In the beginning, was largely confined to humanitarian service on behalf of the victims of war.
2. Later extended to sufferers of natural disasters also.
3. on the work of the Red Cross was extended to other programmes which would prevent human suffering. These comprise service to armed forces, service to war veterans, disaster service, first aid and nursing, health education and maternity and child welfare services.

Active space

Active space

MIST

Washington (USA)

Stockholm (Sweden)

WORLD BANK

shutterstock.com · 2133398343

Rome (Italy)

Active space

DISASTER MANAGEMENT

Definition: >30 require medical help at a time

☞ Nodal ministry: .....

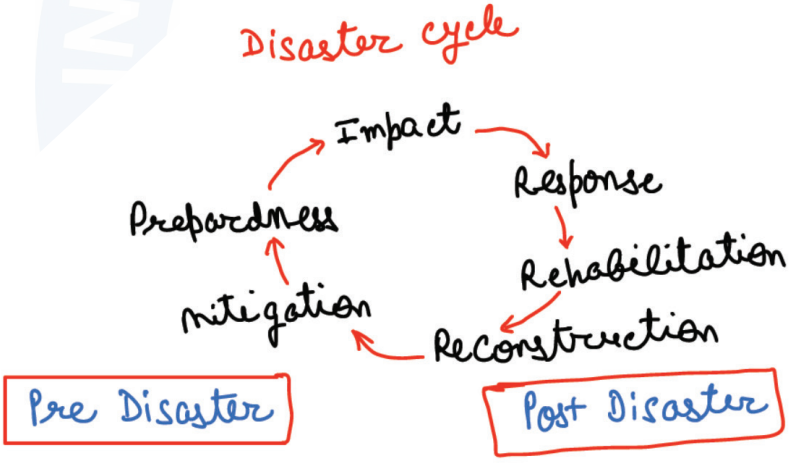
☞ NDMA (national disaster management authority)

chairperson .....

Nodal controlling unit: .....

NDMA Logo

DISASTER CYCLE



Pre Disaster: .....

Post Disaster: .....

TRIAGE AND TAGGING

Segregation of patient based on chances of survival with best possible treatment available Not on the basis of severity of illness

MIST

Active space

4 groups:

1. RED (high priority)

   Lifesaving surgery /resuscitation .....
2. Yellow (medium priority)

   Surgery/resuscitation .....
3. Green (low priority)
4. Black (least priority)

DISEASE IN POST DISASTER PHASE

1. Gastroenteritis
2. Leptospirosis
3. Rickettsial disease
4. Acute RTI

Steps in Post Disaster Management

1st step:

FRC -

ROLE OF VACCINE

- Cholera
- Typhoid
- Hep A
- Measles – .....

- Tetanus

*vitamin deficiency in Disaster .....

Active space

BIOMEDICAL WASTE DISPOSAL

BMW Disposal controlled by
Central pollution control board (Ministry of environment and forest and climate change)

CATEGORIES OF BIO MEDICAL WASTE

1. Yellow
2. Red
3. White
4. Blue

.....

.....

Categories of BMW

YELLOW (NON RECYCLABLE INFECTIOUS WASTE):

1. Human anatomical waste
2. Animal anatomical waste
3. Soiled waste : cotton, gauze piece, bandage
4. Cytotoxic waste (drug, vials / syringe filled with drug)
5. Microbiological waste
6. Chemical waste (solid)
7. Chemical waste (liquid)
8. Discarded bedding/ linen
9. Empty blood bag

.....

RED (INFECTIOUS RECYCLABLE PLASTIC WASTE)

- Gloves
- IV drip set
- IV cannula

Active space

- Syringe
- Foley's catheter
- Uro bag
- Suction catheter

BLUE (PUNCTURE PROOF CONTAINERS)

- Glass item (discarded / broken)
- Metallic body implants

WHITE (PUNCTURE PROOF CONTAINERS) [TRANSLUCENT]

Infectious sharps

- Needle, fixed needle with syringe
- Blade
- Knife
- Scalpel

BLACK (NON INFECTIONS)

- Papper
- Wrappers of surgical item
- Food peel

DISPOSAL OF BIO MEDICAL WASTE

YELLOW :

Procedure .....

Dry oxidation at high temperature

> 1200°C

No pretreatment / pre disinfection

Except: i) Microbiological
ii) Liquid chemical waste

Incinerator

Disposal of liquid chemical waste

- no collection bag required
- disinfection and discard in domestic drain

Active space

MIST

- Advantage:
  - large waste – short time
  - volume – decreased by 80%
- C/I:
  - pressurized gas cylinders
  - PVC contents
  - Heavy metals
  - Radioactive
  - Waste with mercury/ cadmium
- Pre Requisites for Incineration
  - Moisture <30%
  - Combustible waste >60%
- Disadvantages:

  Smoke

Incinerator

Disposal of Red, White, Blue

1. Disinfection /pretreatment :
   - Autoclave
   - Microwave
   - Chemicals : 1% sodium hypochlorite solution
   - Waste saggregation and disifection at point of production
2. Shredding and mutilation
3. Disposal

Red	White	Blue
• Coal tar (road making)	• recycling	• recycling
• Recycling	• land filling	• land filling
• Land filling		

Active space

MIST

4. Secured land filling

➢ To avoid contamination of underground water

Methods of Landfilling

Inertization	Encapsulation
Waste	Put waste in box
Lime	Add immobilizing agent
Cement	Seal from outside
Water	

SCREW FEEDER

- sharp infectious waste
- disinfection by heat (no burning)
- rotatory auger volume decreases by 80%
- wt decreases – 20-35%

Screw Feeder

☞ Disposal of human anatomical waste in rural area with no access to common treatment facility : .....

☞ Disposal of IV drip set and cannula of HIV patient after ART drug therapy : .....

BIOMEDICAL WASTE DISPOSAL IN COVID - 19

GENERAL GUIDELINES:

1. Double layer bags
2. Label with COVID-19 waste
3. Spray with 1% Sodium Hypochlorite ; Trolly, Bin, Container

Active space

OCCUPATIONAL HEALTHS

INTRODUCTION

- Preplacement checkup (..... level prevention)

  Right person gets the right job

- Regular health checkup (..... level prevention)

  Health Checkup At frequent interval

FACTORY ACT (1948)

Space

Safety officer – .....

✦ AGE LIMIT

- <14 year – .....
- 15-18 yr – .....
- >18 yr – .....

✦ WORKING HOURS:

1. Max 48 hr/wk (8-9 hr/day)
2. Max 60 hours/wk (with overtime)

.....

ESI ACT (1948)

❑ **Employee State Insurance Act includes all industries and areas Except:** .....

❑ **ESI corporation committee (ESIC)**

Chairman: .....

Ministry: .....

Active space

MIST

❑ **CONTRIBUTION:**

Employee:

Employer:

Government: state : central

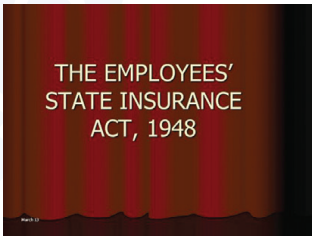
For income Rs < 21,000 /month: facility with contribution

❑ **FACILITIES:**

- Free medical care (worker + family) : IPD + OPD
- Sick leave : .....
- Extended sick leave : .....
- Maternity leave : .....
- Abortion leave
- Pregnancy related complication
- Enhanced leave: vasectomy – .....
- Tubectomy – .....
- Disability benefit :
  - Temporary – leave till recovery (90%)
  - Permanent – pension as per guidelines
- Funeral benefits : 15,000 Rs
- Direct Medical Benefit
  - 1. > 1000 Worker .....
  - 2. < 750 Worker .....
  - 3. Scattered Worker .....
- Indirect Medical Benefit

.....

.....



Active space

OCCUPATIONAL DISEASES

PNEUMOCONIOSIS:

- Dangerous particle : .....
- Size of pneumoconiotic dust : .....
- Most dangerous : .....

SILICOSIS:

- Silica dust [Stone cutting industry, mining]
- 3-5 years IP
- Most common pneumoconiosis
- Follow up for : TB

ASBESTOSIS:

- Asbestos dust (fire proof glass industry, Shipyard industry)
- 6-10 years IP
- Only pneumoconiosis – malignancy

  MC : .....

  Most specific: .....

ANTHRACOSIS:

- Agent ..... [coal industry]
- >10 years IP
- Progressive massive fibrosis even after removal of patient from environment

BAGGASOSIS:

- Agent ..... [Sugarcane, card board industry]
- Thermoactinomycis sacchari
- 2% propionic acid (prevention)
- Non notifiable

Active space

BYSSINOSIS:

- Agent ..... (Textile industry)
- Spinners more affected than weavers

FARMER'S LUNG (Mouldy hay)

- Agent ..... (farmer)
- Mycopolyspora faenni
- 1% mineral oil (prevention)

☞ BRUCELLOSIS : NOT pneumoconiosis
: Occupational disease

CANCERS:

- MC – Occupation Cancer :
- Pneumoconiosis : Causing Cancer
- Agents Causing Cancer

Exposure agent	Cancer
Oil, grease, chemicals	Skin
Asbestos	Lung
Aniline dye	Bladder
PVC	Liver
Benzol preparation	Leukemia

LEAD POISONING:

- Route : 1. Inhalation .....
  2. Ingestion (food/water)
  3. Skin absorption
- Serum lead concentration > for symptoms of poisoning:
- Screening test :

1. Co-proporphyrin level in wine
   - >150 mcg / L of urine suggestive of Lead exposure
2. Amino levulinic Acid in urine
   - > 05mg / L = Lead Absorption
   - > 20mg / L = Excessive
   - > 40mg / L = Dangerous

Active space

- Clinical features:

A. Inorganic compounds (Peripheral)

- ✦ Burtonian lines (blue line on gum)
- ✦ Pallor
- ✦ Basophilic stippling of RBCs
- ✦ Abdominal colic
- ✦ Wrist drop / foot drop

B. Organic compounds (CNS)

- ✦ Insomnia
- ✦ Mental confusion
- ✦ Delirium
- ✦ Convulsion

☞ Treatment: Chelating agents:

1. D-penicillamine
2. Ca-EDTA compounds

NATIONAL PROGRAM RELATED TO OCCUPATIONAL DISEASES

ABHIYAN INDRADHANUSH

- Change of bedsheet in ESI hospital everyday as per – VIBGYOR

Rajiv Gandhi Shramik Kalyan Yojna

- ▪ Unemployment allowance (50%)
- ▪ Max up to 12 months
- ▪ Accidental emergency closure of unit

Active space

SOCIAL AND MENTAL HEALTH

SOCIAL HEALTH

BPL CRITERIA

	RURAL	URBAN
Calorie affordability/person	.....kcal /day	..... kcal/day
Income/day/person (Rang rajan committee, 2014)	.....Rs/day	..... Rs/day

☞ TOTAL BPL FAMILY = ...

SCALE OF SOCIOECONOMIC STATUS CLASSIFICATION

MODIFIED KUPPUSWAMY SCALE :

Criteria:

(i) .....

(ii) .....

(iii) .....

Total score = 29

Class		Score
Upper class :		.....
Middle :	UPPER	.....
	LOWER	.....
Lower :	UPPER	.....
	LOWER	.....

Active space

TYPES OF FAMILIES

1. NUCLEAR :
   - Couple with children
2. New family - <10 yr marriage

3. JOINT FAMILY :
   - >1 couple with children live together
   - Common purse
   - Common kitchen
4. Three generation family

   Couple with parents, children

FAMILY CYCLE

1. Marriage
2. Birth of 1st child
3. Birth of last child
4. First child leave home
5. Last child leave home
6. Death of spouse
7. Death of survives

MISCELLANEOUS POINTS

1. ACCULTERATION:
   - Mixing of culture (culture contact)

     Ex.: Marriage, education, trade
2. TYPES OF LEARNING
   - Cognitive [Knowledege]
   - Affective [Attitude]
   - Psychomotor [Skill]

Active space

3. STATE MEDICINE:
   - Free medical services to community provided by state government
   - Controlled by government

4. SOCIALIZED MEDICINE:
   - Free medical services to community by state government
   - Regulated by professionals (not government)

5 HABIT:
   - Acquired through repetition
   - Way of doing things

6. CULTURE:
   - Socially acquired learned behavior

7. ATTITUDE:
   - Acquired through learning
   - Permanent way of behaving

8. CROWD
   - Group of people coming together temporarily for common purpose, unstable, No leader individuals

9. MOB
   - A crowd with leader who forces members. for action more emotional than crowd, unstable.

10. HERD
   - A crowd with leader, follow orders of leader without question.

     Ex : Tourist guide

11. OPINION
   - Views held by people on a point of Dispute temporary, provisional.

12. BELIEF
   - Views which are permanent, stable, unchangeable

Active space

13. SOCIAL PATHOLOGY

- Relationship between Disease and social condition

14. SOCIETY

- Group of individuals who have organized themselves and follows given way of life.

15. COMMUNITY

- A social group determined by geographical boundaries and / or common values and interest.

16. SOCIOLOGY

- Study of individual as well as groups in a society.

17. SOCIALIZATION

- Process by which on individual gradually acquires culture and become member of a social group.

18. SOCIALISM

- Production and distribution of resources of Country for public welfare.

Consumer Protection Act

a) CPA: 1986

b) Decision in 3-6 month

c) No court fees

d) Person can plead his own case

e) Compensation

- District : up to 20 Lacs Rs
- State : 20 Lac 1 crore
- National : >1 crore Rs

Active space

Economics:

⇨ Gross National Income (GNI) / Gross National Product (GNP)

Gross income generated within country including net income received from abroad.

⇨ GDP (Gross Domestic Product)

Income generated within country excluding net income received from abroad.

⇨ NNP (Net National Product)

GNP minus capital we consume

⇨ NDP (Net Domestic Product)

GDP minus value of depreciation on fixed assets.

Health expenditure in India :

- ❑ Financial year 2023 : 2.1% of GDP
- ❑ As per NHP 2017 : increase up to 2.5% of GDP (2025)
- ❑ Gool of INDIA : up to 3% of GDP
- ❑ WHO recommendation : 5% of GDP

Active space

HEALTH EDUCATION & COMMUNICATION

COMMUNICATION CYCLE

Source (Sender)
↓
Message
↓
Channel
↓
Receiver -

Feedback

Awareness - Interest - Evaluation - Adoption

TYPES OF COMMUNICATION

ONE WAY	TWO WAY .....
Lecture	Group discussion
TV	Demonstration
Radio	Panel discussion
Newspaper	Clinical symposium
Flip charts	Workshop
Flannel graph	Conference

METHODS OF COMMUNICATION

1. Individual : .....
2. Group : .....
3. Mass : .....

Active space

DIADACTIC METHODS:

FLANNEL GRAPH

LECTURE

FLIP CHARTS:

Charts raised in between talk

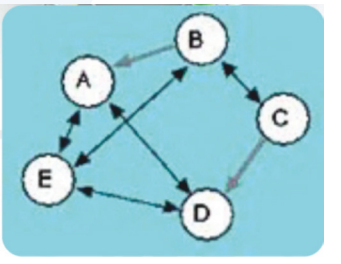
Flannel Graph:

Series of photograph pasted on a piece of cloth in chronological order.

SOCRATIC METHODS

❑ FOCAL GROUP DISCUSSION:

- 6-12 member
- Best method to .....
  .....
  .....
- Sociogram : graphical representation of interaction



- Type of question in FGD
  1. Open ended : Long answer statement
  2. Closed ended : Answer as Yes / No
  3. Double barreled : Question touches more than one issue but allow only one answer
  4. Structured
     i. Type of closed question
     ii. Used in survey for fast & precise answers
     iii. Reducing amount of thinking by participant

Active space

❑ DEMONSTRATION

- Learning by doing
- Ex- ORS preparation, blood grouping, BP monitoring

❑ Pannel Discussion and Clinical Symptosium

PANNEL DISCUSSION	CLINICAL SYMPTOSIUM
4-8 panelist	4-8 speakers
Discuss among themselves	Deliver talk
No prepare talk	Prepared talk
Move in any direction	Fixed order
Audience interaction Simultaneously or end	At end of talk

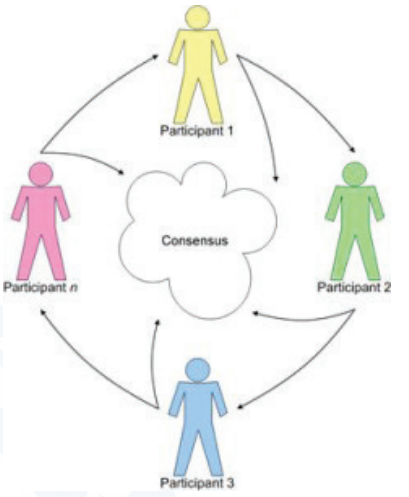
PANNEL DISCUSSION

CLINICAL SYMPOSIUM

Active space

❑ **Delphi Method:**

- Discussion among geographically dispersed experts to make consensus



❑ **Counselling:**

1. Individual :
2. Couple :
3. Family – .....

.....

❑ **Barriers of Communication :**

1. Physiological : Difficulty in hearing, expression.
2. Psychological : emotional Disturbances, neurosis, level of intelligence,
3. Environmental : Noise, congestion, invisibility.
4. Cultural : Illiteracy, custom, belief, religion attitude, difference in social class, level of knowledge and understanding language variation, cultural difficulties

❑ **Health Education Approach :**

1. Regulatory approach :
   - Law/punishment
2. Service approach :
   - Free Services at door Step
3. Health education approach :
   - Slow long lasting
   - Start from younger age

Active space

4. Primary Health care approach :

   - Community participation

   - Inter sectorial coordination

Education	Propaganda or publicity
1. Knowledge and skills actively acquired.	Knowledge instilled in the minds of people.
2. Makes people think for themselves.	Prevents or discourages thinking by ready-made slogans.
3. Disciplines primitive desires.	Arouses and stimulates primitive desires.
4. Develops reflective behaviour. Trains people to use judgement before acting.	Develops reflexive behaviour; aims at impulsive actions.
5. Appeals to reason.	Appeals to emotion.
6. Develops individuality, personality and self-expression.	Develops a standard pattern of attitudes and behaviours according to the mould used.
7. Knowledge acquired through self-reliant activity.	Knowledge is spoon-fed and passively received.
8. The process is behaviour centred - aims at developing favourable attitudes, habits and skills.	The process is information centred - no change of attitude or behaviour designed.

Active space

HEALTH PLANNING AND MANAGEMENT

GOAL, TARGET AND OBJECTIVE

- **Objective :**
  - Planned end point of all activity
  - Concern with problem
- **Target :**
  - Discrete activity with help in measurement of achievements of objective
  - Concern with factors involve in problem
- **Goal :**
  - ultimate end point of all activity
  - No time limit
  - May/may not achieve

STEPS OF PLANNING CYCLE

1. Analysis of Health Situation
2. Establishment of goals and of objective
3. Assessment of resources
4. Fixing priorities
5. Write up formulated plan
6. Programming and implementation
7. Monitoring
8. Evaluation

Active space

HEALTH COMMITTEES

1. BHORE Committee (1946)
   - Social physician
   - PHC concept
   - School health
   - 3 million plan
   - Known as health survey and development committee
2. BALWANT RAI MEHTA committee (1957)
   - Panchayat Raj Institution
3. RENUKA ROY Committee (1961)
   - Functioning of school health services
   - School meal
4. Mudaliar Committee (1962)
   - All India health services
   - Known as health survey and planning committee
5. CHADDAH Committee (1963)
   - Basic health worker – malaria control, F P
6. Mukherjee Committee (1965)
   - Delink of malaria control and F P
7. Jungalwala Committee (1967)
   - No private practice
   - Equal pay for equal work
   - Integration of health services
8. Kartar Singh Committee : (1973)
   - MPW posted at subcenter

     ANM: replaced by MPW – F

     Basic HW: replaced by MPW – M
   - HA M/F at PHC

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MIST

9. Shrivastava Committee (1975)
   - 'ROME' concept

     (Reorientation of Medical Education)
   - Known as group on medical education and support manpower
   - VHG concept
   - Referral services
   - Band of semiprofessional and para professional between community and doctor (school teacher, gram sevak)
   - Concept of 2 tier of health workers between community and doctor

10. Krishna Committee (1983)
    - Concept of urban health centers

11. Bajaj Committee (1986)
    - Concept of UGC
    - National medical and health education policy

12. High level expert group Committee (HLEGC) : 2011-2012
    - Universal health coverage
    - 3 yr Bsc (rural health)
    - 4 yr BRHC (bachelor in rural health care)

HEALTH MANAGEMENT TECHNIQUES

1. Quantitative techniques :

   A. cost – benefit analysis:

   B. cost effective analysis:

   C. system analysis:

Active space

E. network analysis:

Graphical plan of all events leading to end point

(i) PERT (program evaluation and review technique)

(ii) CPM (critical path method):

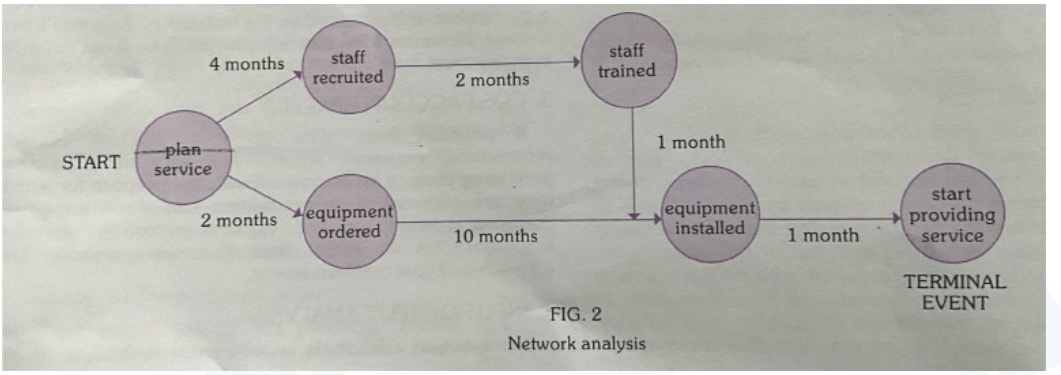


FIG. 2
Network analysis

F. Work sampling

- On doctors, nurses, pharmacist
- Analysis of time spent, amount of work done
- Deciding manpower required there

NITI AYOG

- National institution for transforming India
- Replaced planning commission
- 1st meeting – .....
- Chairman – .....
- Think tank – CM ; governor, experts
- Functioning from April 2017
- Short goal 3 yrs. : .....
- Long 7 yrs. : .....

Active space

MDG – MILLENNIUM DEVELOPMENTAL GOALS

- United nation
- 2000-2015
- Total – .....
- Goal related to health : goal no 4, 5, 6

SDG – SUSTAINABLE DEVELOPMENTAL GOAL

- 2016-2030
- United Nation
- Total goal – .....
  169 target
  304 indicators
- Health related : goal .....

TARGETS OF SDG:

❑ **By 2030,**

- Global MMR- <70/1 LAC LB
- NNMR – 12/1000 LB
- U5 MR – 25/1000 LB
- END Epidemic of HIV, TB, Malaria And Neglected Tropical Disease
- Universal Health Coverage

NATIONAL HEALTH POLICY (2017)

❑ **Goals of National Health Policy**

➢ Elimination of disease:

- By 2025 : TB
- By 2026 : Measles, Rubella
- By 2027 : Filariasis, Leprosy free India
- By 2027 : Leishmaniasis
- By 2030: Malaria

Active space

- LIFE EXPECTANCY – 70(2025)
- TFR – 2.1 (2025)
- U5MR – 23 (2025)
- MMR – 100 (2020)
- IMR – 28 (2019)
- NNMR – 16 (2025)
- SB – SINGLE DIGIT (2025)
- BLINDNESS PREVALENCE – 0.25% (2025)
- DECREASE PREMATURE MORTALITY FROM CVD, DM, CANCER BY 25% (2025)
- INCREASE UTILIZATION OF PUBLIC SECTOR FACILITY BY 50% (2025)
- INCREASE HEALTH EXPENDITURE TO 2.5% OF GDP (2025)
- SUSTAIN ANC COVERAGE AND SKILL BIRTH ATTENDANT >90% (2025)
- FULL VACCINATION BY 1 YEAR >90% (2025)
- ACCESS TO SAFE WATER SUPPLY TO ALL (2020)

Active space

DISEASES

BASIC DEFINITION IN DISEASES

- **Primary Case** – .....
- **Secondary Case** – .....
- **Index Case** – .....
- **Definitive host** – .....
- **Intermediate host** – .....

	Definitive host	Intermediate host
Malaria	.....	.....
Filariasis	.....	.....
Echinococcosis	.....	.....

- **Entry of organism:** → first signs and symptoms of disease .....
  → Max infectivity of disease .....
  → 50% of cases .....

Primary case: → Secondary case .....

- **Quarantine**: Seperation of Healthy exposed till max I.P.

- **Period of communicability:**
  Period during which disease can be transferred from one to another

- **Isolation**: Seperation of DS person till poc

Active space

❑ Biological Cycle:

1. Propagative : .....
   - Aedes mosquito for dengue virus
   - Rat flea for plague bacilli
2. Cyclodevelopmental : .....
   - Microfilaria in culex
3. Cyclopropagative : .....
   - Malaria parasite in anopheles

❑ CFR (Case fatality rate)

$$\text{CFR: } \frac{\text{Total deatls due to specific Ds}}{\text{Total cases of that Ds}} \times 100$$

✦ Importance:

1. Severity of disease
2. Killing power of disease
3. Virulence of organism

✦ Not about – infectivity of disease

❑ SAR (secondary attack rate)

$$\text{SAR} = \frac{\text{Sec cases in single IP}}{\text{Total succeptible}} \times 100$$

✦ Always exclude:

1. Primary case
2. Vaccinated children

✦ Significance-

.....

- ✦ Total children
- ✦ vaccinated children
- ✦ Succeptible
- ✦ 1st case of measle
- ✦ Succeptible
- ✦ Sec cases of measle

Active space

GENERAL CONCEPTS OF DISEASES (IP, POC, AGENT)

DISEASE	IP	POC	CAUSATIVE AGENT
Measles	10-14 days	4 days ← (R) → 5 days	RNA paramyxovirus
Chicken pox	14-16 days	2 days ← (R) → 5 days	Human herpes virus 3
Rubella	14-21 days	1 week ← (R) → 1 week	RNA Togavirus
Influenza	18 hr - 72 hr	1-2 days ← symp → 1-2 days	Orthomyxo-virus
Diphtheria	2-6 days	Symptoms to 14-28 days	C. Diphtheriae
Pertussis	7-14 days	1 week after exposure to 3 weeks after symptoms	B. pertussis
Meningococcal Meningitis	3-4 days	1-2 days ← symp → 1-2 days	N meningitidis
Typhoid	10-14 days	From 1 week after symptom till in stool	S typhi
Hep A	15-45 days	2 wks ← jaundice → 1 week	Enterovirus 72 (picornavirus)
Hep B	45-180 days	Infection → HbsAg in blood	Hepadna virus
Hep C	30-120 days	Infection → Hep C virus in blood	Flavivirus

SOURSE AND RESERVOIR

❑ **Source:**

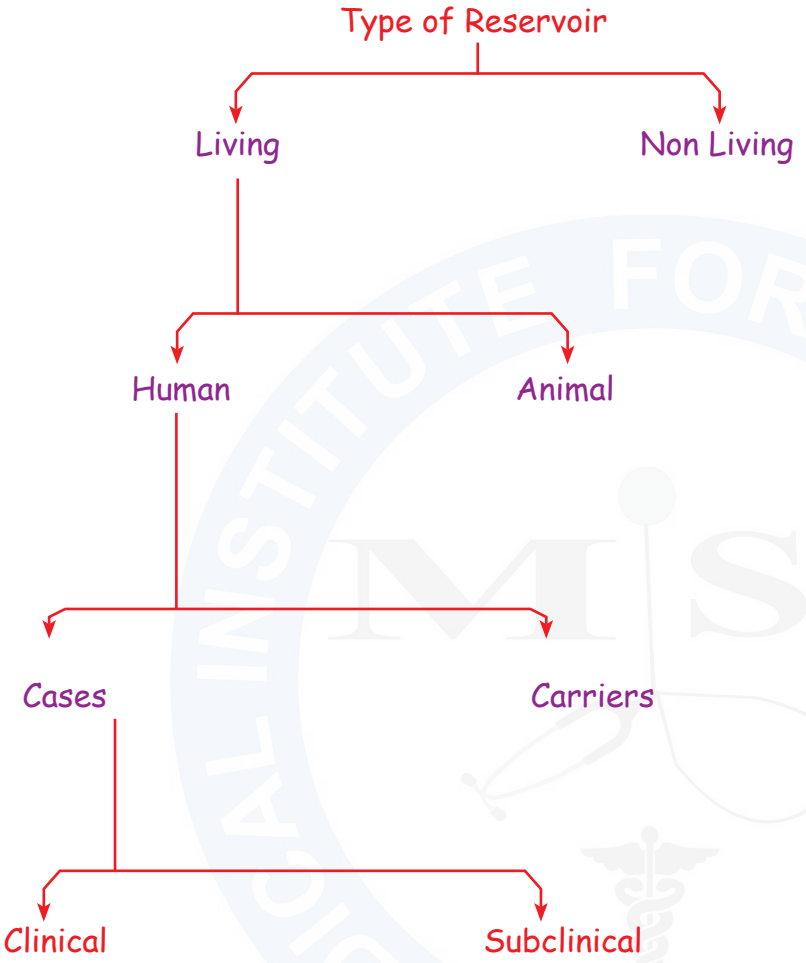
Channel of disease transmission

MIST

Active space

❑ Reservoir:

Where organism persist for survival



❑ Carriers:

Carry infection for others:

1. Healthy carriers : organism + nt, never seen infection (typhoid, cholera)
2. Convalescent carrier : recovery (Typhoid, Cholera)
3. Chronic carriers : > 1 years – Typhoid, Cholera, HIV, Hep B, Hep C
4. Incubatory carrier

Active space

- **Obligate Host :**
  - Only host for Disease
  - Typhoid – human
  - Tetanus: human
- **Opportunistic infection :**
  - Infection due to common organism in presence of low immunity
  - HIV
- **Nosocomial Infection:**
  - Hospital acquired
  - Absent on admission
  - Appears after 48 hr of admission
  - Eg : UTI
- **Iatrogenic Infection:**

  Artificially introduced
  - Post op infection (due to inadequate sterilization)

Active space

VECTOR BORNE DISEASE

❑ NVBDCP:

National Vector Borne Disease control program

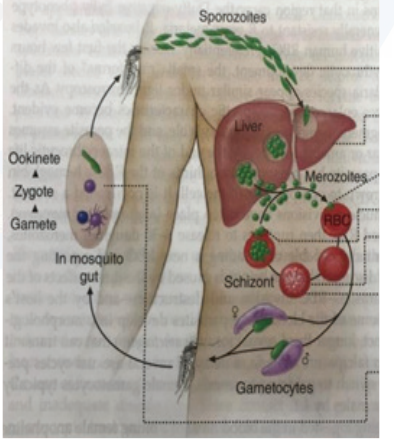
- Malaria
- Filariasis
- Lesihmaniasis .....
- JE
- Dengue
- Chikungunya

राष्ट्रीय वैक्टर जनित रोग नियंत्रण कार्यक्रम
NVBDCP

.....

MALARIA

Agent: Plasmodium
Vector: Anopheles –
Biological cycle: .....
Definitive host: .....
Intermediate host: .....
Incubation period:
Extrinsic: 10-20 days
Intrinsic: 12-14 days



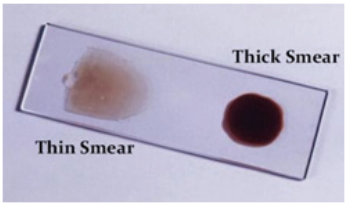
Malaria Cycle

Form of malaria parasite:
A. mosquito bite: Sporozoite –
B. blood transfusion: Trophozoite –

Active space

DIAGNOSIS of malaria

❑ 1. PBS examination



Peripheral Blood Smear

❑ 2. RDT kit

- When report of PBS – .....
- ANTIGEN DETECTED
  - HRP – II (Histidine rich protein)
  - LDH (lactate dehydrogenase)

RDT KIT

TREATMENT:

A General Population

❑ PI VIVEX :

✦ Chloroquine .....

✦ Primaquin : .....

❑ PI Falciparum :

NE states	Other states
.....	.....
Artemether – lumifentrin 3 days	Artesumate – 3 days
	Sulphadoxine Pyremethamin - single dose
Primaquin –	Primaquin – .....

Active space

B. Malaria with Pregnancy:

- **Pl. Vivax :**

- **Pl. falciparum:**
  1. 1st trimester :

     .....

  2. 2nd/3rd trimester

     .....

     Primaquine C/I

PROPHYLAXIS:

1. ..... – doxycycline (100 mg daily) start 1-2 day before
2. .....

   Chloroquine 300 mg/wk (start1 week before)

   Mefloquine 250 mg/wk (start2 wk before)
3. Pregnancy :

   Continue for after leaving area.

INDICATORS:

1. API (Annual parasite incidence)

2. Infant parasite rate

   .....

   .....
3. Annual blood examination rate (ABER)

Active space

4. Slide positivity rate

5. Spleen rate:

   % of individual showing splenomegaly (epidemiological survey)

INFRASTRUCTURE:

1. DDC (drug distribution centers)

   Village (1/1000)

   Distribution of drugs to confirm cases

2. Fever treatment depots :
   - village (1/1000)
   - Slide preparation
   - Treatment of confirmed cases

NATIONAL STRATEGIC PLAN (2023-2027)

Goal : Interrupt local transmission and achieve zero indigenous case through out country by 2027 and prevent restablishment of malaria

Strategy : 1) Malaria surveillance as core intervention
2) Test, Treat, Track
3) Vector control
4) Elimination and attainment of malaria free status
5) Research Promotion

Active space

NATIONAL FRAMEWORK FOR MALARIA ELIMINATION (2016 - 2030)

Goal - Eliminate malaria (zero indigenous case by 2027) through out country by 2030

- Maintain malaria free status and prevent reintroduction malaria

Integrated vector management

(i) Rural : IRS (adult mosquito) Indoor residual spray → Melathion

(ii) Urban : anti larval measures → Pyrethroid

(iii) API ≥ 5 : LLIN

(iv) API 2-5 : IRS + ITBN

URBAN MALARIA SCHEME:

- 1971
- Area : 1) > 50,000
  2) SPR >5%
- Control:
  1. Prevent mosquito breeding
  2. Larvicidal fish
  3. Chemical (larvicidal)

Malaria Vaccine:

- Mosquirix (RTS, S, ASO_1) ; R-21 / MATRIX - M
- WHO approved
- Recombinant vaccine, protein bas
- MOA : block liver infiltration by parasite
- Prevent only pf infection (No known cross protection with other Plasmodium species)

Active space

EXTRA EDGE

Malaria :

1. Pl. Vivex infection treatment

   - In all states → DOC : CQ (3 day) with PQ (14 day)

     ↓ ↓

     25mg/Kg in 3 divided dose 0.25mg/kg/day x 14 day

   - In pregnancy (all trimester, all state) : only CQ (PQ G/I)

Age-wise dosage schedule for treatment of *P.vivax* cases

Age (years)	Tablet Chloroquine (150 mg base)			Tablet Primaquine* (2.5 mg base)
	Day 1	Day 2	Day 3	Day 1 to 14
<1	1/2	1/2	1/4	0
1 – 4	1	1	1/2	1
5 – 8	2	2	1	2
9 – 14	3	3	1 1/2	4
15 and above	4	4	2	6

* Primaquine is contraindicated in infants, pregnant women and individuals with G6PD deficiency. 14 day regimen of Primaquine should be given under supervision.

2. Pl falciparum in North east states

   - ACT AL (Coloured blister pack as per Age) x 3d
   - Singledose Primaquin 0.75mg/kg D2

Co-formulated tablet ACT-AL	5–14 kg (> 5 months to < 3 years)	15–24 kg (≥ 3 to 8 years)	25–34 kg (≥ 9 to14 years)	> 34 kg (> 14 years)
Total Dose of ACT-AL	20 mg/ 120 mg twice daily for 3 days	40 mg /240 mg twice daily for 3 days	60 mg /360 mg twice daily for 3 days	80 mg /480 mg twice daily for 3 days
	Pack size			
No. of tablets in the Packing	6	12	18	24
Give	1 Tablet twice daily for 3 days	2 Tablets twice daily for 3 days	3 Tablets Twice daily for 3 days	4 Tablets Twice daily for 3 days
Colour of the pack	Yellow	Green	Red	White

* ACT-AL is not to be prescribed for children weighting less than 5 kg.

3. Pl falciparum in states other than NE

   - ACT SP (Coloured blister pack as per Age)
   - Artesunate daily 3days
   - Sulfadoxine pyrimethamine singledose D1
   - Primaquine 0.75mg/kg single dose D2

Age Group (Years)	1st day		2nd day		3rd day
	AS	SP	AS	PQ	AS
0-1* Pink Blister	1 (25 mg)	1 (250 +12.5 mg)	1 (25 mg)	Nil	1 (25 mg)
1-4 Yellow Blister	1 (50 mg)	1 (500+25 mg each)	1 (50 mg)	1 (7.5 mg base)	1 (50 mg)
5-8 Green Blister	1 (100 mg)	1 (750+37.5 mg each)	1 (100 mg)	2 (7.5 mg base each)	1 (100 mg)
9-14 Red Blister	1 (150 mg)	2 (500+25 mg each)	1 (150mg)	4 (7.5 mg base each)	1 (150 mg)
15 & Above White Blister	1 (200 mg)	2 (750+37.5 mg each)	1 (200 mg)	6 (7.5 mg base each)	1 (200 mg)

Active space

MIST

4. Pf in Pregnancy

- 1st trimester (all state) : Quinine
- 2nd / 3rd trimester - NE → ACT - AL
  - Other state → ACT - SP
- PQ C/I

CLASSIFICATION OF DISTRICT :

1. Zero case / 1000 popⁿ
2. API : > 0 to <1/1000
3. API : 1 to <2
4. API : 2 to <5
3. API : > 5

Surveillance :

1. Active case detection by PBS by MPW-M
2. Diagnostic services by RDT by ASHA and volunteer workers and prompt treatment at community level
3. Mass screening and treatment (MSAT) campaign in high burden district of Tripura, Mizoram and chattisgarh
4. Decentralization of Lab to PHC for testing of PBS of malaria
5. National API : 0.16/1000 (2023)
6. 34 states/UT has API < 1 except Tripura, Mizoram

Types of malaria active surveillence

A. PROACTIVE → Case detection in high Risk groups not prompted by malaria case detection in area

B. REACTIVE → Case detection in a population which is potentially linked with confirmed case / cluster of cases

Active space

FILARIASIS

Agent: W. Bancrofti (MC)
B Malayi

Vector: MC (culex) (Cu. quincifaciatus)
Mansonian

Can be caused by all types of mosquito

Biological cycle: .....

Definitive host: .....

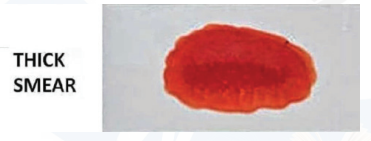
Intermediate host: .....

IP: 8-16 months

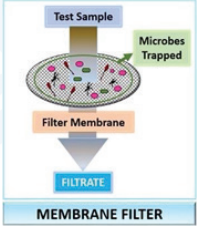
Most dangerous stage: Asymptomatic microfilaria

Occult filariasis: (tropical pulmonary eosinophilia)
No mf in blood
Due to Allergic manifestation

DIAGNOSIS:



Thick Smear



- PBS:

- Membrane Filtration Test :

Active space

Treatment:

- IDA therapy

  1: Ivermectin,

  2: DEC,

  3: albendazole

- Treatment for lymphodema:
  - Adequate lymphatic drainage of part
  - Avoid secondary bacterial infection
  - Surgical correction – hydrocele

LYMPHATIC FILARIASIS ELIMINATION PROGRAM :

Goal : Filariasis free India by 2027

Criteria :

1 Number ofmicrofilaria carrier <1%

2 Children born after initiation of elimination of lymphatic filariasis program 2002 are free from circulating antigeremia (presence of adult filaria worm in body)

Strategy :

1. MDA (mass drug administration)

a) Mapping : microfileremia >1% -> endemic, start MDA

b) MDA : IDA therapy → Ivermectin 150-200by / kg

→ DEC 6mg/kg

→ Albendazole 400mg

C/I : Pregnancy, <2yr, seriously will pt

twice/yr 10th Feb/10thAug >80% Population for 5-6 yr

Active space

c) TAS : (Transmission assessments survey)

- After 5-6 yr of MDA in children (6-7 yr)
- If microfileremia <1% → stop MDA

2. Home based Mx of foot care, lymphoedema cases
3. DEC medicated salt : 1-4gm of DEC/kg salt, 6-9 month
4. Vector control : outdoor fogging
5. High level advocacy (Inter Sectorial Convergence)
6. Innovative approach (Alternate Diagnostic)

Filaria Survey : 1) Routine → 5-7% population

2) Evaluation studies → 20% population

DENGUE FEVER

❑ **Agent:**

.....

❑ **Strains:** Dengue V1, V2, V3, V4

1) No cross immunity
2) Secondary infection with new strain -- (DHF/DSS)

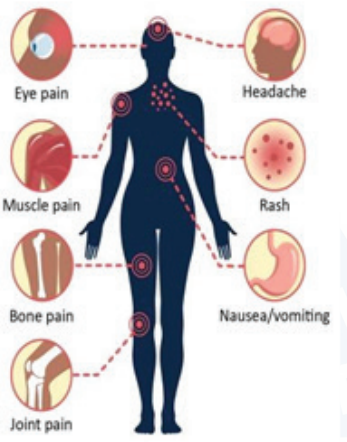
Aedes

❑ **Vector:** .....

Active space

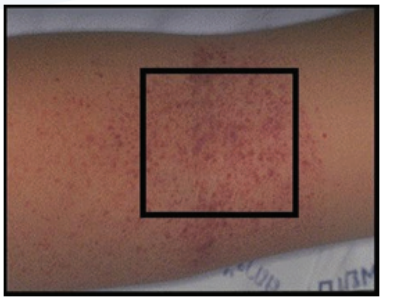
- **IP:** 3-14 days (5-6 days)

- **C/F:**
  1. Undifferentiated fever
  2. Classical dengue fever
  3. Dengue hemorrhagic fever (DHF)
  4. Dengue shock syndrome :



- **Diagnosis:**
  1. Viral isolation : RT PCR
     NAAT
  2. Viral antigen – NS1
  3. Mac ELISA – IgM

     4 fold increase in IgG
  4. Platelet count
     <1.5 lac/mm^3 (DF)
     <1.0 lac/mm^3 (DHF)
  5. Hematocrit value :
     Increase by : 5-10% (DF)
     >20% (DHF)
  6. Tourniquet test
     >10/sq inch Petechia - DF
     >20/sq inch Petechia - DHF
  7. X ray: B/L pleural effusion
  8. USG abdomen: GB edema, Ascites



Tourniquet Test

Active space

MIST

❑ Treatment:

1. Supportive
2. IV fluid
3. Platelet transfusion : <10,000/mm^3 Platlet
   Abnormal coagulopathy
   Abnormal bleeding
4. RBC transfusion
   (a) Loss of >10% of BP volume
   (b) Refractory shock

❑ Prevention:

1. Avoid mosquito breeding .....
2. Mosquito control
3. Vaccine :

a) CYD - TDV Vaccine (Denga Vaxia)

- Live attenuated
- Quadrivalent
- Normal saline diluent 0.9%
- Strain- CYD – TDV
- Route - S/C
- Age - 9-45 yr
- 3 doses at interval of 6 months
- Effective in seropositive individual (Had past dengue infection)

b) TAK - 003 Vaccine (Qdenga)

- Live attenuated
- Quadrivalent
- Diluent : 0.22% Saine
- Route : S/C
- Age : 2-59 yr (Dependon Country)
- 2 doses at interval of 3 months
- Effective in both seronegative and seropositive individual

Active space

MIST

4. Wolbachia method :

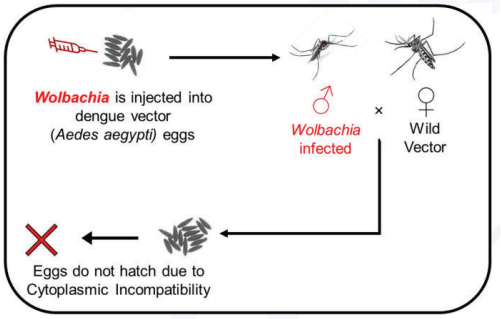
✦ Inject bacteria in mosquito population

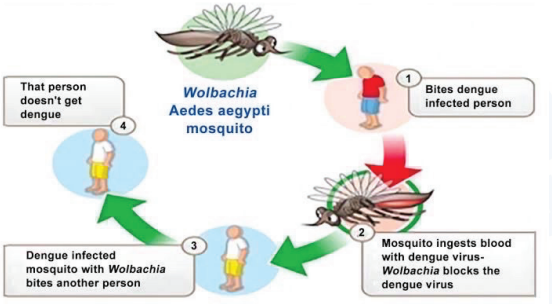
A) Egg injection

B) Larva injection

C) Male mosquito will mate female mosquito, eggs will not hatch

✦ Block virus transmission in adult





❑ Global strategy for dengue prevention and control (2012-2020)

1. Decrease mortality by 50% (2020)
2. Decrease morbidity by 25% (2020)
3. Estimate of true burden of disease (2015)

   Base 2010.

CHANDIPURA VIRUS ENCEPHALITIS

- Member of vesiculovirus genus of family Rhabdo viridae

Vector : Female sand fly

C/F : Rapid onset of fever, vomiting, altered sensorium, diarrhoea, neurological deficit, meningeal irritation, convulsion

Age : Children <15 yr

Rx : Supportive Rx

Prevention : 1. Control of sandfly breeding

2. Bite protection

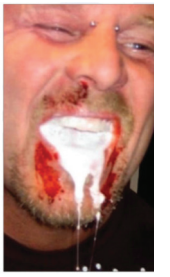
MIST

Active space

ZOONOTIC DISEASE

RABIES

- Agent: .....
  .....

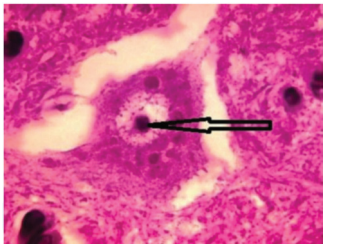


Rabid Animal

Reservoir: .....

- Source: .....
- IP: 3-8 week
- SAR: .....
- CFR: .....
- Role of Herd Immunity: .....
- Rabies free country

- Rabies free states in India
- Diagnosis:
  1. Negri Body
     - After death
     - Intra cytoplasmic eosinophilic inclusion body
     - 1st Site: AMMON's horn of hippocampus
     - 2nd Site: Cerebellum



Negri Body

Active space

2. Alive person (animal)

   Skin biopsy

   Viral antigen by fluorescent antibody test (FRA)

- Grades of Dog Bite:

I: .....

II: .....

III: .....

.....

.....

Pt with immuno compromised ds .....

- Management:

1. GRADE 1 = ...

   .....

2. GRADE 2 = ...

3. GRADE 3 = ...

Rabid Bite

- Local Care:
  1. Thorough cleaning of wound with running water, soap ≥ 15 minute with local Povidone Iodine
  2. No suture –
  3. Local Ig

     Human:

     Animal:

- Vaccine:

  Types:

  1. Chick embryo
  2. Duck embryo
  3. Human diploid cell vaccine
  4. Recombinant vaccine

☞ Nervous tissue vaccine NOT USED NOW

Active space

Schedule:

- ✦ Pre exposure prophylaxis:
  - .....
  - I/M
  - Goal : Antibodies >0.5 IU/ml
  - Repeat booster every 2 yr
- ✦ Post exposure prophylaxis:

A. Vaccinated :

1. within 3 months : .....
2. 3 mth - 2 Yr : .....

I/M: .....

I/D: .....

3. beyond - 2 Yr : .....

B. Non vaccinated

- Essen regiem :

  .....

  .....

- Thai red cross

  .....

  .....

- Zareb regiem :

  .....

  .....

JAPANESE ENCEPHALITIS

Agent :

Vector – Culex (Cu. Tritaneorrhyncus)

BREEDING SITES:

FLY RANGE :

Man: .....

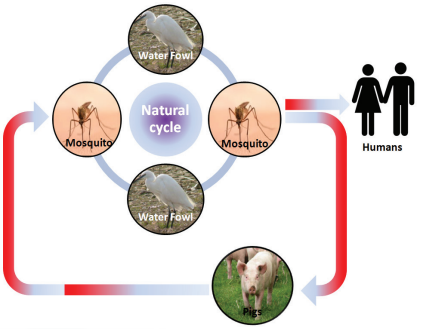
MIST

Active space

Pig: .....
Cattles: .....
Treatment: supportive

❑ **PREVENTION:**

1. Vaccine :
2. Control of mosquito :
   Outdoor fogging
3. Personal prophylaxis
   - Full cloth
   - Mosquito net
   - Mosquito repellant



JE Cycle

RICKETTSIAL DISEASE

	DISEASE	AGENT	VECTOR
A.	Typhus group :		
	Epidemic typhus	R prowazekki	.....
	Endemic typhus	R typhi	.....
	Scrub typhus ESCHAR (wound covered with black scab)	R tsutsugamushi	
B.	Spotted fever		
	Indian tick typhus	R coroni	.....
	RMSF	R rickettsi	.....
	R pox	R akari	.....
C.	Others		
	Q fever : no skin lesion	C. Burnetti	....
	Trench fever	B. Quintana	.....

Active space

MIST

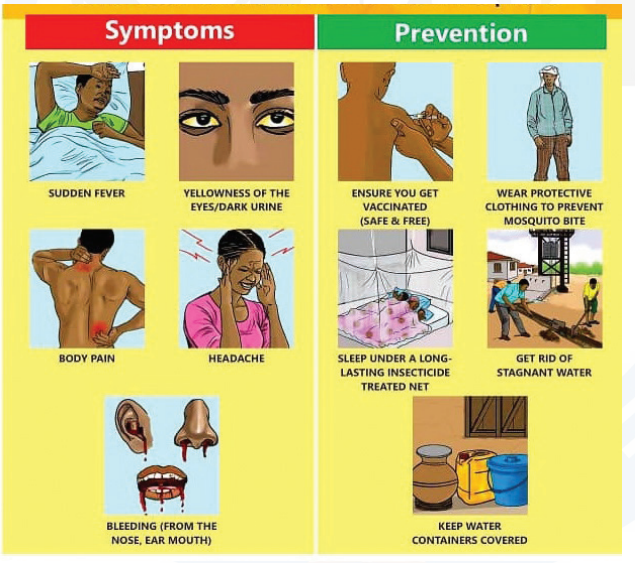
Treatment: DOC: .....

Prophylaxis: .....

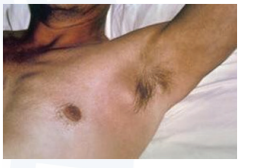
Eschar

R Pox

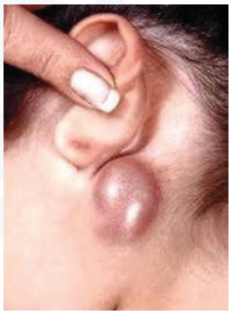
PLAGUE, YELLOW FEVER



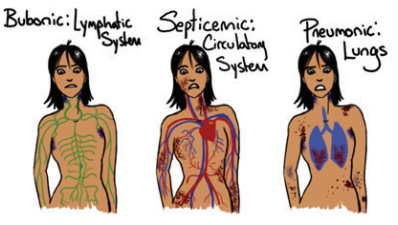
YF



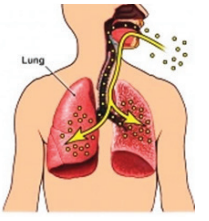
Lymph Nodes



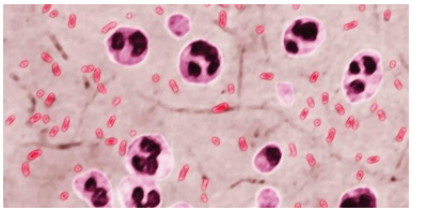
Lymph Node



Types of Plague



Pneumonic Plague



Plague Bacilli

MIST

Active space

FEATURES	PLAGUE	YELLOW FEVER
Agent		
Vector		
Reservoir	Wild rat (tatera Indica)	Wild animals (monkey)
IP	2-7 days 1-3 days (Pneumonic)	2-6 days
Most Common : Most dangerous :		—
Treatment	Injection streptomycin	Supportive
Chemoprophylaxis	Tetracyclin	—
Vaccine :	Killed Inter epidemic (preferred) Can be given in epidemic in larger dose	Live attenuated 17 D 0.5 ml S/C NaCl / NS diluent
Control of vector	Rat flea index to zero within 48 hr	Avoid mosquito breeding EYE strategy Elimination of Yellow fever epidemic

Active space

❑ Control of Insect

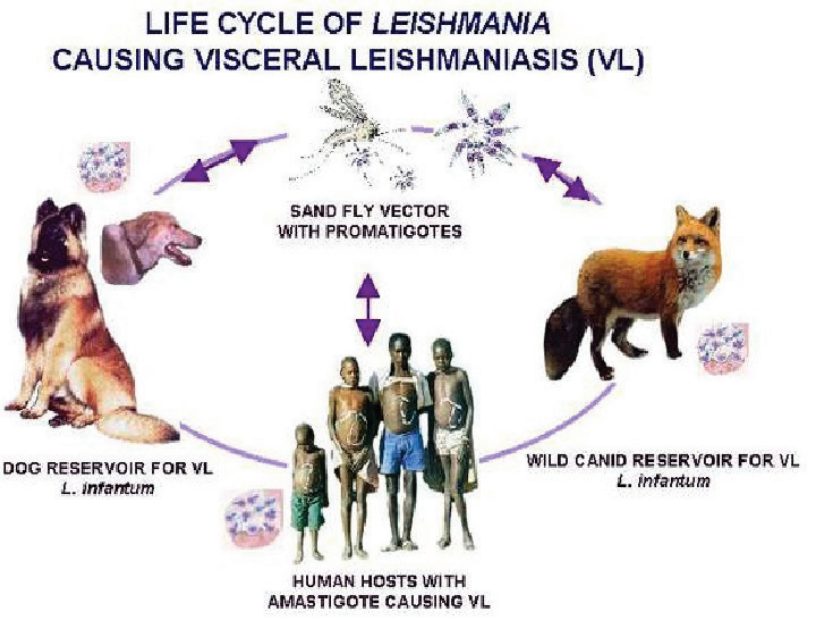
✦ 1. Rat flea ⟶ Never fly only vertical jump for 4 inch

- 10 % DDT dust
- If DDT Resistant ⟶ 5% Melathion
  2% Carbaryl
  - ✦ Spray upto 1 feet Ht.
- Repellent Diethyltoluamide on cloth effective for 1 wk

❑ Rat Control

- Zinc Phosphide
- Fumigation with calcium cyanide cynogas / cymag

LEISHMANIASIS

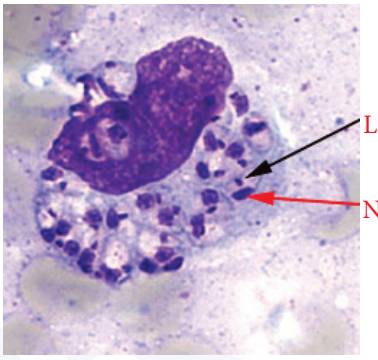


Life Cycle

Active space

C/F

Skin Nodules



LD Body

Agent: L Donovani : Visceral L .....

L Tropica : cutaneous L .....

L Braziliensis : mucocutaneous L

Vector : .....

Reservoir: Wild animals (dog, fox, jackal)

Disease of plain area: .....

IP (sandfly): 4-25 days

IP (human): 1-4 month (100 day- 2 yr)

❑ **Diagnosis:**

✦ 1. RK 39 dipstick test

- Blood, urine
- Antibody detection

✦ 2. Leishmania donovani body (LD)

- Intra cytoplasmic inclusion body
- Liver, spleen, bone marrow, lymph node, skin lesion aspirate
- Non flagellate form .....
- Flaggelar form .....

Active space

✦ 3. Leishmanian test :

Important : immune status of patient

❑ **Treatment :**

I. LAMB :

II. Miltefosin

DOC : 1) VL

2) PKDL

❑ **Vector Control :**

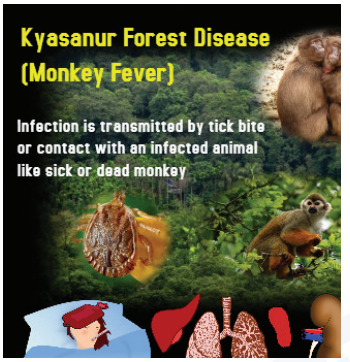
1. Insecticidal spray :
   - DDT (preferred)
   - BHC
   - Upto 6 feet
   - Twice/year

❑ **Goal:** elimination of disease by 2027

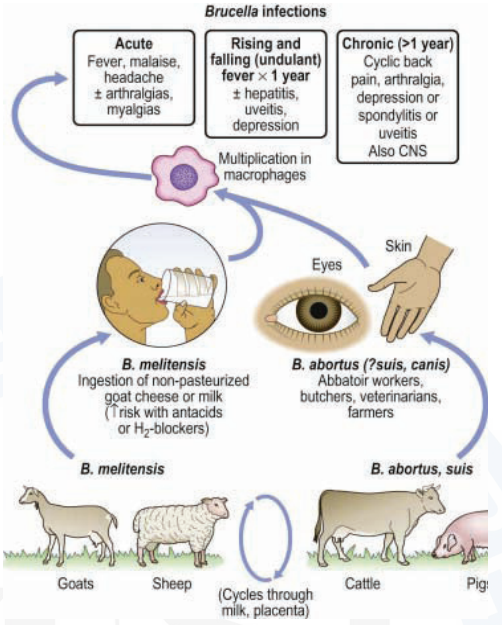
- Criterion : Incidence < 1/10,000
- Active surveillance by ASHA
- Quarterly case search
- KALA AZAR fort night survey
- RK-39 dipstick test
- Single dose of LAMB
- Vector control

Active space

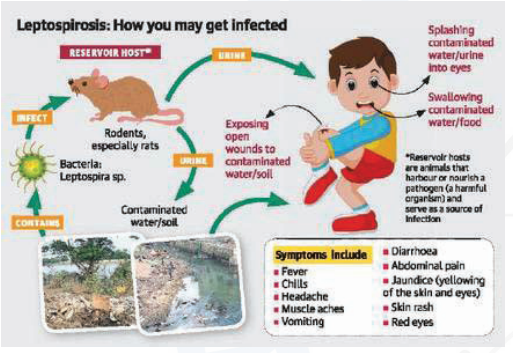
KFD, BRUCELLOSIS, LEPTOSPIROSIS



Life Cycle of KFD



Life Cycle of Brucellosis



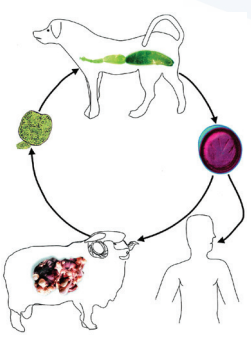
Life Cycle Leptospirosis

Features	KFD	Brucellosis	Leptospirosis
Agent	Group B arbovirus	B mellitensis	L interogans
Vector	Outside India :	Sources : 1. direct contact Infected body tissue, secretion 2. Food 3. Airborne	Urine of infected rat
	India : ..... (Hemophysalis spinigera)		
Reservoir	Monkey Rat Squirrel	Domestic animal	Rat

Active space

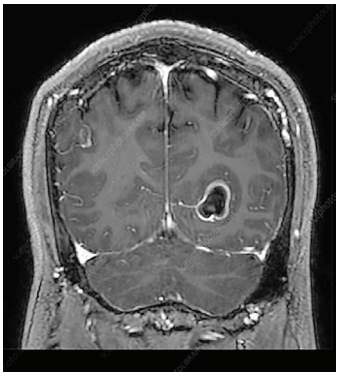
C/F	• Sudden onset of fever • Headache • Myalgia • Bleeding : skin, mucosa, visceral	Acute : • Swinging fever • Sweating • Arthralgia (large) • Headache Chronic : • Recurrence of symptoms for long time	Mild : • Fever with chills • Myalgia • Conjunctivitis • Headache • Renal/pulmonary symptom Severe : • Jaundice • Renal failure • CV collapse
Diagnosis	Viral isolation by RT-PCR Serological test	Bacteria by RT-PCR Rose Bengal test Milk ring test	Leptospira in urine/blood Serological test
Treatment	Supportive	DOC : doxycycline	Mild : doxycycline Severe : penicillin
Prevention	Killed vaccine	Live vaccine • Female cattle • Farmers	Chemoprophylaxis Doxycycline

ECHINOCOCCOSIS, TAENIASIS



Life Cycle of Echinococcosis

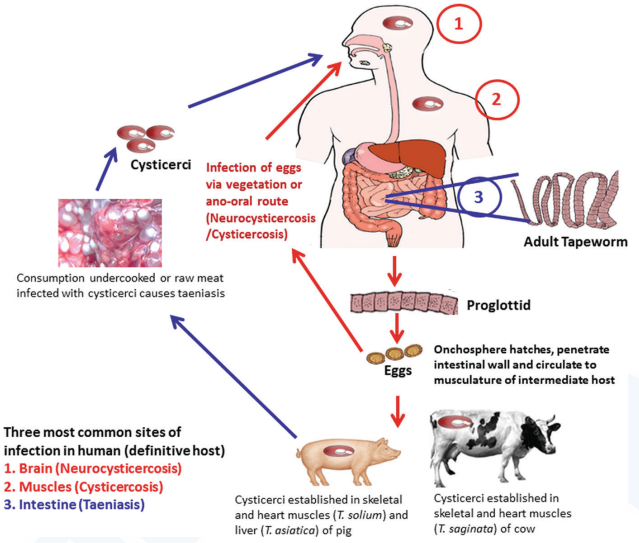
Hydatid Cyst



Neurocysticercosis

MIST

Active space



Life Cycle of Taeniasis

FEATURES	ECHINOCOCCOSIS	TAENIASIS
Agent	E granulosis	T solium (pork) T saginata (beef)
Definitive host	.....	.....
Intermediate host	.....	.....
C/F	Hydatid cyst :	Teniasis (intestine) Cysticercosis (muscle) Neurocysticercosis (brain)
Treatment	Mebendazole Or Albendazole	Praziquantal Or niclosamide with purgative Neurocysticercosis : albendazole + steroid + anti-epileptic

MIST

Active space

NIPAH VIRUS DISEASE

- **Agent:** paramyxoviridae (genus heripavirus)
- **IP:** 4-14 days
- **Mode of Transmission:**
  1. Direct contact with infected bat, infected pigs, infected people
  2. Consumption of raw fruit contaminated with infected bat secretion, excretion
- **C/F :**

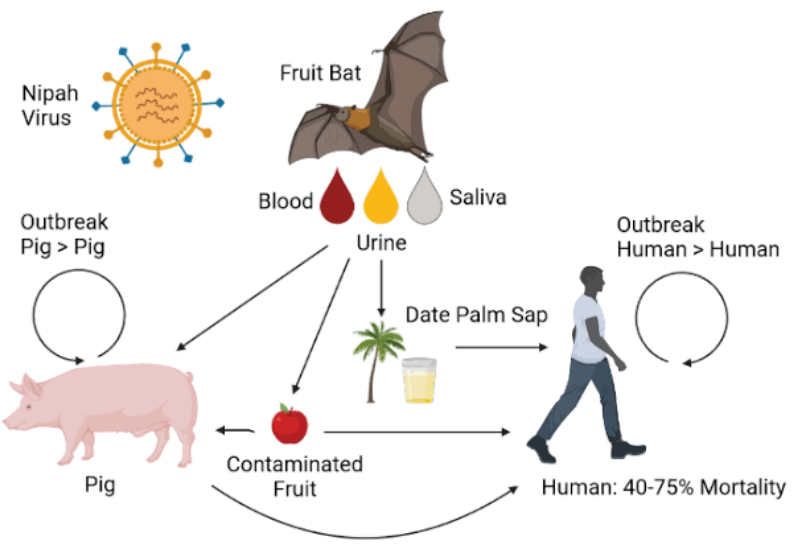
  Asymptomatic

  Acute RTI

  Fetal encephalopathy

  CFR : 40-75%
- **Diagnosis :**
  - RT-PCR (body fluid)
  - ELISA test
- **Treatment :** supportive
- **Prevention :** health education

Life Cycle of Nipah Virus Disease



Active space

SEXUALLY TRANSMITTED DISEASES

NACP (NATIONAL AIDS CONTROL PROGRAM) – 1987

Logo of NACP

Logo of Suraksha Clinic

Syndromic STI/RTI Color Coded Kits

❑ SURAKSHA CLINIC:

- SERVICES .....
- Syndromic approach of management

  Treatment done – based on symptoms (not on diagnosis)

- 7 color kits:

1. Grey : .....
2. Green : .....
3. White : genital ulcers(non-herpetic) – respond to penicillin
4. Blue : genital ulcers – non herpetic (allergic to penicillin)
5. Red : genital ulcer (herpetic)
6. Yellow : .....
7. Black : .....
8. Brown : .....

STI-color coded kits under NACP

Kit	Composition	Targetted Syndrome
Kit-1 (Grey)	Tab. Azithromycin 1000 mg and Tab. Cefixime 800 mg single dose	· Urethral Discharge Syndrome · Vaginal Discharge Syndrome (for cervicitis) · Painful Scrotal Swelling · Presumptive treatment (PT)
Kit-2 (Green)	Tab. Secnidazole 2000 mg and Tab. Fluconazole 150 mg single dose	• Vaginal Discharge Syndrome (for vaginitis)
Kit-3 (White)	Inj. Benzathine penicillin G 2.4 MU and Tab. Azithromycin 1000 mg single dose Disposable syringe 10 ml with 21-gauge needle and Sterile water 10 ml	• Genital Ulcer Disease Syndrome (for Syphilis and Chancroid)
Kit-4 (Blue)	Tab. Doxycycline 100 mg (28 capsules as twice/day dose for 14 days) and Tab. Azithromycin 1 g single dose	• Genital Ulcer Disease Syndrome (for Syphilis and Chancroid when unavailability or history of allergy to BPG)
Kit-5 (Red)	Tab. Acyclovir 400 mg (21 tablets as three times/ per day dose for 7 days)	• Genital Ulcer Disease Syndrome (for Herpetic Ulcers)
Kit-6 (Yellow)	Tab. Cefixime 800 mg single dose and Tab. Metronidazole 400 mg (28 tablets as twice/day dose for 14 days) and Cap. Doxycycline 100 mg (28 capsules as twice/ day dose for 14 days)	• Lower Abdomen Pain • PID
Kit-7 (Black)	Tab. Doxycycline 100 mg (42 capsules as twice/day dose for 21 days)	• Inguinal Bubo under Genital Ulcer Disease Syndrome • LGV Proctitis under Anorectal Discharge Syndrome
Kit-8 (Brown)	Tab. Cefixime 800 mg STAT dose and Tab. Doxycycline 100 mg (14 capsules as twice/ day dose for 7 days)	• Anorectal Discharge Syndrome

Active space

❑ INTERVENTIONAL STRATEGY FOR CASE DETECTION

1. Screening of high risk cases :
   a) Sex workers
   b) Transgenders
   c) MSM
   d) IV drug users
   e) Long distance truck drivers
   f) Single male migrant
2. Contact tracing :
   - Testing of close contact of screening + ve cases
3. Cluster testing :
   - Screening / testing of all individuals moving in same socio sexual background

✦ SURVEILLANCE TYPE

HIV & AIDS

- Agent : HIV 1, 2
- IP :
- MC age: .
- Prevalence of disease :

❑ WINDOW PERIOD:

❑ ROUTE OF TRANSMISSION:

➢ Sexual - 1 : 10,000 [0.01%]

➢ Parenteral :

- Blood transmission - 95%
- Needle prick - 1 : 300 [0.33%]
- Needle sharing - 1 : 150 [0.66%]

Active space

- Mother to child :
- Breast feeding :

STAGES

CLINICAL FEATURES:

1. INITIAL INFECTION :
   - Signs and symptoms of viral infection
   - Antibody after window period
2. Asymptomatic carrier stage
3. AIDS related complex
   - Diarrhea > 1 month
   - Fever > 1 month
   - Weight loss > 10%
4. Opportunistic infection

OPPORTUNISTIC INFECTIONS BASED ON CD4 COUNT

- No Infection : .....
- MC (India) : .....
- MC (World) : .....
- CD4 < 50 : .....

MALIGNANCY WITH HIV

- Kaposi sarcoma
- Non-Hodgkin's lymphoma
- Cervical cancer

Lab diagnosis of HID & AIDS

- Elisa : Screening test for HIV infection (Sn > 99.9%)
- Western blot : Confirmatory test for HIV (Sp > 99.9%)
- CD4 count : Predictor of HIV progression to AIDS
- CD4 percentage : Predictor of HIV progression to AIDS
  (CD4 percentage better than CD4 count)
- HIV viral load : Ds progression and response to ART

Active space

- P_{24} antigen Early detection of HIV (earlier than ELISA) in window period.

❑ TREATMENT:

✦ HIV treatment initiation

1. Same day ART : ART on D1 of diagnosis
2. Rapid ART : ART within 7 days of diagnosis

1. HIV + ve / AIDS patient

- HAART : .....
- When : .....
- Whom : .....
- How long : .....
- FDC : ⟶ Once/day (> 10yr, > 30kg)

1st line: Tenofovir 300mg
Lamivudine 300mg
Dolutegravir 50mg

2nd line: Zidovudine
Lamivudine
Lopinavir + Ritonavir
or
(Atazanavir + Ritonavir)

Active space

2. POST EXPOSURE PROPHYLAXIS

When: .....

Duration: .....

Test: .....

Drug ⟶ once / day (> 10yr, >30kg)

- Tenofovir
- Lamivudine
- Dolutegravir

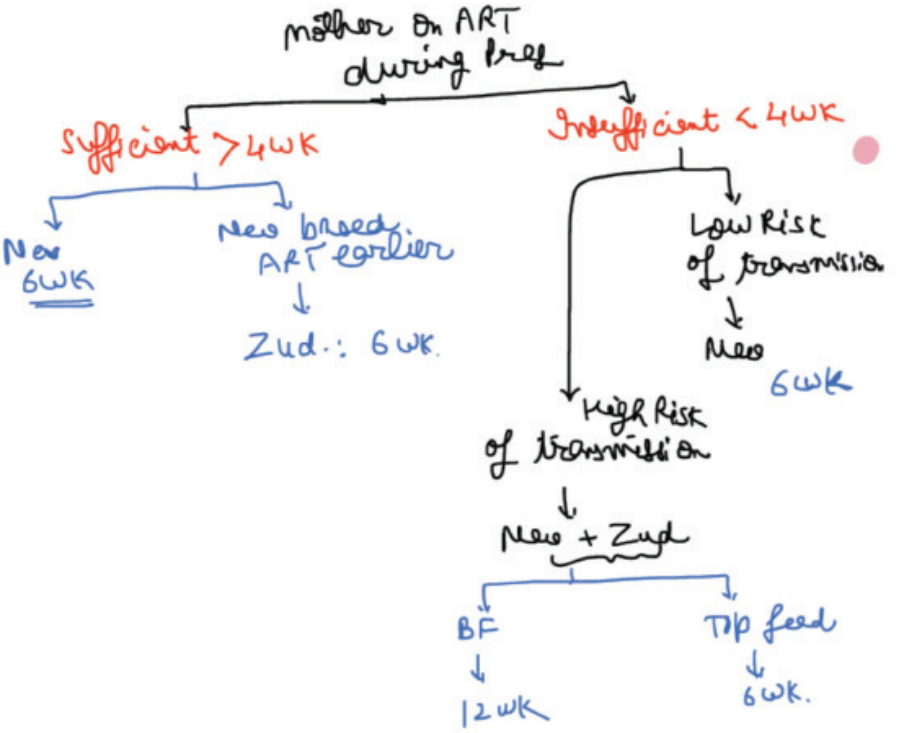
3. PPTCT (Prevention from parents to child transmission)

✦ Mother

1. Known case of HIV : .....
2. Diagnosed during ANC screening : .....
3. Continue during pregnancy delivery & breast feeding, life long

✦ New born prophylaxis

- When : .....



Active space

- Breast feed : A: developing: .....

  B: developed: .....

- Follow up : 6 week, 6 month, 12 months, 18 months

  3 month after stopping breast feeding

❑ PREVENTION OF CHILD TRANSMISSION :

Method — Net Transmission

1. Nevirapin:
2. Zidovudine:
3. Caesarean section:
4. Caesarean section + ART:
5. ART [mother + baby]:

❑ COTRIMOXAZOLE PROPHYLAXIS in HIV Individual

1 HIV exposed infant

- Start at 6 wk and continue till 18month
- Final diagnosis at 18month

+Ve -Ve (stop)

Continue till 5yr

(Irrespective of CD4 count)

CD4 at 5yr

< 350 : Continue > 350 : Stop

Active space

2) Adult (Include pregnant women)

- Start : a) severe or advanced HIV clinical ds (WHO stage 3 or 4)

  b) CD4 < 350

- Discontinue when clinically stable with immune recovery and viral suppression

3) Infant, Children, Adolescent

a) Children <5yr : Till 5yr irrespective of CD4 count or clinical stage

b) Children > 5yr

a) severe or advanced HIV clinical ds (WHO stage 3 or 4)

b CD4 < 350

TB Preventive therapy in PLHIV

1) 6H	: Daily Isoniazid	6month
2) 3HP	: Weekly Isoniazid and Rifapentine	3month
3) 1 HP	: Daily Isoniazid and Rifapentin	1 month
4) 6 Lf	: Daily Levofloxacin	6 month
5) 4R	: Daily Rifampicin	4 month

Active space

❑ CLASSIFICATION OF STATES AND DISTRICT

CLASSIFICATION OF STATES			CLASSIFICATION OF DISTRICT
STATE	HRG	ANC	
High prevalence	.....	.....	A
Moderate prevalence	.....	.....	B
Low prevalence	.....	.....	C (With few hotspots)
	.....	.....	D (No hotspots)

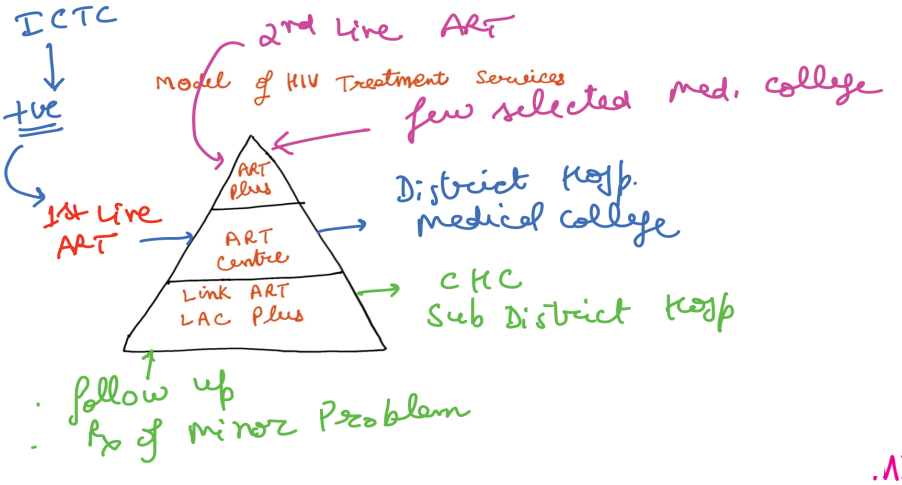
❑ ICTC CENTERS

- Integrated counseling, testing centers
- From district hospital onwards

ART CENTERS- .....

ART PLUS- .....

LINK ART/ LAC PLUS- .....



Active space

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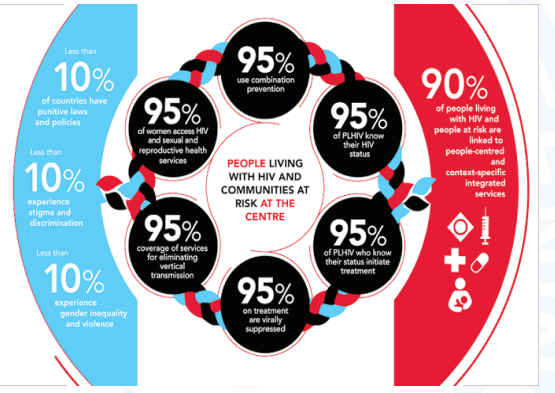
❑ NATIONAL AIDS CONTROL PROGRAM ⟶ NACP Phase V

Goal :

1) Reduce annual new HIV infection, AIDS related death by 80% by 2025-2026 (Base line 2010)
2) Eliminate vertical transmission of HIV, syphilis
3) Eliminate HIV / AIDS related stigma discrimination
4) Promote universal access to quality STI / RTI services

Objective, Target

95 : 95 : 95 : 95 : 95 : 95



UNAIDS GLOBAL STRATEGY

1) 'Fast Track'
2) end AIDS epidenic by 2030

SAMPOORNA SURAKSHA
SURAKSHA SE SAMPOORNA SURAKSHA TAK

Strategic Intervention

1) Sampoorne Suraksha Kendra (SSK)

   Prevention ⟶ Test ⟶ Treat and care continum to keep HIV free

2) One stop centers (OSC)

   Promotion of integrated service delivery

3) I SHTH Clinic ⟶ Integrated centers for Syphilis, HIV, TB, Hep B C diagnosis

4) Decentralization of services : DISHA

   District Integrated strategy for HIV / AIDS

MIST

Active space

5 ICT C, ART, ART plus Link ART, LAC plus center

6 Safe blood transfusion

7 Suraksha Clinic

8 Red Ribbon club, express train, run

9) Link worker scheme

10) Sex education in class IX, XI : 16 hr

11) Sankalp Project

- Launched by ESIC
- Organize awareness Camp
- Prevent fresh HIV infection among members of ESI

12) Project Sun Rise

- HIV transmission in I/V drug addict in NE state
- ↓ Promote safe sexual practice

13) Ab Nahi chalega campaign

14) Health education

15) Use of male female condom

16) Needle syringe program

17) ART as PrEP, PEP

18) Voluntary Medical male circumcision (VMMC)

19) Opiod substitution therapy with methadone or buprenorphine (↓ Injection risk)

Active space

EXTRA EDGE

FIRST LINE ART AS PER BODY WT

Particulars	Recommended Regimen
1. Age less than 6 years and body weight less than 20 kg	Abacavir + Lamivudine + Lopinavir/ritonavir (AL+LPV/r)
2. Age between 6 and 10 years and body weight between 20 kg and 30 kg	Abacavir + Lamivudine + Dolutegravir (ALD)
3. Age more than 10 years and body weight more than 30 kg	Tenofovir + Lamivudine + Dolutegravir (TLD)

ALTERNATE FIRST LINE ART IN SPECIAL SITUATION

Condition	Alternate First-line Regimen
PLHIV with body weight <30 kg	ABC 600 mg + Lamivudine 300 mg, one tablet + DTG (50 mg) once daily in the morning or any fixed time every day as per patient's convenience
All patients with high (above ULN for laboratory) serum creatinine values (Calculate Creatinine clearance)	ABC 600 mg OD, Lamivudine (as per creatinine clearance**) and DTG 50 mg once daily in the morning or any fixed time every day as per patient's convenience
PLHIV on Rifampicin-containing ATT regimen	Tenofovir (300 mg) + Lamivudine (300 mg) + Dolutegravir (50 mg) - FDC one tablet once daily (in the morning or any fixed time every day as per patient's convenience) + Additional dose of DTG 50 mg to be provided (12 hours after taking their regular dose) until 2 weeks after completion of ATT
Women of childbearing potential who do not wish to take DTG-based ART after adequate and optimal counselling***	Tenofovir (300 mg) + Lamivudine (300 mg) + Efavirenz (600 mg) If Efavirenz is contraindicated (HIV-2/HIV-1&2/prior NNRTI exposure) then Tenofovir (300 mg) + Lamivudine (300 mg) + [Lopinavir (200 mg) + ritonavir (50 mg) twice daily]

MONITORING AND FOLLOW UP

Monitoring Tool	When to Monitor
Body weight Height / length in children	Every visit
Treatment adherence	Every visit
Clinical monitoring and T-staging	Every visit
4-symptom TB screening	Every visit
Screening for common NCD; Hypertension, Diabetes mellitus	Every 6 months or symptom directed
Laboratory evaluation based on ART regimen	Every 6 months or symptom directed
CD4 Count	CD4 must be done every 6 months*
Viral load	At 6 months, 12 months and then every 12 months**

Active space

MIST

- CD4 : discontinued in PLHIV (except for HIV-2 infection) when count > 350 and viral load < 1000 copies (if both test done together)
- If Pt on 2nd / 3rd line ART, continue viral load testing every 6month

POST EXPOSURE PROPHYLAXIS

Exposed Person	Preferred Regimen for PEP Drugs and Dosages	Alternate Regimen (If the Preferred Regimen is not available or Contraindicated)
Adolescents and Adults (>10 years of age and >30 kg weight)	Tenofovir (300 mg) + Lamivudine (300 mg) + Dolutegravir (50 mg) (FDC: One tablet OD)	Tenofovir (300 mg) + Lamivudine (300 mg), (FDC: One tablet OD) + Lopinavir (200 mg)/Ritonavir (50 mg) (two tablets BD), OR,Tenofovir (300 mg) + Lamivudine (300 mg) + Efavirenz (600 mg), (FDC: One tablet OD)
Children (>6 years of age and >20 kg weight)	Zidovudine + Lamivudine (Dosage as per weight band) + Dolutegravir (50 mg) (One tablet OD)	If Hb <9 g/dl: Abacavir + Lamivudine (dosage as per weight band) +, Dolutegravir (50 mg), (One tablet OD)
Children (<6 years of age or <20 kg weight)	Zidovudine + Lamivudine + Lopinavir/ritonavir (Dosage as per weight band)	If Hb <9 g/dl: Abacavir + Lamivudine + Lopinavir/ritonavir (Dosage as per weight band)

PRE EXPOSURE PROPHYLAXIS

- ✦ Indication : Non HIV at Higher Risk
- ✦ Drug : Tenofovir + Emtricitabine daily FDC
- ✦ Duration
  - A) As long as needed
  - B) PrEP on demand : 2 - 1 - 1
    - 2 Tab 2-24hr before sex
    - I Tab 24 after 1st dose
    - 1 Tab 24 hr after2nd dose

Indications for PrEP Therapy

Men who have sex with men	• HIV-positive sexual partner • High number of sex partners • Anal intercourse with inconsistent or no condom use • Commercial sex work • Recent STI with gonorrhea, chlamydia, or syphilis
Heterosexual men and women	• HIV-positive sexual partner • High number of sex partners • History of inconsistent or no condom use • Commercial sex work • Recent STI with gonorrhea or syphilis • High HIV prevalence area or network
Individuals who inject drugs	• HIV-positive injecting partner • Sharing any injection equipment

MIST

Active space

SURFACE INFECTIONS

TRACHOMA

Problem: .....

DOC: .....

Prevention:

1. SAFE strategy
   - S: .....
   - A: .....
   - F: .....
   - E: .....
2. GET strategy

   .....

TETANUS

Agent : Cl Tetani

Ip : 6-10 days

Reservoir : soil

Tetanus

Treatment :

DOC : .....

Ig : human (Tet globe) – .....

Animal (ATS) .....

Prevention :

1. Active Immunization :

   Goal : antitoxin >0.01 IU/ml

Active space

2. Post Exposure Prophylaxis:

IMMUNIZATION STATUS	Type of wound	
	Minor	Others
	Clean	
	<6 hr	
	Non penetrating	Rest others
	Negligible damage	
Booster <5 yr	.....	.....
Booster 5-10 yr	.....	.....
Booster >10 yr	.....	.....
Non immunized	.....	.....

NEONATAL TETANUS

- Elimination by India
- Criteria : Incidence of NNT .....
- Elimination possible due to
  - Coverage of pregnant women with 2 doses of Td .....
  - Delivery are attended by skilled person by 5 clean methods with Use of DDK .....

Neonatal Tetanus

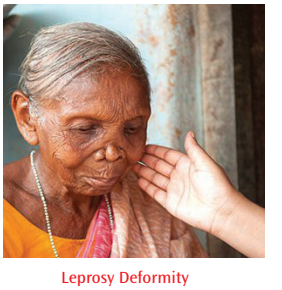
LEPROSY

- IP : 1-20 yrs (5 yrs)
- Nerve damage :

  MC – .....

  Least affected –
- Plantar ulcer

  Secondary impairment.....



Leprosy Deformity

MIST

Active space

Primary impairment – .....

- Loss of sensation :

  .....

- Source of infection

  1. Case of MBL (Nasal secretion) :
  2. Close skin contact of leprosy patient
  3. Breastfeed
  4. Use of tattoo needles

LEPROMIN TEST

- Agent : Dharmendra's antigen
- Reading :

VALUE OF LEPROMIN TEST

- ➢ Evaluate immune response
- ➢ Classify type of leprosy
- ➢ Prognosis of disease

☞ Not for diagnosis of disease

❑ **Infrastructure:**

- ➢ SET (survey, education, treatment) :

  .....

- ➢ ULC (urban leprosy centers) :

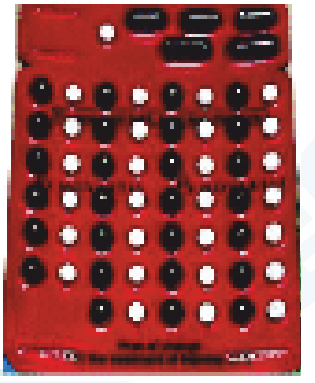
  .....

- ➢ LCU (leprosy control unit) : .....

Active space

❑ Types of Leprosy:

Criteria	Paucibacillary	Multibacillary
1. No of skin lesion		
2. Nerve affected		
3. Skin smear		
4. TREATMENT	1. RIFAMPICIN :	SAME
	2. Dapsone :	SAME
	3. Clofazamine :	SAME
5. Duration of treatment		
6. Follow up		



Pack of Leprosy Treatment

Active space

DIAGNOSIS OF LEPROSY

- Hypopigmented / reddish skin lesion with sensory deficit
- Involvement of peripheral nerves

  Thickening with loss of sensation and / or weakness or paralysis of corresponding muscle of hand, feet or eyes.
- M. laproe demonstration in lesion

☞ Cure rate

PBL =

MBL =

❑ **Accompanied MDT:**

Any responsible person from family/ village can collect medicine on behalf of patient

❑ **DPMR (Disability prevention & medical rehabilitation)**

- Microcellular rubber footwear
- Ulcer treatment kit
- Reconstruction surgery

  Pt incentive = 12,000 Rs

  Institution = 5000 Rs / surgery

Post exposure prophylaxis

- Inclusion criteria for PEP
  - A person living/working/having social activity for >3 month and 20 hr/ week with a newly detected leprosy case in last 1 year
  - Age ≥ 2 year
- Single dose of rifampicin (to close contact of patient) DORS :
- Dose :

  >35 kg = 600 mg

  20-35 kg = 450 mg

  <20 kg = 10-15 mg/kg

Active space

❑ GLOBAL LEPROSY STRATEGY (2021-2030)

TOWARD ZERO LEPROSY by 2030

Vision : Zero New infection zero disease

Zero disability stigma discrimination

Goal : Leprosy elimination by interruption of transmission

❑ NATIONAL LEPROSY ERADICATION PROGRAM (NLEP)

Goal : leprosy Mukt BHARAT by 2027

Objective, Target

1. Prevalence at subnational, districtlevel : < 1 / 10,000
2. Annual New Case Detection Rate (ANCDR) : <10 / 1 Lac
3. Grade II disability among new cases at National level : <1%
4. GradeII disability cases at National level : <1 /10 Lac
5. Disability among New child (<15yr) cases : Zero
6. Stigma and discrimination among affected cases : Zero

INDICATOR

1) Most Important epidemiological indicator : ANCDR

2) Most Important operational Indicator : New case with grade II Disability

3) Prevalence : - Case load
- Planning of Treatment services

4) Incidence : Effectiveness of program

Monitoring success of control Program

Active space

Leprosy Vaccine

MIP : mycobacterium Indicus Pranii

- Vaccine With Single dose of rifampicin 600 mg oral
- Close contact of newly detected case

MIST

Preventive Activity

- Leprosy case detection campaign – ASHA (ABSULS) ASHA based surveillance for leprosy suspects.
- SPARSH leprosy awareness Campaign (Gram Sabha)
- SET Scheme (NGO)
- SAPNA - Leprosy mascot
- Integration of leprosy screening of children under RBSK (0-18yr, RKSK (10-19 Yr)
- Integration of leprosy screening in ≥ 30yr with other activities of comprehensive Primary health care under AYUSHMAN BHARAT

Lepra Reaction

Type I (Reversal)	Type II (Errythema-Nodosum leprosum)
Cell-mediated immunity	Humoral immunity
Both PBL, MBL	Only MBL
Existing –Leison	.....
Nerve -	.....
General symptom –	....
hyper sensitivity	hyper sensitivity
Rx	

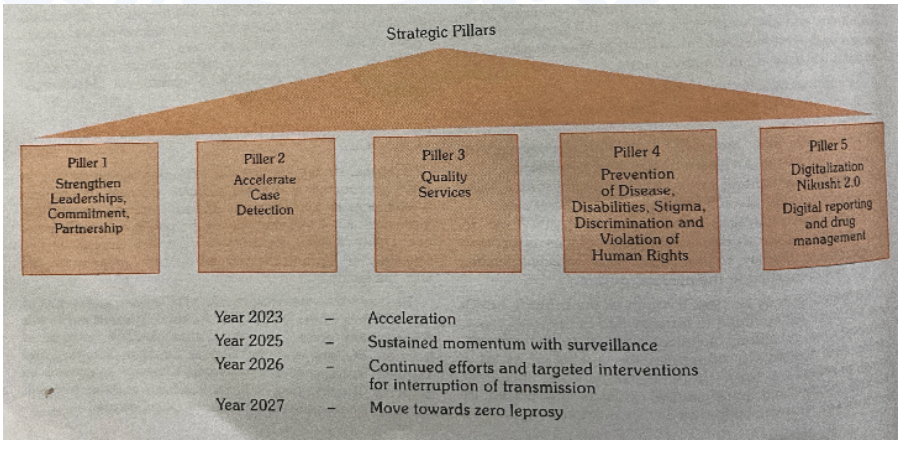
Active space

EXTRA EDGE

DEFINATION

1) Case of leprosy : Pt with clinical sign of leprosy with or without bacteriological confirmation, not completed full course of MDT
2) Adequate treatment : Treatment completion in resonably short time (PBL 6month MDT in 9month)
   (MBL 12month MDT in 18month)
3) Regular treatment : Received MDT for 2/3 of month in any time interval
4) New case : case of leprosy who had never taken MDT in past
5) Defaulter : leprosy pt on MDT, who had not collected medicine for 12 consecutive month
   Name should be removed from register
6) Relapse : New Sign and symptom ofds after completion of adequate course of MDT

STRATEGIC PILLARS OF NLEP



Active space

REVISED TREATMENT REGIMEN OF LEPROSY IN INDIA:

S. No.	Types		Duration of treatment	Dapsone	Clofazimine	Rifampicin
1	Pauci-Bacillary (PB)	Adult	6 months	100 mg Daily	300 mg once a month & 50 mg daily	600 mg once a month
		Child (9-14 years)	6 months	50 mg Daily	150 mg once a month & 50 mg alternate day	450 mg once a month
2	Multi-Bacillary (MB)	Adult	12 months	100 mg Daily	300 mg once a month & 50 mg daily	600 mg once a month
		Child (9-14 years)	12 months	50 mg Daily	150 mg once a month & 50 mg alternate day	450 mg once a month

LEPROSY DISABILITY GRADE

Examination of parts	WHO Disability Grades	Sensory testing	Voluntary muscle testing	
Hands	0	Sensation present	Muscle power normal	
	1	Sensation absent	Muscle power normal	
	2	Sensation absent	Muscle power weak or paralyzed	
Feet	0	Sensation present	Muscle power normal	
	1	Sensation absent	Muscle power normal	
	2	Sensation absent	Muscle power weak or paralyzed	
Eye		Vision	Lid Gap	Blinking
	0	Normal	No lid gap	Present
	2	Cannot count fingers at 6 metres	Gap present/ red eye/ corneal ulcer or opacity	Absent

Active space

INTESTINAL INFECTIONS

DIARRHOEA

ORS COMPOSITION :

❑ **Osmolarity (low osmolar)** ..... ❑ **WEIGHT (20.5 gm)**

- Na + : 75 mmol/L
- K + : 20 mmol/L
- Cl- : 65 mmol/L
- Citrate : 10 mmol/L
- Glucose : 75 mmol/L

Dose : (Mild - Moderate diarrhoea)

.....

.....

☞ Preparation of ORS (1 sachet + 1 liter of boiled cooled water)

☞ Method of health education: .....

☞ Breast feed with diarrhoea : .....

Other fluid:

- Coconut water
- Rice water
- Lemon water with sugar + salt

☞ OPV not C/I with diarrhea

.....

.....

IV FLUID :

I/V fluids in severe diarrhoea

100 ml/kg in divided dose

Active space

MIST

Age	First 30 ml/kg	70 ml/kg
<1 yr		
>1 yr		

❑ **Fluids**

1. diarrhea treatment solution
2. ringer lactate
3. normal saline

MAINTENANCE FLUID

Mild :

Severe :

Zinc supplementation

<6 month :

>6 month :

Vaccine preventing diarrhea

1. Rotavirus
2. Measles

☞ Vit deficiency with diarrhea :

POLIOMYELITIS

✦ Elimination on 27th March 2014

NPEP (National Polio elimination program)

❑ **1. Routine vaccination :**

❑ **2. PULSE POLIO IMMUNIZATION :**

- Started 1995

Pulse Polio

- Mass vaccination of children ..... by ..... doses of ..... every year on some days irrespective of vaccination status

Active space

❑ **3. Intensive pulse polio immunization**

Vaccination at fixed booth

House – house search – vaccination

❑ **4. AFP surveillance (Acute Flaccid paralysis)**

Age - .....

Any case of acute flaccid paralysis should be considered as polio paralysis until diagnosed otherwise

Diagnosis : .....

Adequate stool sample

1. Within 14 days of paralysis
2. 2 sample 24 hrs apart
3. Each – 8 gm
4. Transport : + 2°C to + 8°C (Reverse cold chain)
5. Reach lab within 48 hours

Report available within 4 weeks

Indicators of AFP surveillance:

1. Non polio AFP rate : >1/1,00,000
2. Reported AFP cases with 2 stool sample <14 days of paralysis (>80%)

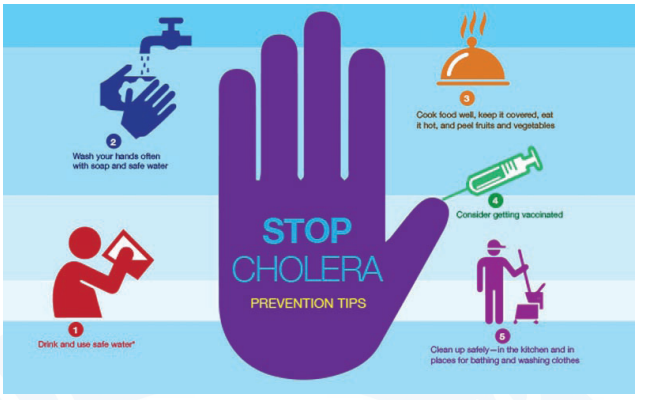
❑ **ORI (outbreak response Immunization)**

Single dose of OPV to All susceptible (0-5 yrs) around suspected case

Active space

CHOLERA

Rice Water Stool



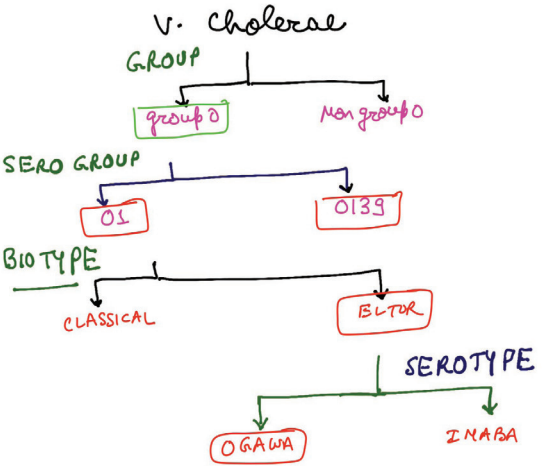
Prevention of Cholera

- Father of Public Health .....
- IP = 1-2 days [few hours- 5days]
- Sub clinical cases and carriers [yes Ice berg phenomenon]

❑ **Carriers in Cholera**

- Pre clinical [Incubatory] 1-5 days
- Convalescent 2-3 week
- Contact [health] less than 10 days
- Chronic carrier more than 1 year

❑ **Strains of V. Cholerae**



Active space

☞ DS under IHR .....

☞ Most effective prophylactic measure .....

☞ Correction of Dehydration by .....

❑ **DOC (Treatment)**

- Adult DOXYCYCLIN
- Children AZITHROMYCIN
- Pregnant AZITHROMYCIN

❑ **Chemoprophylaxis:** Tetracyclin

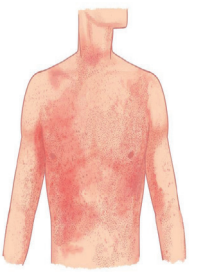
☞ Disinfection [stool/vomits] - .....

☞ Chlorination of well at time of cholera epidemic - .....

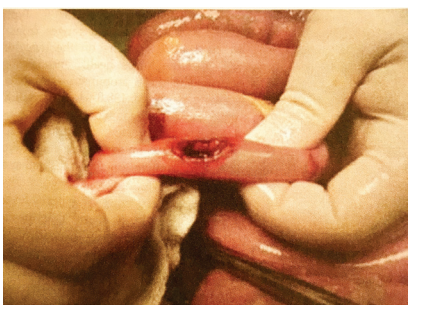
❑ **CHOLERA VACCINE**

1. Oral Killed Vaccine [MC]
   a. Monovalent [O1/O139 group]
      Dukoral (WC-rBS)
   b. Bivalent [O1 + O139 group]
      - SANCHOL/EUVICOL
      - 2 doses [gap of 1-6 week]
      - Booster after 2 years
2. Oral Live Vaccine
   - O1 Sero Group
   - CVD 103- Hg R [Vax chora]
   - Single Oral Dose

TYPHOID FEVER



Typhoid Rash



Intestinal Perforation

Active space

- Index of general sanitation of a country
- Sub clinic cases and carriers [Ice berg phenomenon present]
- Carriers
  - Convalescent carrier 6-8 weeks
  - Chronic > 1 year
- Reservoir – man[only]
- Most common age- 5-14 years
- DIAGNOSIS:

  B- 1st week – .....

  A 2nd week- .....

  S- 3rd week- .....

  U- 4th week- .....
- Treatment:
  - ✦ 1. CASES :
    - Cefixime
    - Quinolone
  - ✦ 2. RESISTENT CASE:
    - Ceftriaxone
    - Azithromycin
  - ✦ 3. CARRIERS:
    - Ampicillin / Amoxycillin + Probenecid
    - Cholecystectomy

☞ DISINFECTION [Feces, urine, vomitus]

- Vaccine:

1. TYPHORAL [Ty 21a]
   - Live
   - Oral
   - ≥ 5 years age
   - Cap [Day 1,3,5]
   - Repeat every 3 year

new protocol

[Cap 1, 3, 5, 7]

Repeat every 5 year

Active space

2. TYPHI-M VACCINE
   - Killed
   - VI Polysaccharide
   - ≥ 2 years
   - 0.5 ml, I/m single dose
   - Repeat every 3 year
3. TCV - Typhoid VI Polysaccharide Conjugate Vaccine
   - Conjugate, Killed
   - Single dose 0.5ml I/m
   - 6 month - 45 Yr.
   - life long immunity

HEPATITIS

- Hepatitis c̄ chlorine carrier state .....
- Hepatitis c̄ liver cancer .....
- Hepatitis c̄ chronicity .....
- Hepatitis in pregnancy with maximum Fulminant stage .....

Vaccine

Hepatitis A

1. KILLED
   - MC
   - 2 doses [1. At 1 year, 2. After 6 months]
   - 0.5ml I/m
2. LIVE VACCINE
   - Not commonly used
   - H2 and L-A-1 strain

Active space

Hepatitis B [Killed, Recombinant]

1. 3 doses of Hep B
   - Preferred 0,1,6 month
   - current [Hospital Delivery 4 Doses- Birth, 6, 10, 14 week]
   - [Home Delivery 3 doses 6, 10, 14 week]
   - 0.5ml I/ m
2. PMTCT [Prevention from Mother to Child Transmission]
   a. Presence of HB-s antigen + HB-e antigen

   .....

   1st dose- 0.05- 0.07ml/kg within 48hours of delivery [Preferably within in 6 hours]

   2nd dose- Ig- After 30 days

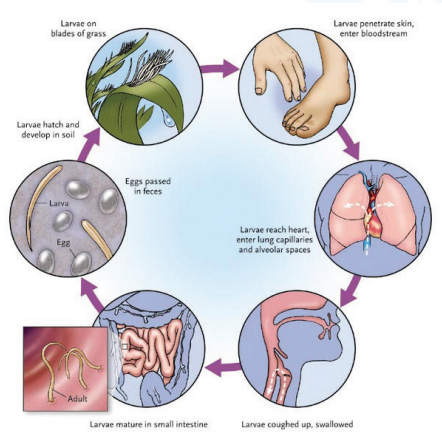
   b. Presence of only HB-s Antigen [HB-e antigen absent]

   .....

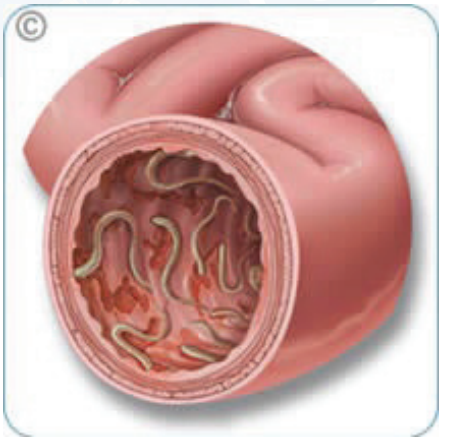
   c. HB-s + ve, HB-e status not known

   .....

WORM INFESTATION



Hook Worm Cycle



Hook Worm in Intestine

Active space

Round Worm in Intestine

Round Worm Expulsion from Mouth

- Most common worm infestation : .....
- Worm causes anemia : .....
- Worms related to step wells : .....
- National Deworming Day : 10th February

CHANDLER INDEX

- WORM:
- DEF:
- CACULATION:
- NON-SIGNIFICANT:
- POTENTIAL DANGER:
- MINOR HEALTH PROBLEM:
- MAJOR PUBLIC HEALTH PROBLEM:

☞ DOC: ALBENDAZOLE

NATIONAL DEWORMING SCHEDULE

- Twice/ year [February, August]
- Target- 1- 19 years
- Albendazol- [1-2-year ½ tablet [200mg]
  [2-19 years 1 tablet [400mg]
- UIP PLUS ACTIVITY
  Linking of vit A c̄ Albendazol in UIP

Active space

RESPIRATORY INFECTIONS

MEASLES [RUBEOLA]

....

AGENT: - RNA PARAMYXOVIRUS

I.P: -

POC:

SAR: -

SOURCE: - case of measle [Respiratory route]

DS of 3C:

- .....
- .....
- .....

❑ ..... Bluish white spot-on Red base on Buccal mucosa opposite 2nd molar teeth (Table salt crystal)

.....

❑ **COMMON COMPLICATIONS**

- .....
- .....
- .....

Active space

❑ **RARE COMPLICATIONS**

- .....

❑ **VITAMIN DEFICIENCY ASSOCIATED WITH MEASLES**

- .....

❑ **RX** Supportive, Isolation, vit A supplementation if Deficiency symptom (3 dose Day 0, 1, 14)

❑ **PREVENTION**

✦ 1. Pre-Exposure

- Measles Vaccination

✦ Post Exposure

1. Good Immunity

.....

.....

2. Poor Immunity

❑ **GOAL** Elimination of Measles by 2026

Criteria : Absence of transmission > 12 months

1. AMMRS [WHO- UNICEF]

.....

.....

High population coverage of 2 doses of MCV

>90%	NATIONAL
>80%	DISTRICT

2. WHO MEASLES ELIMINATION STRATEGY

.....

.....

Active space

❑ GLOBAL MEASLE & RUBELLA STRATEGY PLAN [2021-2030]

WORLD FREE FROM MEASLES RUBELLA

Strategies :-

1. 95% coverage of Measle, Rubella vaccination
2. Case based measle-Rubella surveillance with adequate Laboratory support
3. Linkage with other child Health interventions
4. Increased public confidence and demand for immunization

CHICKEN POX

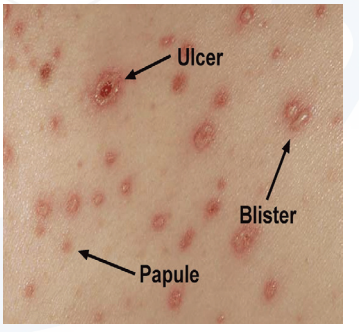
Agent: - VZ Virus [Human Alpha Herpes Virus-3]

IP: -

SAR: -

POC:

SOURCE: - case of varicella [Respiratory]



❑ RASH CHARACTERISTICS

1. .....
2. .....
3. .....
4. .....

❑ REYE'S SYNDROME

- Acute encephalopathy ass. c̄ fatty liver degeneration due to aspirin

Active space

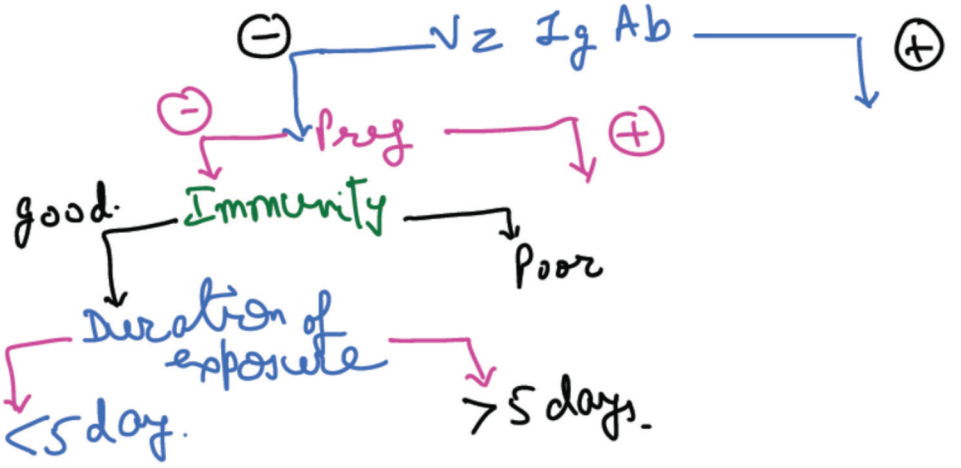
❑ TREATMENT

Supportive

❑ PREVENTION

- ✦ Vaccine
  - Live attempted, OKA, 0.5ml S/c
  - 2 doses
    - 1st : At 1 year, 2nd : At pre school

❑ POST EXPOSURE PROPHYLAXIS



❑ VARICELLA Ig TO NEW BORN

- if Mother has infection 5 days before to 2 days after delivery
- Ig to New Born

  Within 72 hours of exposure

  12.5 unit/kg I/M

❑ CONGENTITAL MAL FORMATION

- ✦ Maternal Infection
  - < 13-week Pregnancy : 0.4%
  - 13-20-week Pregnancy : 2.0%

Active space

Congenital Varicella Syndrome

1. Skin ⟶ Cutaneous scar, atrophied limb
2. LBW, microcephaly
3. Eye ⟶ Cataract, microphthalmia, chorio retinitis
4. Ventriculomegaly
5. Deafness
6. CNS ⟶ Mental retardation, cerebro cortical atrophy

INFLUENZA

- Agent: Ortho Myxo virus
- IP: .....
- TYPES: A, B, C, D (Type A most common)
  - Human Influenza
    - TYPE A : H1N1, H3N2
    - TYPE B : VICTORIA, YAMAGATA
  - BIRD FLU : H5N1, H7N9, H5N9, H7N1

❑ ANTIGENIC VARIATION

- SHIFT — Genetic Recombination
  Rearrangement
- DRIFT — Point Mutation

❑ CATEGORIES

- UN COMPILACTED : .....
- COMPLICATED : .....
- PROGRESSIVE : .....
  - .....
  - .....

Active space

❑ TREATMENT

- OSELTAMIVIR .....
- ZANAMAVIR [Nasal]
- PARAMIVIR [IV]

✦ Doses Of Oseltamivir

- ≥ 40kg = ...mg BD
- 24-39 kg = ...mg BD
- 15-23kg = ...mg BD
- <15kg = ...mg BD
- <3 month age = ...mg BD
- 3-5 month age = ...mg BD
- 6-11 month age = ... mg BD

5 Days

❑ PREVENTION

1. Mask N99 – N95
2. Chemoprophylaxis
3. Vaccination

Type of Vaccine	LIVE	KILLED
• ROUTE	.....	.....
• STRAIN	Quadrivalent	Monovalent/ Quadrivalent
• FREQUENCY	Single RPT every year	2 dose RPT every year

❑ VACCINATION ON PRIORITY ORDER

❖ Pregnant women
❖ >6-month age with chronic medical DS
❖ 15-49 years [Healthy Adult]
❖ Healthy children
❖ Healthy adults 49-65 years
❖ Healthy adult >65 years

Active space

MENINGO COCCAL MENINGITIS

- High number of sub clinical cases and carriers
- High CFR .....
- Early diagnosis and treatment .....

Prevention Strategy

Vaccination → Chemoprophylaxis → Early Diagnosis & Treatment

Sero Type : A, B, C, W135, Y (B-non immunogenic)

.....

.....

.....

I.P: - 3-4 days

Agent: - N. MENINGITIDIS [Gram -ve Diplococci]

❑ TREATMENT

1. Cases
   - PENICILLIN
   - CEFTRIAXONE
   - CEFOTAXIME
2. Carriers
   - CIPROFLOXACINE
   - CEFTRIAXONE
   - RIFAMPICIN

❑ POST EXPOSURE CHEMOPROPHYLASIS OF CLOSED CONTACTS

- Ciprofloxacine
- Rifampicin
- Ceftriaxone [Pregnancy]

Active space

❑ MASS CHEMOPROPHYLAXIS

- Close contact
- Health community team members
- Drugs
  - Ciprofloxacin
  - Minocyclin
  - Spiramycin
  - Ceftriaxone

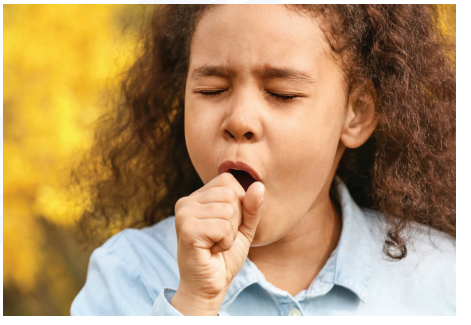
❑ PROPHYLAXIS BY VACCINE

- Meningococcal vaccine
- TYPE : Polysaccharide
- SERO TYPE : Quadrivalent [A, C, W135, Y]
- DOSE : Single dose, I/m, >2 years age
- Booster every 3 year [in epidermic prone area]
- Mass Vaccination [can be given]

DIPHTHERIA AND PERTUSSIS

Pseudo Membrane

Bull Neck



Pertussis

	DIPHTHERIA	PERTUSSIS
Agent	C. DIPHTHERIA	B. PERTUSSIS
IP	2-6 days	7-14 days
POC	From Infection till 14-28 days after	1 week ← (P) → 3week

MIST

Active space

C/F

- Bull neck (Swelling in neck)
- Difficulty in breathing,
- Fever
- High number of sub clinic cases And carriers
- Paroxysm of cough
- Apnoea
- Fits
- only clinical cases
- Bleeding/Hemorrhage

❑ DOC FOR TREATMENT

	DIPHTHERIA	PERTUSSIS
A.	CASES	CASES
	Penicillin	Erythromycin
	Anti-toxin	
B.	CLOSED CONTACT	
	Erythromycin	
	Penicillin	
C.	CARRIER	CLOSE CONTACT
	Erythromycin	Erythromycin

❑ PREVENTION

✦ DPT VACCINE

Goal : Antibody (Anti toxin) > 0.03 IU/ml

☞ No effect against Carrier stage

✦ Prevention of Contact [Diphtheria]

A. Received primary dose/ booster

- Within 5 years : .....
- > 5 years back : .....

B. Non-Immunized

- Erythromycin
- Anti-toxin 1000-2000 unit
- Vaccine

Active space

❑ **CASE**

1. 5yr baby with diphtheria infection, Protection of 3yr baby who had received all vaccination as per UIP till present

Ans. .....

2. 3yr baby with diphtheria infection, Protection of 6yr baby who had received only primary doses of Pentavalent vaccine

Ans. .....

✦ Prevention of PERTUSSIS CONTACT

- Erythromycin
- Protection of new born

1. Booster of DPT to sibling before delivery
2. T dap vaccine to mother at 30weeks of pregnancy

MUMPS

Agent: - RNA Paramyxovirus

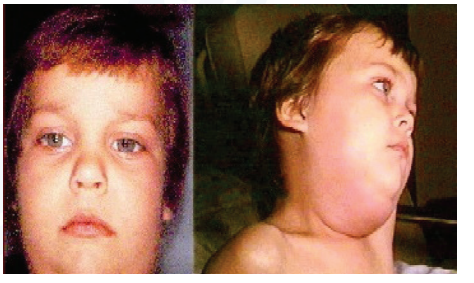
IP: - 2-4 week

POC: - 1week ← symptoms → 1 week

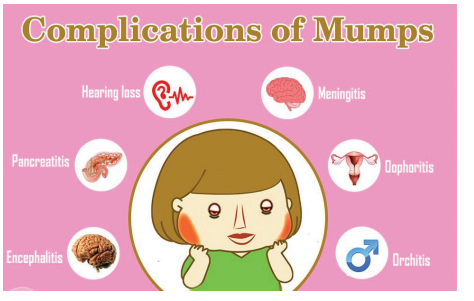
AGE: - 5-9 years

C/F:

- Asymptomatic [30-40%]
- Swelling over salivary glands
- Pain in ear
- Difficulty in mouth opening



Clinical Features of Mumps



Active space

- **COMPLICATION:**
  - Pancreatitis
  - Oophoritis/ orchitis
  - Encephalitis/ Meningitis
- **RX:** Symptomatic
- **PREVENTION:** Vaccination [MMR]

RUBELLA [GERMAN MEASLE]

Agent: - RNA Toggavirus

IP: - 2-3 Week

C/F: -

- Asymptomatic [60-65%]
- Prodromal symptoms
- Lymphadenopathy
- Rash : Discrete, (Maculo papular)
- Forchheimer spots [soft palate]

- **CRS [Cong Rubella Syndrome]**

.....

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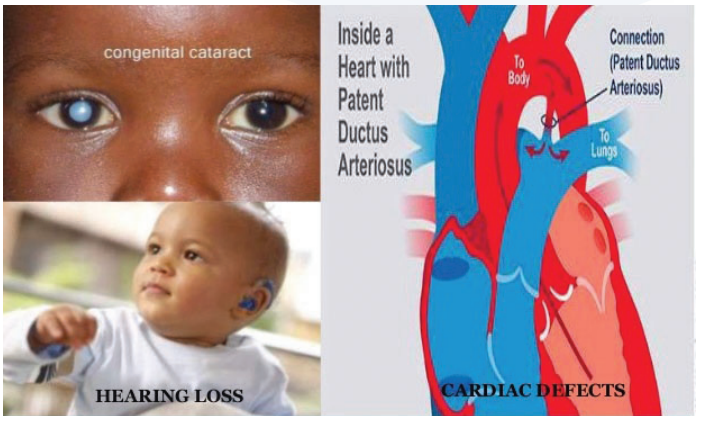


Cong Rubella Syndrome

Active space

- [most serious eye complication]
- [MC single complication]
- [PDA> PS> ASD]

☞ Most dangerous period .....

☞ After 20 weeks No birth defect

❑ **PREVENTION**

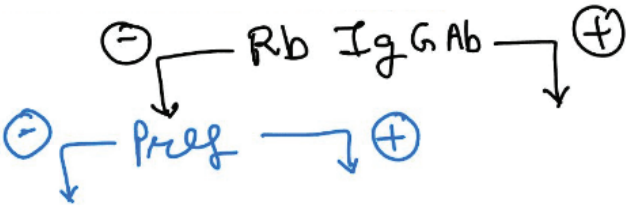
1. PRE-EXPOSURE PROPHYLEXIS

[Vaccine {MMR/MR}]

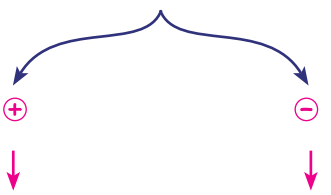
✦ PREFERENCE

- Rep age woman 15- 49 years
- 1- 14 years
- Routine vaccination

2. POST EXPOSURE PROPHYLAXIS



3. **Screening Ig G**



Active space

PNEUMONIA

Global action plan for prevention of Pneumonia and diarrhoea

❑ Goal for 2025

1. Mortality
   - Pneumonia [<5 years] : < 3/1000 lb
   - Diarrhoea [<5 years] : < 1/1000 lb
2. Incidence
   - Severe pneumonia [<5 years] : ↓ by 75% [base 2010]
   - Severe diarrhoea [<5 years] : ↓ by 75% [base 2010]

❑ NORMAL RR/minute

- 0-2 month .....
- 2-12 month .....
- > 12 months .....
- **CLASSIFICATION OF PNEUMONIA**

	No Pneumonia	Pneumonia	Severe / Very Severe
1. RR	.....	.....	.....
2. URTI Symptoms	.....	.....	.....
3. Accessory Resp muscle Involvement	.....	.....	.....
4. Features of Shock	.....	.....	.....
5. IMNCI category	.....	.....	.....
6. Treatment			

Active space

❑ VACCINATION TO PREVENT PNEUMONIA

- .....
- .....
- .....

TUBERCULOSIS

NTEP [January 2020]

1. SPUTUM EXAMINATION

✦ SPUTUM SAMPLES -

- Two Sample
  - .....
  - .....

✦ TRANSFER: - [Triple Packing]

✦ Result

.....

.....

..... †

2. X RAY chest
3. Other test : Culture, molecular assay, line probe assay, DST

Active space

LATENT TB [TB INFECTION]

1. TUBERCULIN SKIN TEST [TST]

   Given by : Von Pirquet (1907)

- Antigen : .....

- Reading : .....

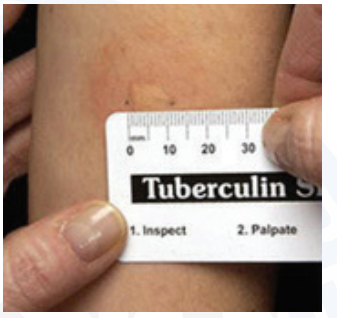
  ≤ 5mm = ...

  6-9 mm = ...

  ≥ 10 mm = ...

Tuberculin Skin Test [TST]

- Neg TST
  - HIV
  - Malignancy
  - Military TB
  - Malnutrition
  - Viral infection



Tuberculin Skin Test [TST]

2. IGRA TEST [Interferon Gama release assay]

CASE FINDING

- Passive : Suspicious patient goes to Health worker, Health centre
- Intensified : search for patients in clinic

  Suraksha [HIV], NCD [DM, Cancer]
- Active : search by health worker [in Vulnerable population]
- Population based screening

  a) Vulnerability mapping

  b) Periodic screening of vulnerable pop^{n}

  c) Leveraging Outreach of other health Programme

CATEGORY OF TB PATIENT

- Drug sensitive TB : Sensitive to Rifampicin
- Mono resistant TB : Resistance to one first line ATT drug (Other than Rifampicin, INH)

Active space

- Poly drug resistant TB : Resistance to > 1 first line ATT (Other than Rifampicin, INH)
- Multi drug resistant TB : Resistance to both INM, Rifampicin
- Pre XDR : Resistance to INH, Rifampicin and any of flouro quinolones
- XDR TB : Pre XDR + Resistance to any group A drugs (Bedaquillin Linezolid)

TREATMENT (Daily Dots Regime)

1. DRUG Sensitive TB

IP	CP	DURATION
.....	.....	.....

2. MDR / RR TB
   a) Shorter oral MDR Rx
      (4-6) (Bdq (6month) Lfx Cfz, Z, E, H^h, Eto) → IP
      (5) (Lfx, Cfz, Z, E,) → CP

   b) Longer oral MDR / XDR TB RX
      (18-20) Lfx, Bdq, (6 month or more) LZd, cfz, CS

   c) B PaLM (MDR) / BPaL (Pre XDR)
      B PaLM → 6month (26 wk) in 7 month
      B PaL → 9month (39wk) in 10 month.

☞ NEW DRUG

- BEDAQUILINE
- DELAMINID
  - .....
  - .....

Active space

FOLLOW UP

- Drug Sensitive : - .....
- (MDR/XDR:)

  IP: .....

  CP: .....

☞ FDC (Fixed Dose combination drugs)

☞ WT BANDS:

.....

.....

☞ Reporting to district level -

.....

'NIKSHAY' (TB):

IT based software for tracking of TB patient

NIKSHAY POSHAN YOJNA

- .....
- Every patient
- Till treatment

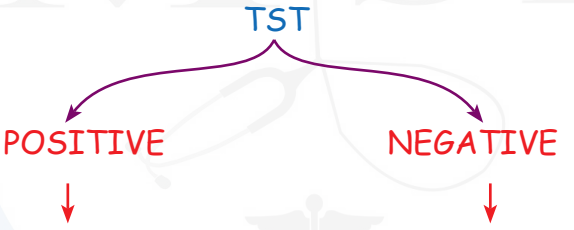
Active space

DOTS 99: ICT BASED ADHERENCE SUPPORT SYSTEM

Dots 99

PROPHYLAXIS OF NEWBORN

- Drug :
- Dose :
- Duration :
- After 3 months, do TST



TB WITH HIV

- First test for Diagnosis of TB in HIV pt : CB NAAT
- Dx of TB & HIV to gether
  - First start ATT for 2 wk
  - After 2wk add ART to avoid IRIS

Immune reconstitution inflammatory syndrome

Active space

- Isoniazid preventive therapyfor PLHIV

  all people living with HIV (Adult, children > 12month) without active TB, should receive preventive INH for 6month (10mg/kg/day)

TB IN PREGNANCY

1) Drug sensitive TB → same as non pregnant stage
2) MDR TB
   a) If MTP possible (24wk) → MTP → start B PaLM
   b If MTP not possible (Not willing, > 24wk) start oral short MDR, or Long oral MDR Rx with modification

PREVENTION OF TB

BCG Vaccination → As per UIP
BCG Vaccination → Adult BCG

END TB STRATEGY (2016-2035)

- Goal: End global TB epidemic by 2035
- Vision : world free from TB
- TARGET : 90 : 90 : 90
  - Dx 90% of TB pt and start Rx
  - Reach 90% of key population
  - 90% of treatment success
- STRATEGY : Test, treat, completed treatment
- Pillars :
  1. Integrated Pt centered care & prevention
  2. Bold policies and supportive system
  3. Intensified research and innovation

Active space

NATIONAL STRATEGIC PLAN FOR TB ELIMINATION (2017-2025)

Vision/goal : TB elimination by 2025 (TB free India with zero death, ds and poverty due to TB)

STRATEGY

The key strategies are as follows :

1. Private sector engagement
2. Active case finding
3. Drug resistant TB case management
4. Addressing social determinants including nutrition
5. Robust surveillance system
6. Community engagement and multi-sectoral approach

TARGET

1. 80% reduction in TB incidence (i.e. reduction from 211 per lakh to 43 per lakh)
2. 90% reduction in TB mortality (i.e. reduction from 32 per lakh to 3 per lakh
3. 0% patient having catastrophic expenditure due to TB

STRATEGIC PILLAR (DTPB)

Detect → Treat → Prevent → Build

Active space

INFRASTRUCTURE

1. DMC .....
   - .....
   - .....
   - .....
   - .....
2. TU .....
   - .....
   - .....
   - .....
   - .....
   - Senior TB Lab Supervisor (STLS)
   - .....

Indicators

1. ARI
   - .....
   - .....

Active space

EXTRA EDGE

PRESUMPTIVE TB CASE

- Fever ≥ 2week
- Cough ≥ 2week
- Significant weight loss > 10%
- Night sweat
- Haemoptysis
- Abnormal Xray chest

PRESUMPTIVE PAEDIATRIC TB

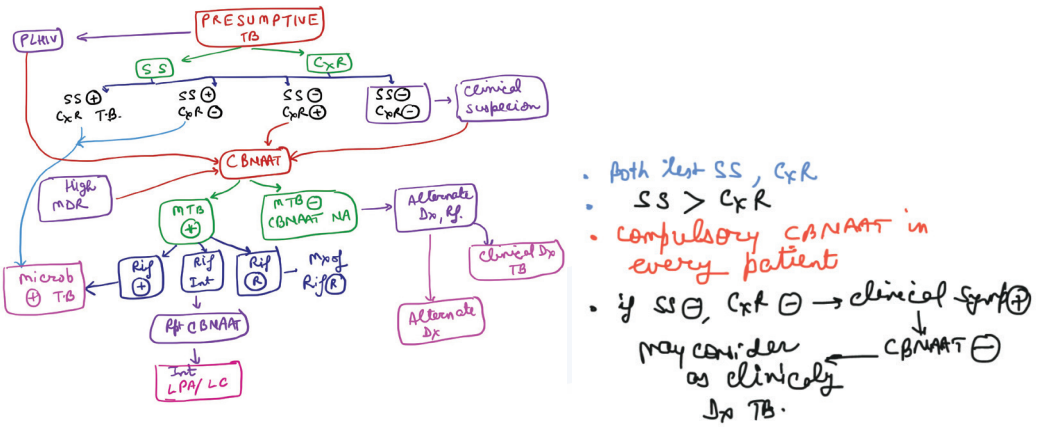
- Fever ≥ 2 week
- Cough ≥ 2 week
- Weight loss of ≥ 5% in last 3 month
- No weight gain in past 3 month
- H/o Contact WITH TB case

PRESUMPTIVE DR TB

- New TB With HIV
- All newly reported TB cases
- Previously treated case
- TB with contact of DR TB
- Paediatric TB – not responding to Rx
- Patient with SS + ve on follow up during 1st line drugs

Active space

ALGORITHUM OF PRESUMPTIVE TB CASE



Algorithum of Presumptive TB Case

ALGORITHUM OF PRESUMPTIVE PEDIATRIC CASE

Algorithum of Presumptive Pediatric Case

Pharmacovigilance in TB Control Program

science and activities relating to detection, assessment, understanding and prevention of adverse effects or any other drug related problem

DT3C (Difficult to treat TB clinic)

Special clinic at state and national level to treat difficult TB cases

Active space

Definitions

- Presumptive case : Pt with Sign symptoms suggestive of TB
- Pulmonary TB : Case of TB involving lung parenchyma or tracheo- bronchial tree
  - ✦ Miliary TB -> Pulmonary TB
  - ✦ Tuberculous intra thorasic lymphadenopathy -> Extra Pul TB
  - ✦ TB pleural effusion -> extra pul TB
  - ✦ Both Pulmonary + Extrapulmonary TB -> Pulmonary TB

- New case : Pt who had never taken ATT or for < 1 month
- Previously treated Patient : Pt taken ATT > 1 month

a) Relapse : Previously treated for TB now had recurrent episode of TB (relapse / re infection)
SS +ve any time after treatment completion

b) Treatment after failure
- SS +ve anytime during treatment after previous SS -ve report
- SS +ve at 5th month of treatment

c) Treatment after loss to follow up
taken ATT > 1 month but donot know outcome

Cured : Taken full course of ATT and SS -ve in end of treatment and one previous report

Active space

Treatment completed :
Taken full ATT treatment but report of sputum smear not available

High Risk group for TB

Vulnerable	Underserved	At Risk Population
↑ TB exposure due to work or living	limited access to quality TB services	At Higher Risk of TB infection
• Prisoners • Sex workers • Miners • Hospital visitor • HCW	• Migrant labour • Children • Refugee • Illegal miners • Women c̄ gender disparity • Undocumented migrants	• PLHIV • DM • Smoking • Alcohol use • Drugs addict • Mal- Nourished • Steroid Rx

B Pa LM regiemen

Dosing of component drugs for adults and adolescents (aged ≥14 years) for BPaLM

Drug	Dose
Bedaquiline (100 mg tablet)	400 mg once daily for 2 weeks, then 200 mg 3 times per week afterwards OR 200 mg daily for 8 weeks, then 100 mg daily
Pretomanid (200 mg tablet)	200 mg once daily
Linezolid (600 mg tablet)	600 mg once daily
Moxifloxacin (400 mg tablet)	400 mg once daily

Eligibility Criteria

1) Pulmonary or Extra PulTB except CNS TB, miliaryTB, osteoarticular TB
2) Age ≥ 14 yr
3) No known allergy to BPaLM component
4) No evidence of resistance to BPaLM component
5) All person regardless of HIV status
6) NO XDR TB

Active space

7) No Pregnancy, Lactation

8) QtcF ≤ 450 ms in males and ≤ 470 ms in females

9) Person with exposure of less than one month intake of Bdq, Lzd and / or Pa in the past

10) Person with exposure of more than one month intake of Bdq, Lzd and/or Pa and documented sensitivity to these drugs

TB Preventive Treatment (TPT)

A) Expanded eligible group

1) People living with HIV (All age)

2) All Household contact of Pulmonary TB pt

B) Expansion to other risk group

Children / Adults on immunosuppressive therapy, silicosis, antiTNF Rx dialysis, transplantation

C/I

- Active TB pt
- Acute/chronic Hepatitis
- Regular/Heavy alcohol intake
- Use of Hepatotoxic drug (Nevirapin)
- S/S of peripheral neuropathy
- Allergy to drug of TPT

Test for Dx of TB infection :

1. Tuberculin skin test
2. IGRA
3. A I enabled handheld x ray

Rx

1.	6H : Daily isoniazid	6 month
2.	3HP : Weekly isoniazid and Rifapentine	3 month
3.	1 HP : Daily isoniazid and Rifapentin	1 month
4.	6 Lf : Daily levofloxacin	6 month

Active space

- Nikshay Mitra

  Adoption of TB patient by individual, organization, corporates for providing nutritional, diagnostic and vocational support

- Dots Providers Incentive
  - Completion of Drug sensitive TB Rx : 1000/-
  - Completionof MDR / XDR TB Rx : 5000/-

    2000 Rs endof IP, 3000 Rs end of CP
  - Referral of Presumptive TB pt to public Health facility and on DX as TB : 500/-

- Nikshay Sampark

  Toll free Helpline 1800116666

- TB AROGYA SAATHI

  Mobile app for TB symptoms, TB information

- 4 S screening :

  Cough, fever, Loss of wt, night sweat

- Neglecting TB notification

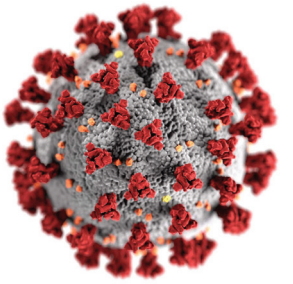
  IPC 269, 270

- TB travel reimbursement

  750/- if coming from notificable tribal area

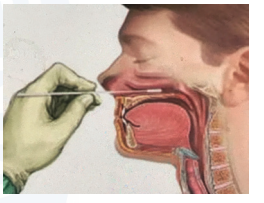
Active space

CORONA DISEASE [COVID- 19]



- Pandemic stage
- High number of Sub clinical cases and carriers
- I.P .....
- Quarantine period .....
- Route of Transmission
  - Droplets
  - Contact of articles having covid virus
  - Fomites
- SAR
- CFR

DIAGNOSIS



1. RT-PCR Test
2. Rapid Antibody kit test
3. Rapid Antigen Test

TYPES OF CONTACT

1. High Risk
   - Who live in same house hold
   - Anyone in close proximity [Less than 3 feet]
   - Touched body fluid of patient
   - Touched or cleaned linen, dish
   - Direct contact of patient without PPE Kit
   - Passenger with symptomatic patients (< 3 ft) (> 6 hrs)
2. Low Risk
   - Travelled in same environment
   - Shared same space

Active space

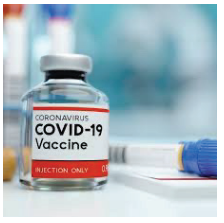
CATEGORY OF PATIENTS

- **1. Very mild and mild cases**
  - s/s of fever, URI, spo2 95% Or more, R/R : < 24/ min
- **2. Moderate cases**
  - Spo2 90- 94%, R/R – 24-30/mins, S/S of LRTI [Pneumonia] not very severe
- **3. Severe Cases**
  - S/S of lower respiratory disease, Spo2 less than 90% [room air]
  - RR > 30/min

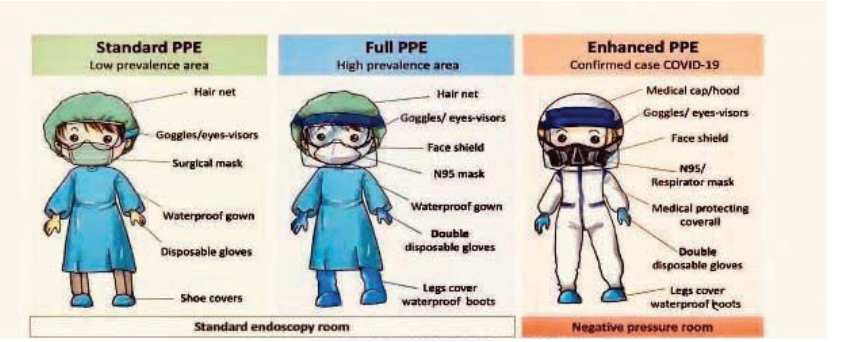
HEALTH CARE CENTERS

- **1. Covid Care Centers [CCC] (Mild, very mild case)**
  - School, Hotel, Stadium, Lodges, Quarantine Centre, Railway coaches, CHC
- **2. Dedicated Covid Health Centers (Moderate cases)**
  - District Hospital, Private hospitals, Nursing Home
- **3. Dedicated Covid Hospital [with ICU Facility] (severe cases)**
  - Medical college, Super specialty hospital

HOW TO PREVENT COVID-19



Active space



2. ENVIRONMENTAL MEASURES

3. SOCIAL DISTENCING

DEAD BODY MANAGEMENT

- ✓ Plug oral, nasal orifices
- ✓ Body pack in leak proof plastic bag
- ✓ Disinfect bag with 1% Sodium hypochlorite
- ✓ No autopsy

VACCINE

Contraindications of Vaccines

1. Hypersensitivity to previous dose during trial

Active space

Sided effects

1. Minor side effects at site of injection
2. S/S of viral infection
3. Rare - demyelinating disorders

AVAILABLE VACCINE: -

Name	Age	Type	Dose	Tempera-ture	Efficacy
• COVISHIELD • [AstraZeneca + SII]	≥ 18Yr	Viral vector	2 I/m 28 days	+ 2- + 8°c	70-80%
• COVAXIN [Bharat Biotech ICMR]	≥ 15Yr	killed	2 I/m 28 days	+ 2- + 8°C	70-80%
• SPUTNIK [Gameliya]	≥ 18Yr	Viral vector	2 I/m 21 days	+ 2- + 8°C	92%
• MODERMA	≥ 18Yr	mRNA	2 I/m 28 days	-20 -50°C	95%
• PFIIZER [Bio tech]	≥ 16Yr	mRNA	2 I/m 21 days	-70°C	95%

MIST

Active space

• COR BEVAX	12-14 yr ≥ 18Yr Boos-ter	Protein sub unit	2 I/m 28 days	+2-+80°C	>90%
• COVOVAX	≥ 18Yr Boos-ter	Recom-binant	2 I/m 21 days	+2-+80°C	>90%
• INCOVACC	≥ 18Yr Boos-ter	Viral Vector	2 dose nasal 28 days	+2-+8°C	90%

1. COVISHIELD

➢ ChAdOx1n COV-19

2. COVAXIN

➢ NIV - 2020-770

Myths of Vaccination

1. Covid vaccine after ..... wk of covid recovery
2. Covid vaccine after ..... wk of monoclonal Antibody, plasma
3. Blood donation after ..... of vaccine or recovery
4. Covid vaccine after ..... wk of recovery with co morbid diseases

MONKEY POX

Agent : Monkey Pox virus double stranded DNA virus orthopox virus (genum) poxviride family

Host : Primate, squirrel, Rat

I.P : 6-13 days (5-21 days)

Active space

POC : 1-2 day ← (R) → Complete fall of scabe

CFR : 3-6%

SAR : 10%

MOT : 1. Direct contact with body fluid / leison material

2. Droplet

3. Contaminated cloth / linen

C/F

1. Prodromal symptom : fever, headache, muscle-ache, lymphadenopathy, sore throat

2. Rash : Pleomorphic (Vesicopustular)

: Centrifugal destribution (face to arm, leg, palm, sole)

Dx : RT-PCR

Complication : Secondary infection pneumonia,

sepsis, encephalitis

corneal ulcer - loss of vision

Active space

Mx :

- Isolation till fall of scab.
- Supportive Rx
- Rehydration
- Good Nutrition
- Cover Rash c long sleeve shirt and pant
- Antiviral (TECOVIRIMAT)
- Vaccinia Ig

Prevention by Vaccination

- Type - JYNNEOS ACAM 2000

A. JYNNEOS : live vaccine, modified vaccinia Ankara - Bavarian Nordic (MVA-BN) high Risk ≥ 18 years 2 doses at gap of 4 wk s/c Route, safe in immuno compromised

B. ACAM 2000 : live attenuated small Pox vaccine effective against monkey pox

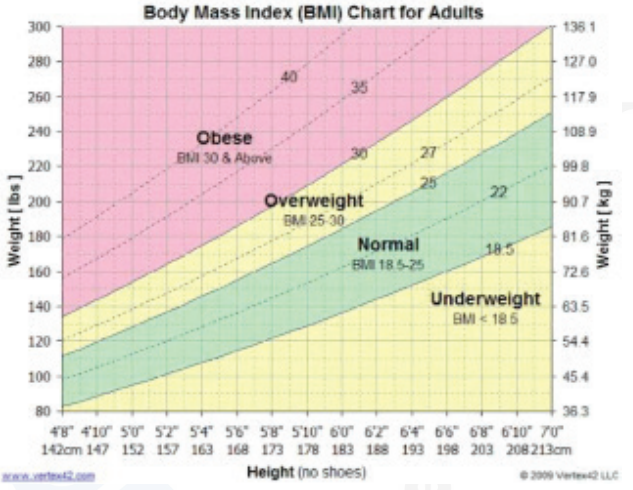
Active space

NON COMMUNICABLE DISEASES

OBESITY

Index :

❑ **Quetlet index (BMI)**



BMI Chart

- Most important WHO obesity index
- BMI = ...

	GLOBAL	INDIAN
Normal		
Over weight		
Obesity		

Active space

BASED ON PERCENTILE

- ✓ < 5th percentile :
- ✓ 5th - 85th percentile :
- ✓ 85th - 95th percentile :
- ✓ > 95th percentile :
- ✦ Undernutrition :

  Mild : .....

  Moderate : .....

  Severe : .....

- ❑ **Ponderal Index**
- ❑ **BROCA index**

  .....
- ❑ **Corpulence Index**

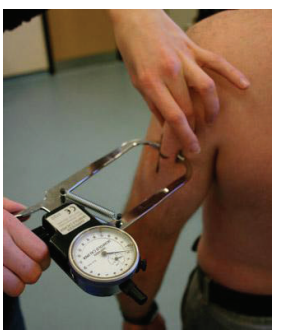
☞ MORBID obesity : .....

Waist hip ratio

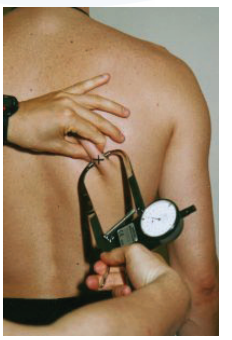
W : H → > 0.9-

→ > 0.85 -

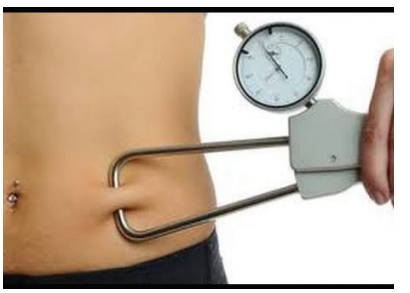
Skin fold thickness



Herpenden Skin Caliper



Skin Fold Thickness



Skin Fold Thickness

Active space

- Instrument Herpenden Skin Caliper
- Most important site : .....

  Other : mid biceps

  Anterior abdominal wall

  Interscapular

  Infrascapular
- Mid triceps >32 mm

  >18 mm

  Waist height ratio

  > 0.5 – risk of CVD

CARDIO VASCULAR DISEASES

❑ **Role of BP :**

- Very important risk factor of CVD – BP
- Systolic BP more important

❑ **Role of smoking :**

- Very important risk factor
- Synergistic with other
- Depends on :

  Duration of smoking

  Number of smoke /day
- No role of filtered cigarette
- Cessation effect

❑ **Important study done to know Dietary risk factor of CVD**

- .....
- .....

Active space

❑ STEP APPROACH

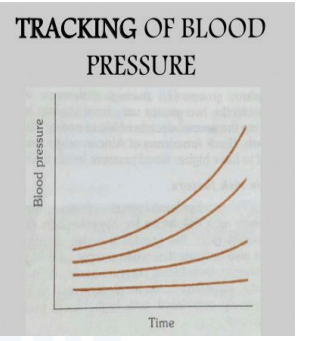
✦ .....

Step 1. .....

Step 2. .....

Step 3. .....

❑ TRACKING PHENOMENON



High BP in childhood – .....

Low BP in childhood – .....

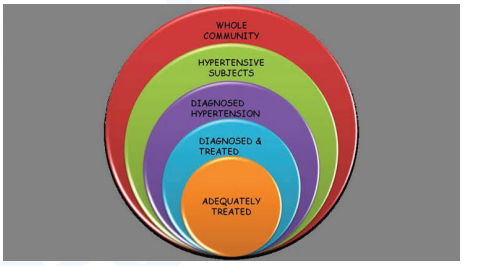
❑ Rule of ½ FOR HT

½ of community = ...

½ of HT = ...

½ diagnosed = ...

½ treatment = ...



❑ PRUDENT DIET

- Total fat energy : .....
  .....
- SFA : .....
- UFA : .....
- Transfat : .....
- Daily salt : .....
- Daily cholesterol : .....

Active space

CANCERS

❑ **Most common cancer:**

1. World – .....
2. Male (World) – .....
3. Male (India) – .....
4. Elderly male – .....
5. Young male – .....
6. Female (world) – .....
7. Female (India) – .....
8. Female in rural area – .....
9. Occupational in India – .....

Most common cause of cancer death :

	Global	India
Male	.....	.....
Female	.....	.....

BLINDNESS

❑ **Cutoff for blindness:**

❑ **MCC of blindness:**

1) All age :.....

2) Children : .....

Active space

CATEGORY OF BLINDNESS

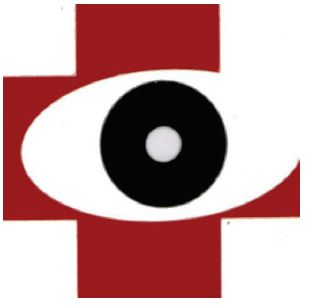
Category	New Classification	Vision	Old Classification
0	normal vision (no visual impairment)	6/6-6/18	normal vision
1		<6/18-6/60	
2		<6/60-3/60	
3		<3/60 -1/60	
4		<1/60 – P/L +	
5		<P/L + - P/L-	
9			

National program for Prevention and control of blindness and visual impairment (NPCBVI)

Vision 2020

- Avoid preventable cause of blindness by 2020
- Goal : Reduce Prevalence of blindness – to 0.3% by 2020
- Diseases included under vision 2020

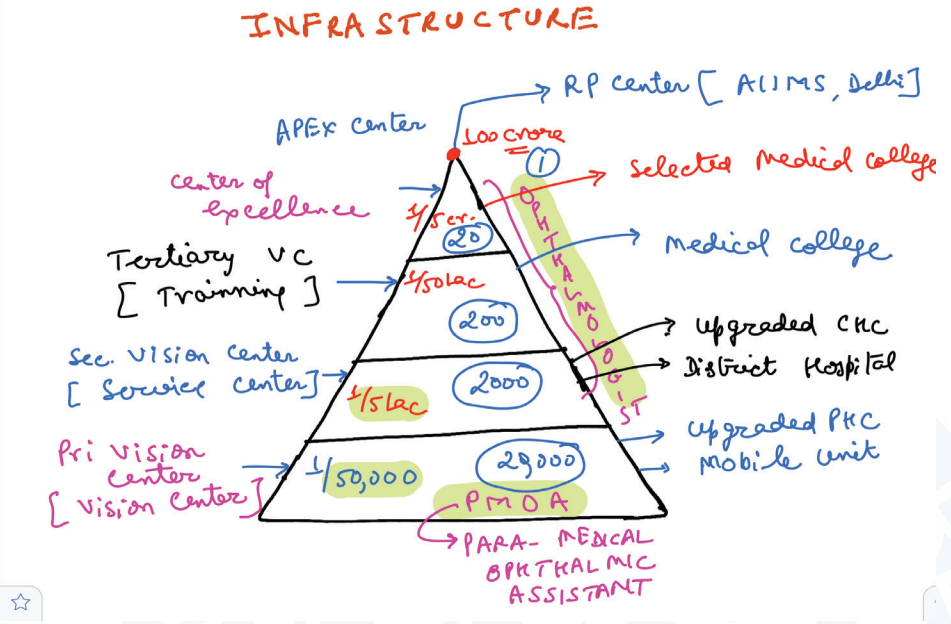
Global	India
Cataract	cataract
Refractory error	refractory error
Childhood blindness	childhood blindness
Trachoma	trachoma



Logo of NPCBVI

Active space

INFRASTRUCTURE ,LEVEL, STAFF POSTED, POPULATION COVERAGE



- **Vaccine to prevent blindness**
  - .....
  - .....
- **Nehru dristi yojna**
  - Promote eye donation
  - Corneal extraction

Active space

EXTRA EDGE

1. Rheumatic Heart Disease

A) Primary Prevention

- Prevent first attack of RF by identifying all patients of streptococcal throat infection and treat them with penicillin
- Single I/m injection of 1.2 million unit of Berzathine benzyl penicillin for adult 6,00,000 unit for children or oral Penicillin for 10 days
- MCH and school Health services are utalized for this purpose

B) Secondary prevention

- Identifying those who have had RF and giving them one intramuscalar injection of benzathine benzyl penicillin (1.2 million units in adults and 600,000 units in children) at intervals of 3 weeks. This must be continued for at least 5 years or until the child reaches 21 years whichever is later.

- For patients with carditis (mild mitral requrgitation or healed carditis) the treatment should continue for 10 years after the last attack, or at least until 40 years of age, which ever is longer

- More severe valvular disease or post-valve surgery cases need life-long treatment

C) Non medical measures :

- Improve living conditions break cycle of poverty-disease-poverty

D) Evaluation

- Know prevalence of RHD in school children from periodic survey (6-14yr Age group at 5yr interwal in sample size of 20,000 - 30,000)

Active space

2. National Sickle cell Anaemia Elimination Mission

- Goal :
  - Eliminate sickle cell disease as a public health problem in India before 2047.

- Objectives :
  - Provision of affordable, accessible, and quality care to all SCD patients.
  - To reduce the prevalence of SCD and sickle cell trait.

- Strategies
  - Primary prevention : awareness generation, counsellig
  - Secondary prevention : screening for detection
  - Holistic management & continuum care

- The program shall be a part of NATIONAL HEALTH MISSION.
- Sickle cell cards
  - For purpose of pre-marital and pre-conceptional counselling by matching.

3. National Action plan for prevention and control of snakebite envenoming (NAP-SE)

- AIM :

  To prevent and control snakebite envenoming in order to halve the numbers of deaths and cases of disability that it causes by 2030.

- Strategic Actions:

• Human Health : Provision of anti snake venom at all health facilities, strengthening surveillance of snakebite cases and deaths in humans:

Active space

- Strengthening of emergency care services at District Hospitals/ CHCs including services for ambulances, institutionalisation of Regional Venom Centre's

4. National Suicide prevention strategy:

AIM : Reduction in suicide mortality by 10% by 2030

STRATEGY :

- ❑ Establish effective surveillance mechanisms for suicide
- ❑ Establish psychiatric OPD
- ❑ Integrate a mental well-being curriculum in all educational institutions within the next eight years.
- ❑ It envisages developing guidelines for responsible media reporting of suicides, and restricting access to means of suicide.

5. ZooWin:

- ❑ A digital platform aimed at real time monitoring of Anti-Rabies Vaccines and Anti Snake venom.
- ❑ ZooWin operates similarly to the Co-Win and U-win platforms
- ❑ The platform will provide real-time tracking of vaccine stocks and ensure timely administration to victims of animal bites and snake bites.
- ❑ It aims to increase the availability of anti-rabies vaccines and anti-snake venom in health facilities.

Active space

NATIONAL HEALTH PROGRAMS

INDIA NEW BORN ACTION PLAN

- Started in 2014
- INAP(India newborn action plan) – Replacement of ENAP (Every newborn action plan)
- Goal – single digit NNMR, SBR by 2030

Logo of INAP

NATIONAL PROGRAM FOR HEALTH CARE OF ELDERLY (NPHCE)

- Provide separate, specified comprehensive health care to senior citizens at various levels of health care

RASHTRIYA VAYOSHRI YOJANA

- Providing physical aids and assisted living devices for senior citizens of BPL family

NATIONAL PROGRAMME FOR PREVENTION & CONTROL OF NON-COMMUNICABLE DISEASES (NP-NCD)

- Screening for BP, DM ≥ 30 yrs, pregnant women at subcenter

Logo of NPCDCS

MIST

Active space

NATIONAL MENTAL HEALTH PROGRAMME (NMHP)

- Mental clinic (district hospital)
- All types of mental illness
- National mental health policy (2014)

Logo of NMHP

AYUSHMAN BHARAT

National health protection scheme (AB-NHPS)

- Known as PM-JAY – PM Jan Arogya Yojna
- Started in 2018
- Benefit – 5 lac Rs / family / year
- PM Arogya Mitra (Health service provider)
- Target: cover 10 Cr family (50 Cr population)
- Private Hospital >10 bed can enroll
- School Health program and Health and wellness centers (Ayushman Arogya mandirs) are now part of this program

Logo of PM Jay

UJJAWALA (2007)

Prevention of human trafficking

- Rescue
- Rehabilitation
- Reintegration
- Repartriation

PM UJJWALA YOJNA (2016)

- Provide LPG connections to women from BPL family
- Protect from indoor air pollution

Active space

UJALA

- LED bulb affordable to community

NATIONAL PROGRAM FOR PREVENTION AND CONTROL OF DEAFNESS (2006)

- Avoid, manage deafness

  2nd DALY (hearing loss)

  1st DALY – depression

Logo of NPPCD

SWACH BHARAT MISSION

- Make urban, rural areas free from open defecation
- 100% scientific management of municipal solid waste

  2 components: Swach bharat mission – rural

  Swach bharat mission – urban

Logo of Swach Bharat Mission

SABLA SCHEME

❑ **Empowerment of adolescent girls and women**

- Nutritional supplementation
- IFA tablet
- Regular health checkup and referral
- Nutritional, health education
- Counseling on family welfare, ARSH, child care practice, home management
- Life skill education
- Vocational training

Active space

BETI BACHAO, BETI PADAO YOJNA

- Survival, protection, education of female child
- Improve sex ratio

Logo of Beti Bachao, Beti Padao Yojna

PM JAN AUSHADHI PARIYOJNA

- Provide quality medicine to poor at
- affordable rate

Logo of PM Jan Aushadhi Pariyojna

KAYAKALP

- Promotion of cleanliness in public area
- Awards to public health services
- Winners : 70% of score

DH: 50 lac	3 lac
SDH: 10 lac	1 lac
CHC: 15 lac	
PHC: 2 lac	50,000

Logo of Kayakalp

URBAN HEART

Urban health equity assessment and response tool (urban heart)

- Decision support tool to identify reduced health inequities in cities
- Enables local communities progressive managers and municipal national authorities to :

1. Better understand the unequal health determinants, unequal health risks, unequal health outcomes faced by people belonging to different socioeconomic group within a city

Active space

2. Use evidence for planning intervention
3. Participate in intersectoral collaboration
4. Apply a health equity lens in policy making

Logo of Urban Heart

JEET (JOINT EFFORTS FOR ELIMINATION OF TB) :

- Public private partnership in TB
- Collaboration by doctors for doctor promote awareness about
  1. Diagnostic modality
  2. Incentive
  3. Treatment option

MERM SYSTEM

- Medication event reminder monitor
- Digital box
- Reminder of pill intake at fixed time
- Remote medication adherence on Nikshay portal

AMRIT

- Affordable medicine and reliable Implants for treatment
- ↓ Expense by patient on cancer Rx, Heart ds
- Make them available, accessible, affordable

Affordable Medicines and Reliable Implants for Treatment

Active space

SAANS

- Social awareness and action plan to neutralise pneumonia successfully
- Incidence <3/1000 LB by 2025

Objective →

1) Community Awareness
2) Early Detection and Rx
3) Remove myths
4) Behavior change.

MONICA PROJECT

- WHO project
- Monitoring trends and determinants in cardio-vascular DS.

LEPROSY MASCOT

- ..

Sapna (India)

Meena (UNICEF)

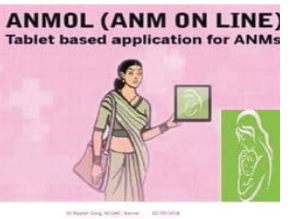
NIKUSHT

- Software for reporting and data Mx of Leprosy

Active space

ANMOL

- ANM on line
- Help in telemedicine in rural area.



SNEHA SHIVER

- Non Profit organization
- Work for women, children for their well being.

SWASTHYA SLATE

- Point of care
- Conduct 33 test in rural area
- Connected to tablet like Structure
- Get Plan of action
- Data monitoring

NATIONAL HEALTH PORTAL

- Single window access to health information and services delivered by government of india

NATIONAL HEALTH PORTAL

SAARTHI VAN

- Mobile van for FP Promotion
- Information & services at doorstep
- main focus on 20-24 yr

Active space

AYUSHMAN VAY VANDANA CARD

- Health insurance card for elderl more than 70 years under AB-PMJAY.
- Scheme is regardless of economic status.
- Senior citizens will be eligible for free medical treatment up to Rs 5 lakh under this scheme.

TELE - MANAS : TELE MENTAL HEALTH ASSISTANCE AND NETWORKING ACROSS STATES

- Free tele-mental health services all over the country.
- NIMHANS being the nodal centre.
- Toll-free, 24x7 helpline number (14416) has been set up across the country

OTHER E PORTEL

1. e-Hospital : Software for managing hospital in government sector
2. eRakt kosh : Digitise blood banks
3. eSanjeevani : Web based telemedicine solution
4. Mera aspataal : Patient Feedback on service quality
5. mDiabetes : For prevention and care of diabetes
6. m-cessation : Quit tobacco

Mera Aspataal (My Hospital)

e-Rakt Kosh
Blood Cell NHM

eSanjeevani
National Telemedicine Service

Active space

JAY VIGYAN

- Deeper undestanding of RF and RHD

INCENTIVE OT ASHA

1. Contraception
   - Delay of 1st Child for 2 Yr - 500/-
   - Gap 3yr in 2 baby - 500/-
   - Permanent method after 2nd Baby - 1000/-

2. Home delivery of contraceptive
   - Rs 1 : Pack of 3 Condom
   - Rs 1 : OCP
   - Rs 2 : ECP (ezypill)

3. HBNC
   - 250 Rs. for all visit (on 45th day)
   - Twin (500 Rs.)
   - Triplet (750 Rs.)
   - Visit at 2 Places Shared by 2 ANM if visit ≥ 3
   - Additional 50 Rs. /mth
     1. LBW baby follow up x 2yr
     2. SNCU discharge baby x1yr

4. HBYC
   - 250 Rs. for 5 visit

5. Immunization
   - Full immunization after 1yr - Rs. 100/-
   - Complete immunization till 2yr - Rs.75/-

6. Mobilizing and attending VHSND
   - 200 Rs/Session

Active space

7. Incentive to TB care providers on Rx completion
   - Drug Sensitive - 1000/-
   - MDR/XDR - 5000/- (Rs. 2000/- after IP, Rs. 3000/- after CP)
   - Referral of suspected TB patient to public health facility after TB diagnosis - 500 Rs.

8. Leprosy
   a) Detect new case c̄ out disability → 250 Rs.
   b) Detect new case c̄ disability → 200 Rs.
   c) Successful Rx completion → PBL → 400 Rs.
      → MBL → 600 Rs.

9. Kala AZAR
   1. Case detection, complete Rx follow up of Kala AZAR, PKDL → 500 Rs.
   2. One time wage loss incentive to pt → 4000/- PKDL
      → 5000/- VL.

Active space